



MEC Manager User Guide v4.0



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Notice

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Any dispute arising out of or in connection with this proposal or any contract that arises out of this proposal, including any question regarding its existence, validity, or termination, shall be referred to and finally resolved by arbitration in Singapore in accordance with the Arbitration Rules of Singapore International Arbitration Centre ("SIAC") for the time being in force, which rules are deemed to be incorporated by reference into this clause. The tribunal shall consist of a single arbitrator to be appointed by agreement between the parties and, if no agreement has been reached within twenty-one (21) days after a proposal as to the identity of the arbitrator is made by a party, then by the Chairman of SIAC. The language of the arbitration shall be English. The arbitration shall be administered by SIAC.

Version Control

Author	Summary	Release Date
Augustine Chan	v4.0 of MEC Manager User Guide.	31/05/2024
Augustine Chan	Updated captions for screenshots.	21/10/2024
Sharlotte Teo	Updated screenshot descriptions	13/11/2024

Introduction

This user guide is designed to provide you with an overview of the platform's features and functionalities, empowering you to navigate and utilize the platform effectively. It aims to educate you on the key aspects of the platform and enable self-service for future use.

The values in this user guide are just illustrative. Other values can be used by users based on their preferences or requirements.

Please note that certain features are accessible only by administrators. If you encounter these features without functionality, it's because they're exclusively available for admin-level access.

For any specific instructions or additional information on using particular features, refer to the platform's detailed documentation and resources provided within the platform itself. Should you



encounter any difficulties or require further assistance, please reach out to the designated support channels for prompt resolution.

We hope this user guide empowers you to understand the MEC Manager better, enabling self-service capabilities and facilitating a seamless user experience.

MEC Manager

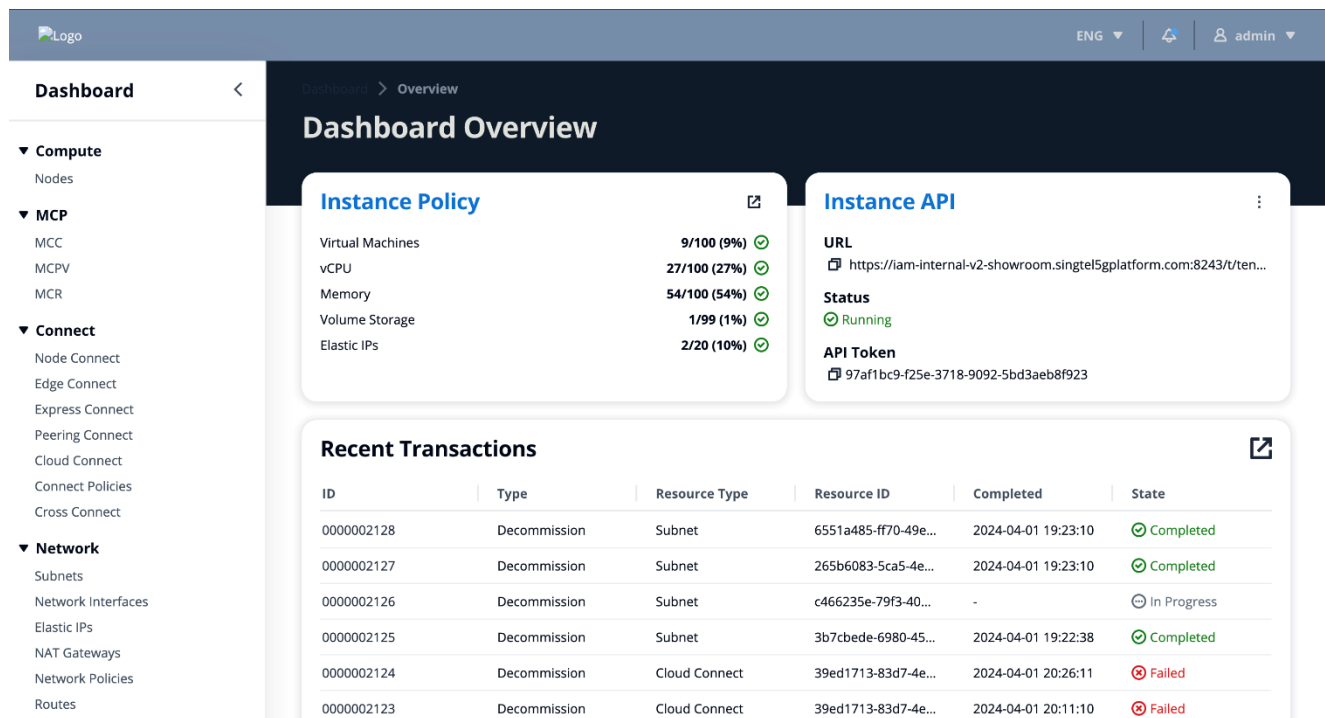


Figure: Dashboard - Overview

When users log in, they can find the side menu expanded, offering key categories like Compute, MCP, Connect, Network, Object Storage, Storage, Services, and Admin.

In the Dashboard Overview section, users can easily check their Instance Policy to see the usage/capacity of the existing virtual machines, including vCPU, Memory, volume storage, and

Elastic IPs, all represented in percentages. users can also access Instance API information, such as the URL (which can be copied), status (e.g., running), and API token (also copyable) at a glance.

In the Recent Transactions section, a table displaying columns like ID, Type, Resource Type, Resource ID, Completed (timestamp), and State will be seen.

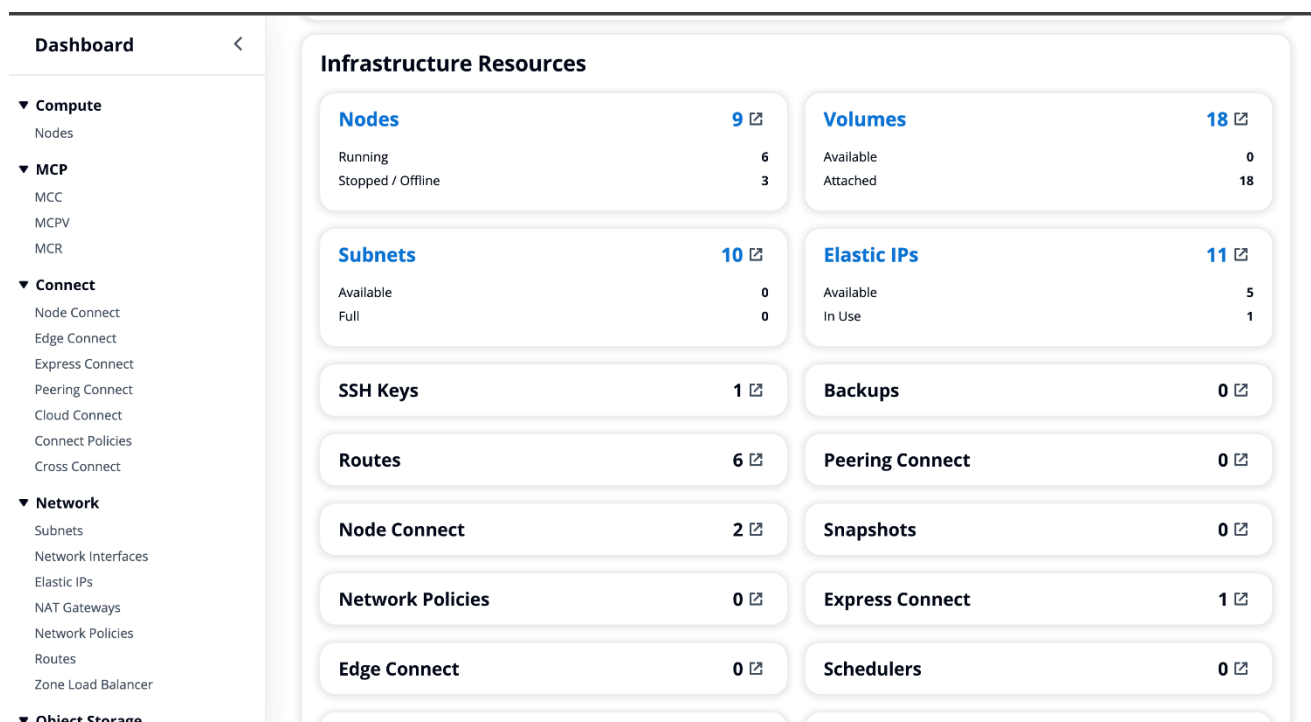


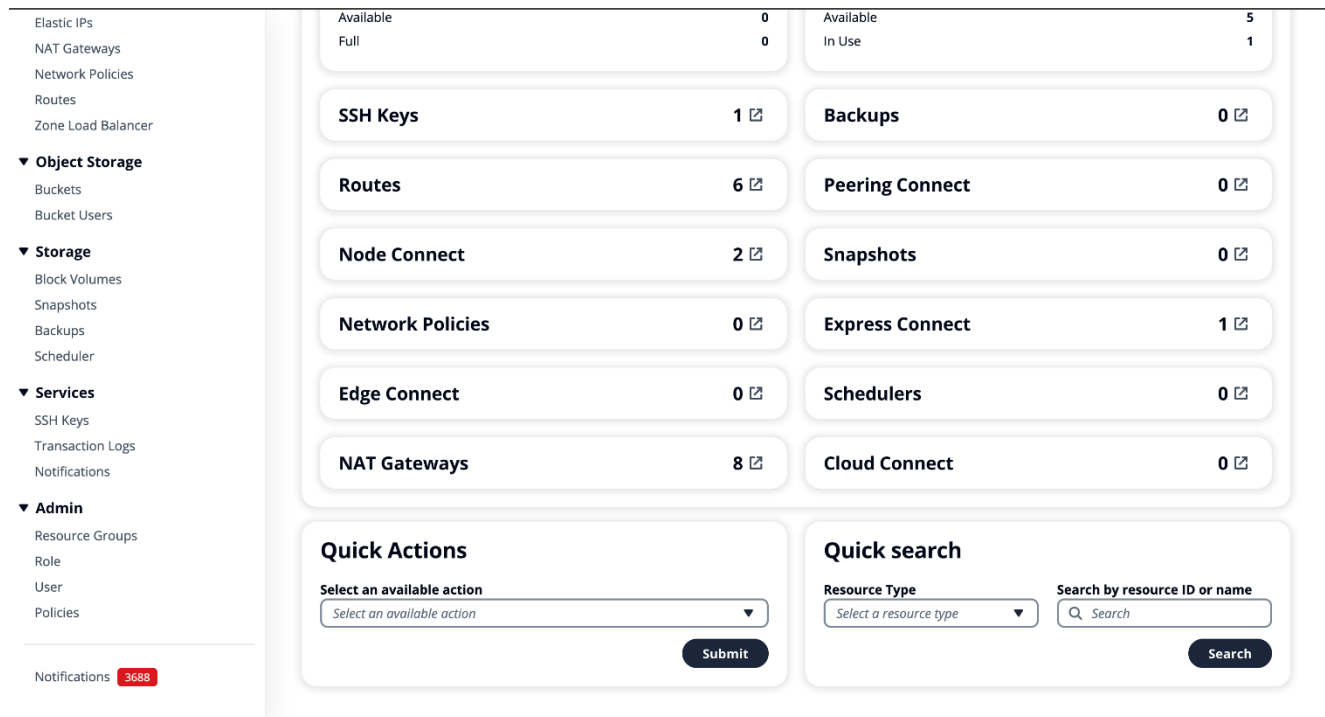
Figure: Dashboard – Infrastructure Resources

Within the “Infrastructure Resources” section, there are hyperlinks for these categories:

- Nodes
- Volumes
- Subnets
- Elastic Ips
- SSH Keys
- Backups
- Routes
- Peering Connect
- Node Connect
- Snapshots

- Network Policies
- Express Connect
- Edge Connect
- Schedulers
- NAT Gateway
- Cloud Connect

Clicking on any of these tabs, will route the user to its respective page.



The dashboard features a left-hand navigation menu with the following categories:

- Object Storage**
 - Buckets
 - Bucket Users
- Storage**
 - Block Volumes
 - Snapshots
 - Backups
 - Scheduler
- Services**
 - SSH Keys
 - Transaction Logs
 - Notifications
- Admin**
 - Resource Groups
 - Role
 - User
 - Policies

At the bottom of the navigation menu, there is a notification badge: **Notifications 3688**.

The main content area displays resource counts for various services:

Resource Type	Count	Link
Elastic IPs	0	
NAT Gateways	0	
Available Full	0	
Available In Use	5	1
SSH Keys	1	
Backups	0	
Routes	6	
Peering Connect	0	
Node Connect	2	
Snapshots	0	
Network Policies	0	
Express Connect	1	
Edge Connect	0	
Schedulers	0	
NAT Gateways	8	
Cloud Connect	0	

At the bottom of the dashboard, there are two sections:

- Quick Actions:** A dropdown menu labeled "Select an available action" with a "Submit" button.
- Quick search:** A search bar with a "Resource Type" dropdown and a "Search by resource ID or name" input field, with a "Search" button.

Figure: Dashboard – Quick Actions & Quick Search

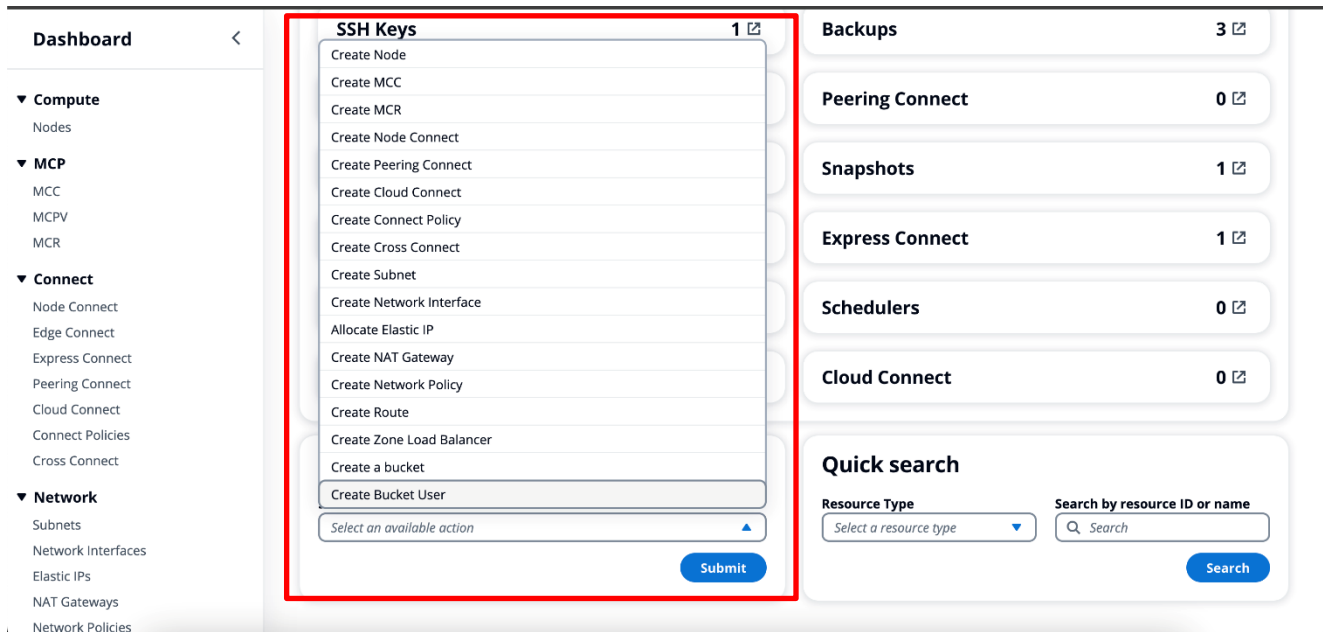


Figure: Dashboard – Quick Actions

Users have the option to utilize shortcuts within the "Quick Actions" section. By selecting from the various types of actions or processes such as creating a route, a node, a bucket, etc., from the dropdown menu, users will be directed to the initial step of the selected process upon clicking.

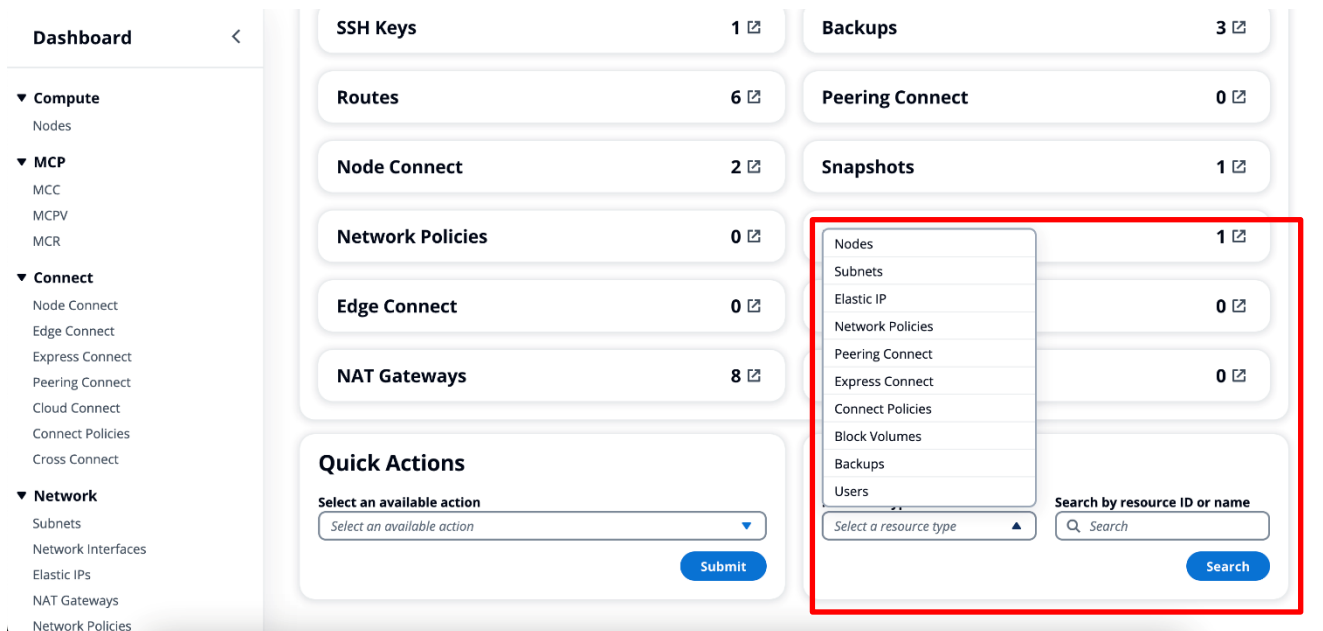


Figure: Dashboard – Quick Search

Users have the option to utilize the quick search bar within the "Quick Search" section. Firstly, by selecting the resource type and providing the resource ID or name in the field beside it.

2.0 Notifications

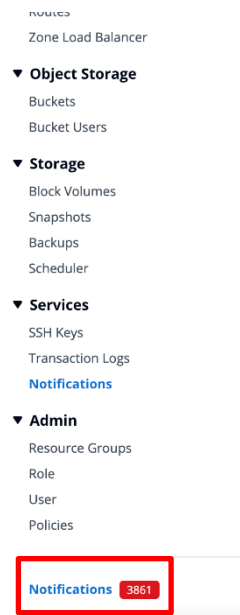
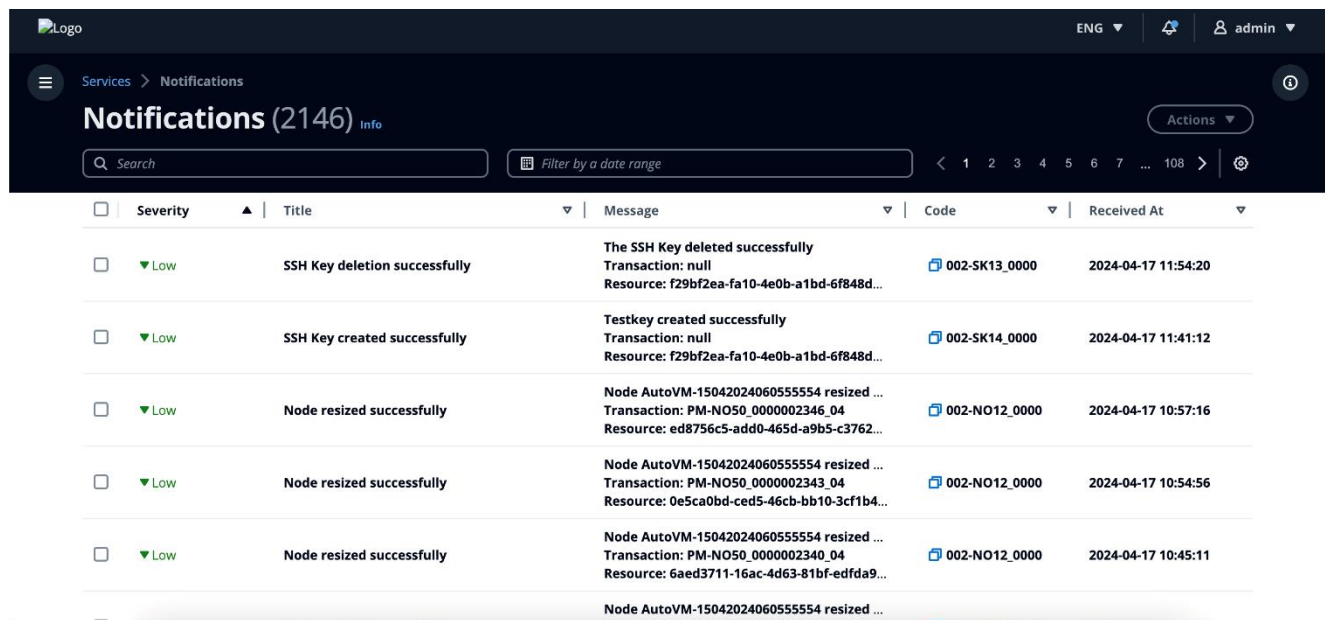


Figure: Viewing Notifications – Method 1

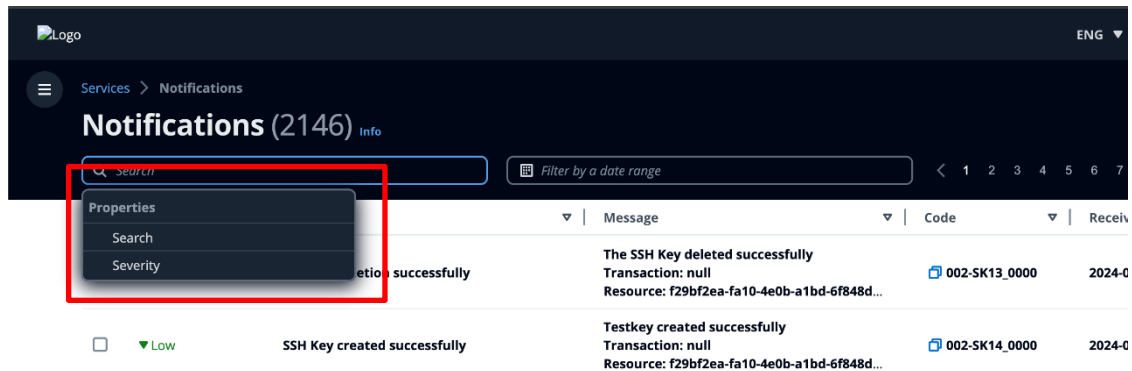
Users can navigate to the notifications section by expanding the side menu and clicking onto the “Notifications” label.



Severity	Title	Message	Code	Received At
Low	SSH Key deletion successfully	The SSH Key deleted successfully Transaction: null Resource: f29bf2ea-fa10-4e0b-a1bd-6f848d...	002-SK13_0000	2024-04-17 11:54:20
Low	SSH Key created successfully	Testkey created successfully Transaction: null Resource: f29bf2ea-fa10-4e0b-a1bd-6f848d...	002-SK14_0000	2024-04-17 11:41:12
Low	Node resized successfully	Node AutoVM-15042024060555554 resized ... Transaction: PM-NO50_0000002346_04 Resource: ed8756c5-add0-465d-a9b5-c3762...	002-NO12_0000	2024-04-17 10:57:16
Low	Node resized successfully	Node AutoVM-15042024060555554 resized ... Transaction: PM-NO50_0000002343_04 Resource: 0e5ca0bd-ced5-46cb-bb10-3cf1b4...	002-NO12_0000	2024-04-17 10:54:56
Low	Node resized successfully	Node AutoVM-15042024060555554 resized ... Transaction: PM-NO50_0000002340_04 Resource: 6aed3711-16ac-4d63-81bf-edfda9...	002-NO12_0000	2024-04-17 10:45:11

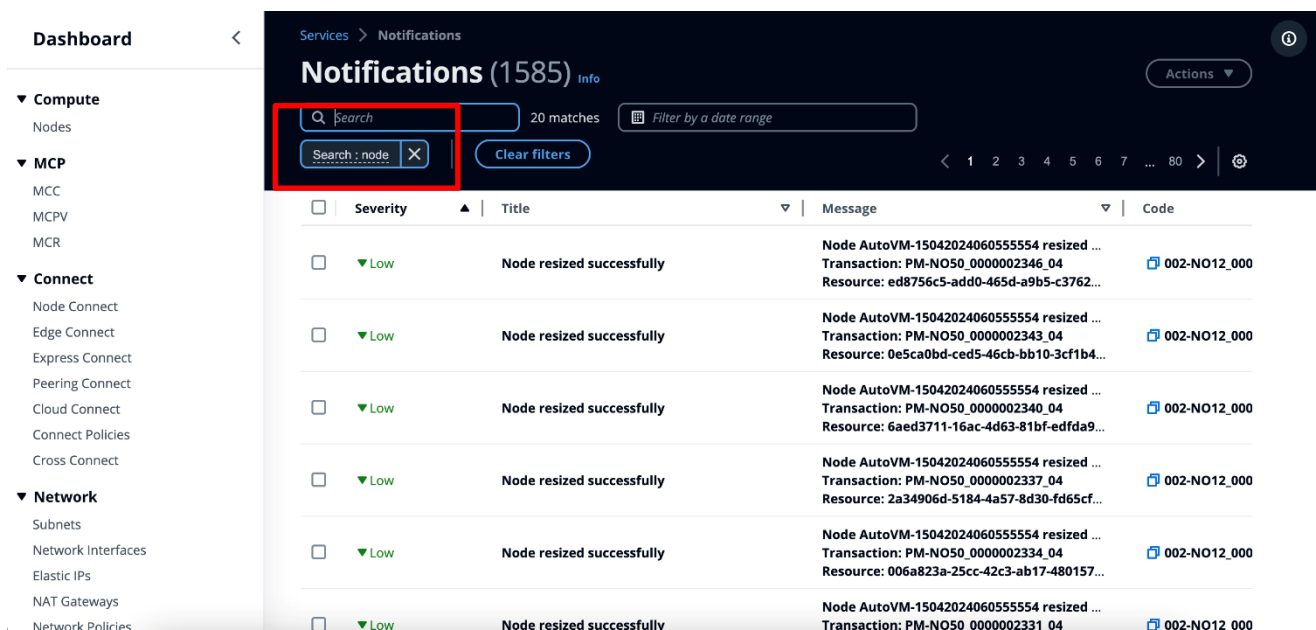
Figure: Notifications – Dashboard

In the “Notifications” page, users can opt to sort the records based on severity, title, message, code, or the “Received At” timestamp.



The screenshot shows the 'Notifications (2146)' page. A search bar is highlighted with a red box, and a dropdown menu is open, showing 'Properties', 'Search', and 'Severity'. The table below shows notifications with columns for Message, Code, and Received. The first notification is 'The SSH Key deleted successfully' with code '002-SK13_0000' and received date '2024-0'. The second notification is 'Testkey created successfully' with code '002-SK14_0000' and received date '2024-0'.

Figure: Notifications – Search/Severity



The screenshot shows the 'Notifications (1585)' page. A search bar is highlighted with a red box, and the value 'node' is entered. The table below shows notifications with columns for Severity, Title, Message, and Code. The first notification is 'Node resized successfully' with severity 'Low', title 'Node AutoVM-15042024060555554 resized ...', message 'Transaction: PM-NO50_0000002346_04', and code '002-NO12_000'. The second notification is 'Node resized successfully' with severity 'Low', title 'Node AutoVM-15042024060555554 resized ...', message 'Transaction: PM-NO50_0000002343_04', and code '002-NO12_000'. The third notification is 'Node resized successfully' with severity 'Low', title 'Node AutoVM-15042024060555554 resized ...', message 'Transaction: PM-NO50_0000002340_04', and code '002-NO12_000'. The fourth notification is 'Node resized successfully' with severity 'Low', title 'Node AutoVM-15042024060555554 resized ...', message 'Transaction: PM-NO50_0000002337_04', and code '002-NO12_000'. The fifth notification is 'Node resized successfully' with severity 'Low', title 'Node AutoVM-15042024060555554 resized ...', message 'Transaction: PM-NO50_0000002334_04', and code '002-NO12_000'. The sixth notification is 'Node resized successfully' with severity 'Low', title 'Node AutoVM-15042024060555554 resized ...', message 'Transaction: PM-NO50_0000002331_04', and code '002-NO12_000'.

Figure: Notifications - Searching using values.

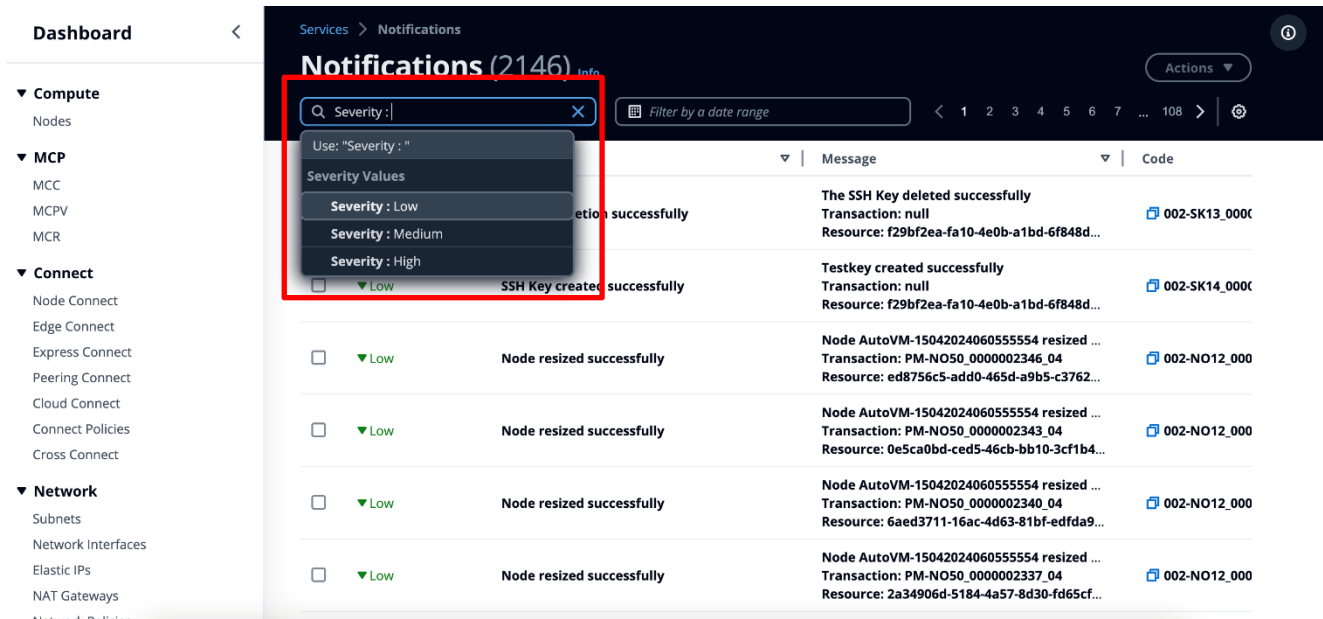


Figure: Notifications – Searching based on Severity.

Users can also toggle between these filters:

1. Search: This will allow user to provide values and then the search will scan through records containing those values.
2. Severity: Selecting a Low, Medium, or High value.

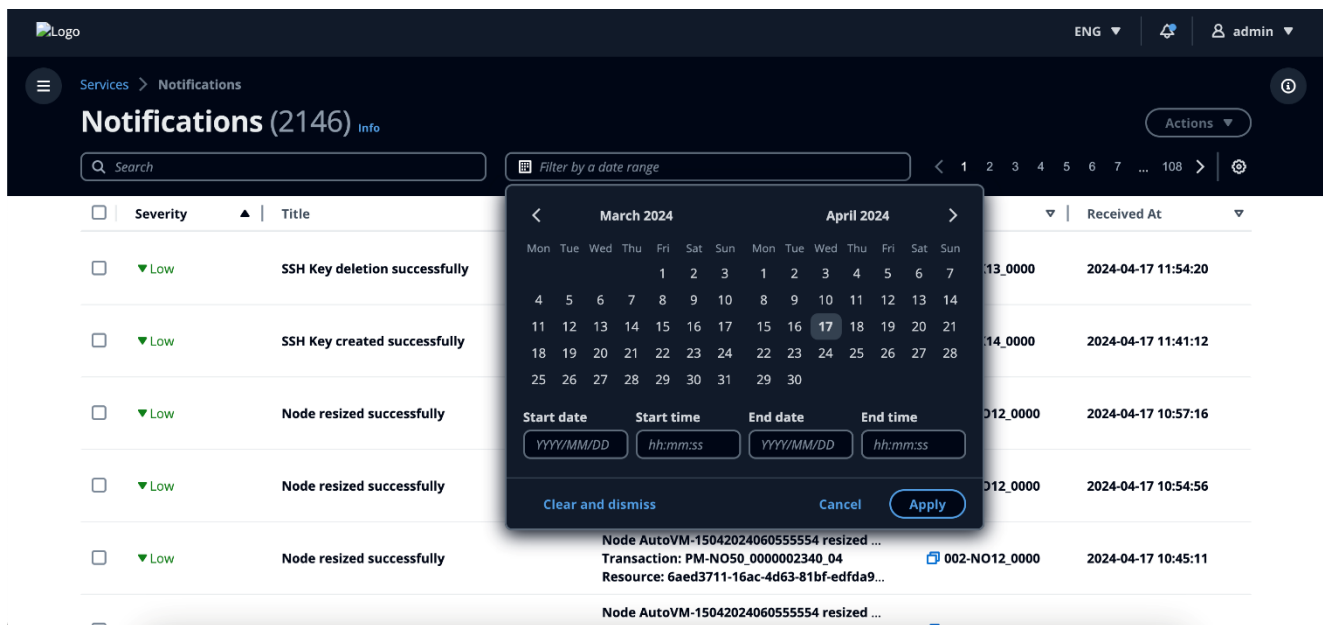
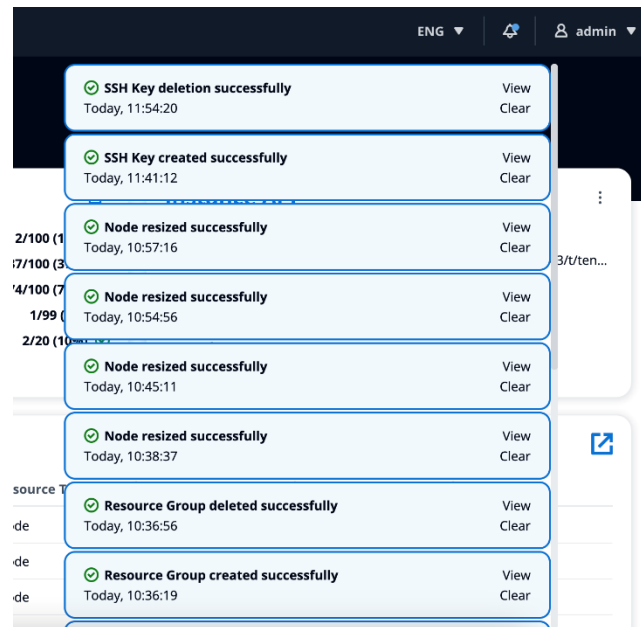


Figure: Notifications – Selecting a date range.

Users can opt to select a date range from the available date picker and apply the filter on the list of notifications.

Figure: Notifications



Alternatively, users can have a quick glance on the latest notifications by clicking the bell icon on the top right of the browser beside the profile menu.

3.0 Admin Controls

3.0.1 Policy Management

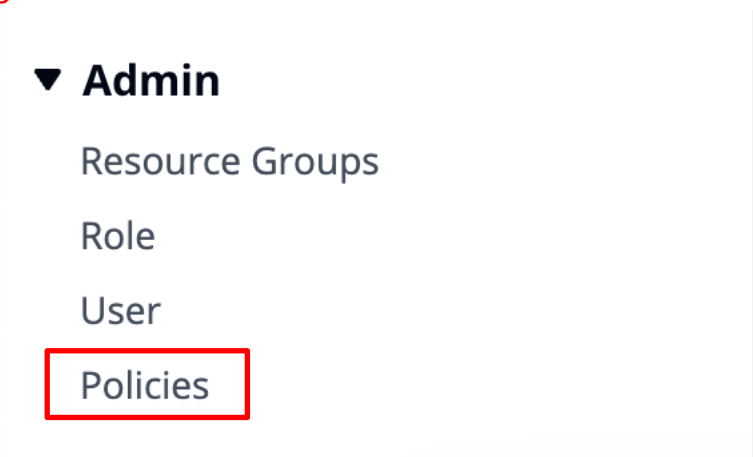
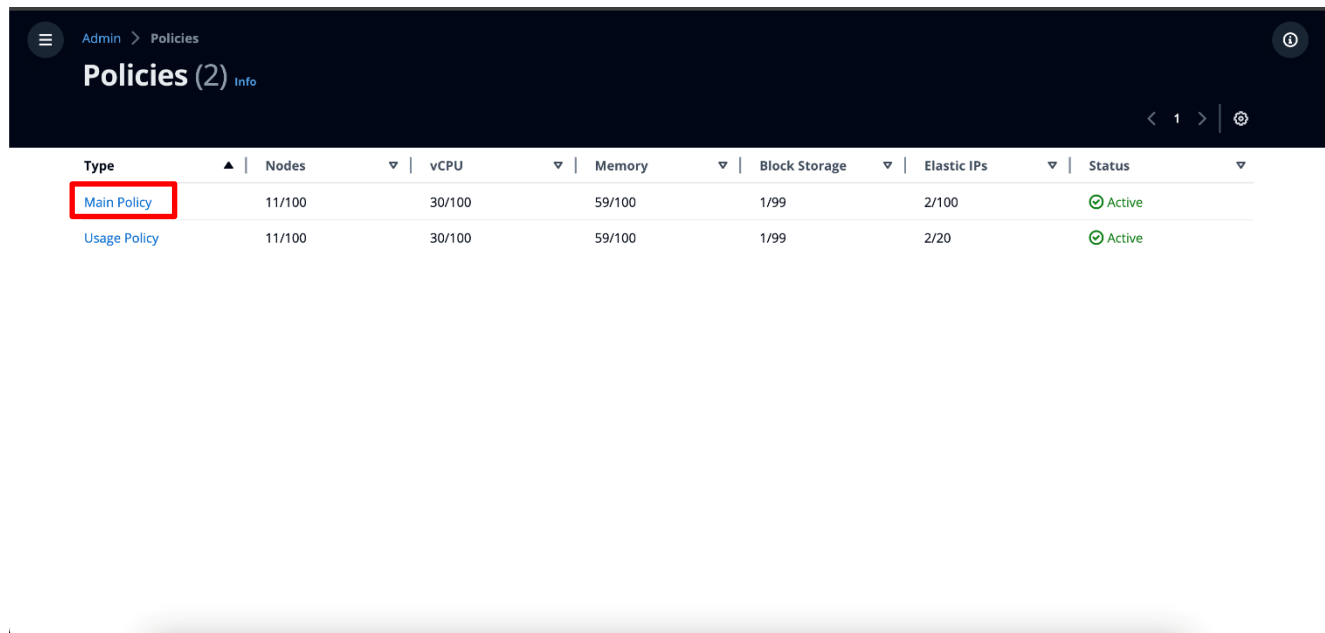


Figure: MEC Manager – Policies

To start navigating, click on the “Admin” tab and select “Policies” to land on this page.

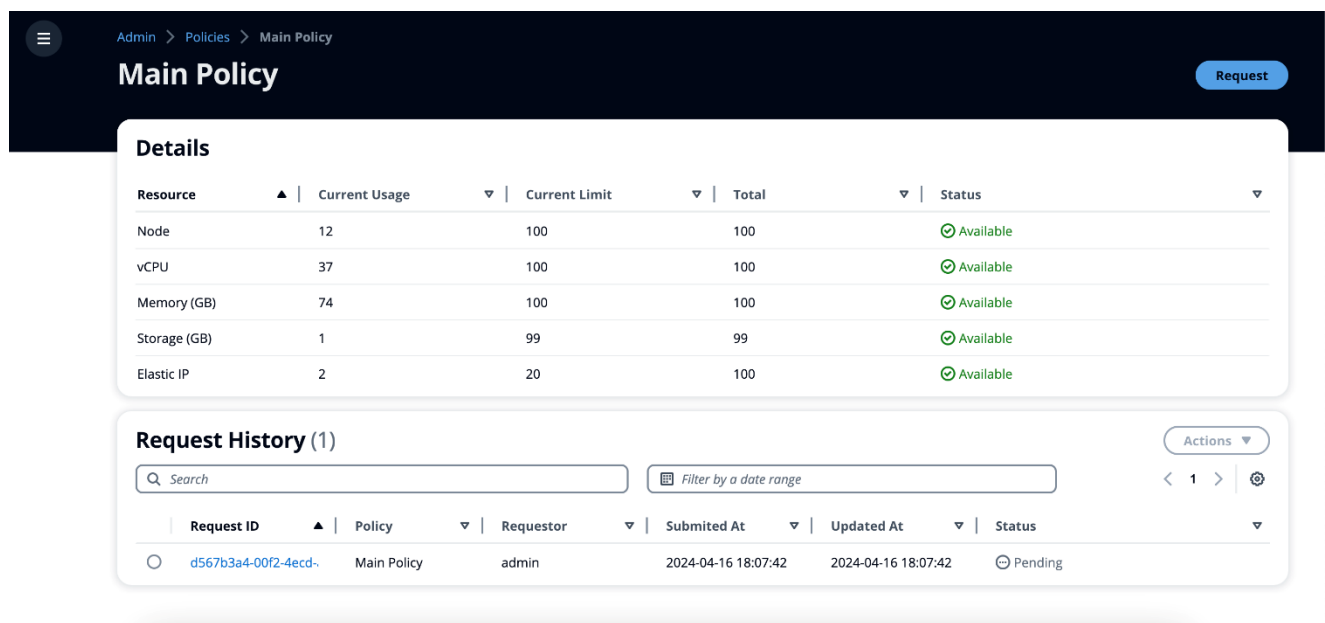
Click on the desired policy to view more details.



Type	Nodes	vCPU	Memory	Block Storage	Elastic IPs	Status
Main Policy	11/100	30/100	59/100	1/99	2/100	Active
Usage Policy	11/100	30/100	59/100	1/99	2/20	Active

Figure: MEC Manager – Policies

When you click on the policy, you can see an overview of the policy, the type of resource being used, current usage, current limit, total limit, and status. Additionally, there is a request history section displaying the request ID, policy details, requester information, submission time, update time, and status.



Resource	Current Usage	Current Limit	Total	Status
Node	12	100	100	Available
vCPU	37	100	100	Available
Memory (GB)	74	100	100	Available
Storage (GB)	1	99	99	Available
Elastic IP	2	20	100	Available

Request ID	Policy	Requestor	Submitted At	Updated At	Status
d567b3a4-00f2-4ecd-	Main Policy	admin	2024-04-16 18:07:42	2024-04-16 18:07:42	Pending

Figure: MEC Manager – Policy Request History

For a more detailed view of a specific policy request, navigate to the actions tab under the corresponding record in the request history view. Click on "View Detail," and a pop-up display will appear on the same screen, providing information about that particular request.

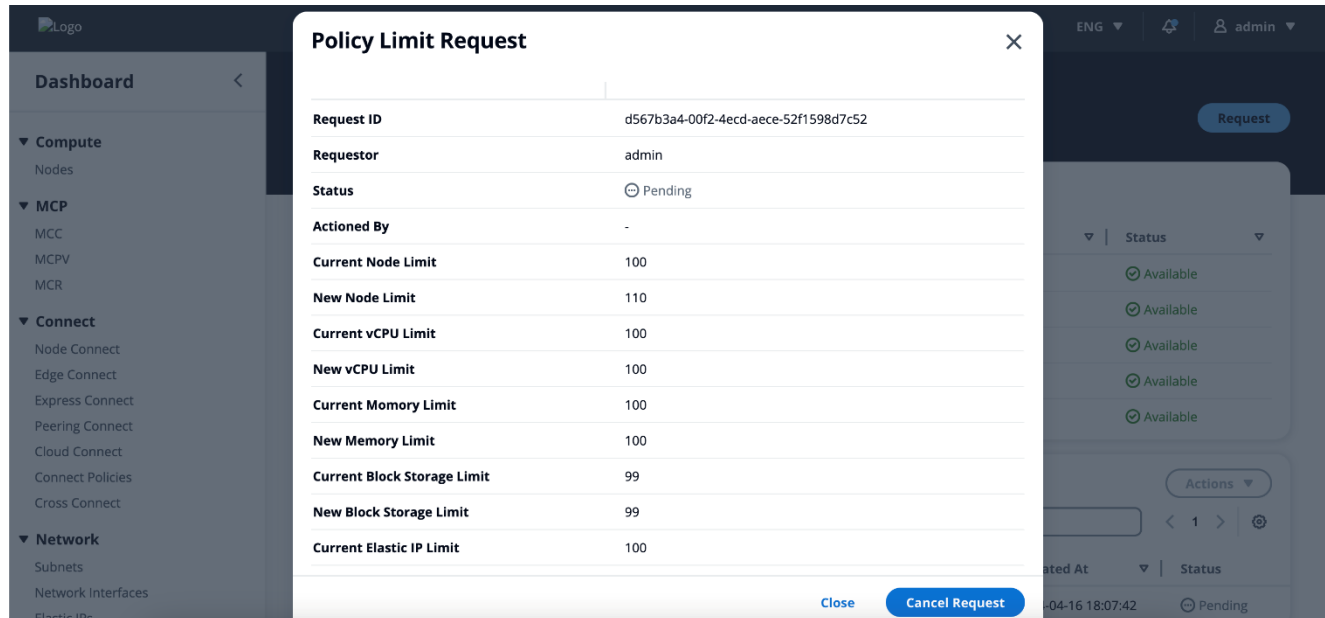


Figure: MEC Manager – Policy Limit Request

Within this pop-up, you will find key details of the request, including information such as the request ID, requester's details, status updates, current and new limits for various resources like node, vCPU, memory, block storage, and elastic IP. Additionally, timestamps for when the request was submitted and updated will be available, providing a complete overview of the request's evolution and specifics.

3.0.2 Viewing and Creating Resource Groups

By allocating various resources to a group known as a Resource Group, users may manage those resources more effectively.

The use of Resource Groups is optional and has no bearing on how the resources function. When the instance is created, it will come with a default resource group; users are free to add more if users so choose. When a resource is created, it is possible to designate it to a group.



Figure: MEC Manager – Resource Groups

To access the resource groups overview, go to the "Admin" tab, then select the "Resource Groups" tab.

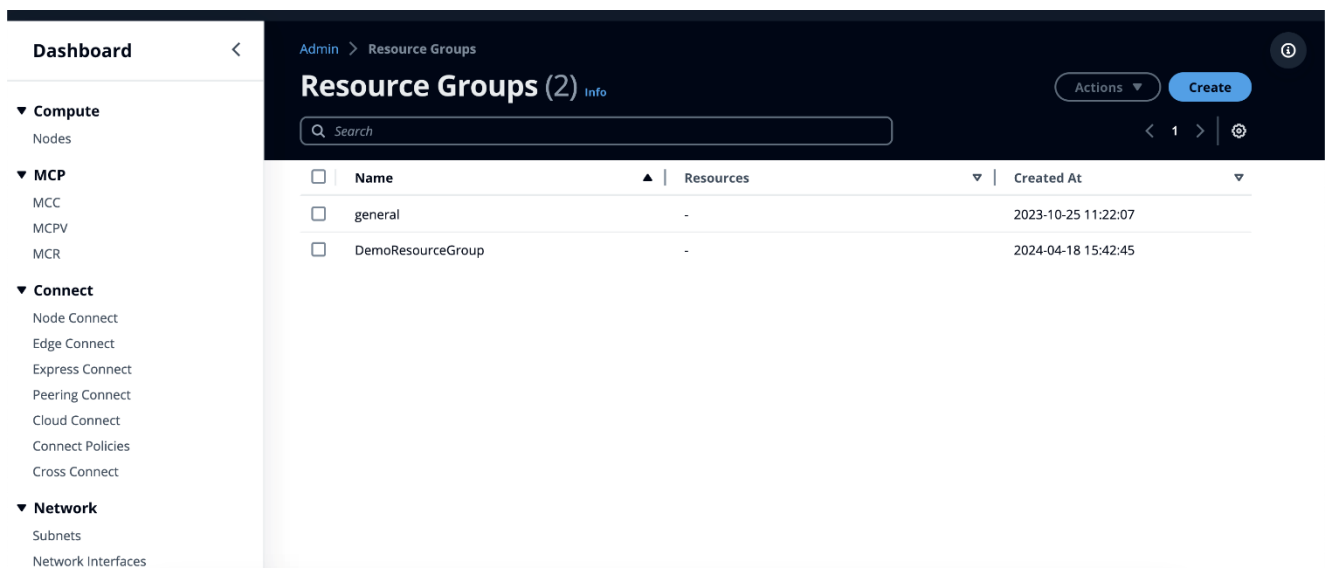


Figure: Resource Groups – Overview

Within this interface, you'll find the list of resource groups along with details like name, provider, and action options such as editing or viewing the specific group details.

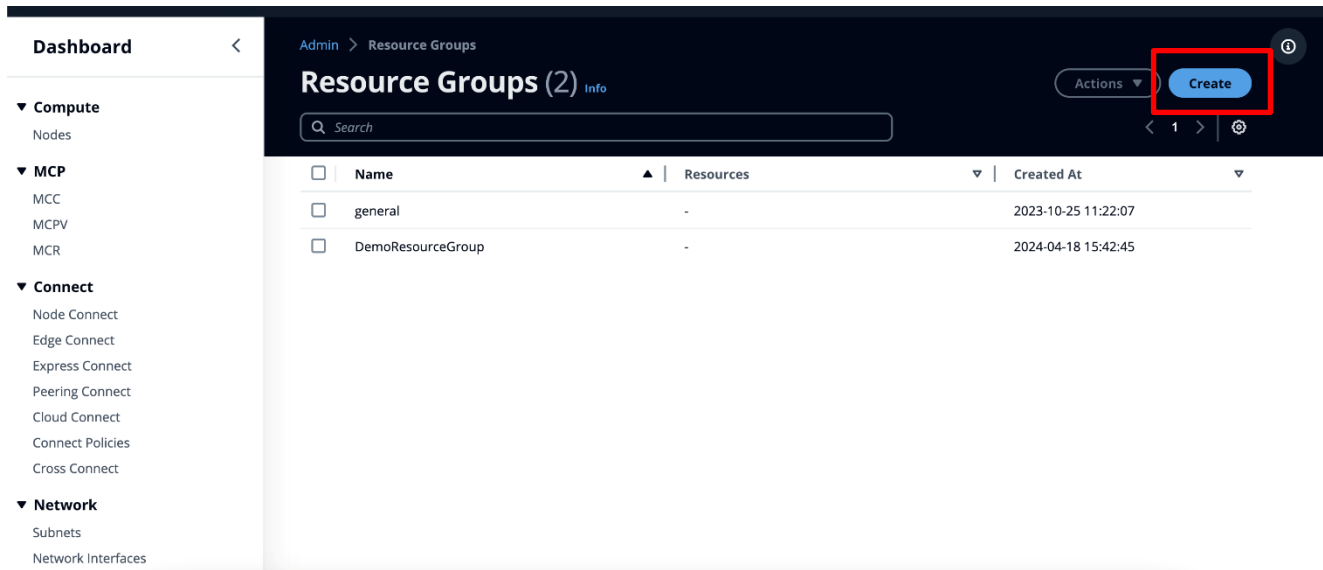


Figure: Resource Groups – Creating a Resource Group

You can initiate the creation of a resource group by clicking the "Create" button located on the main overview page of resource groups.

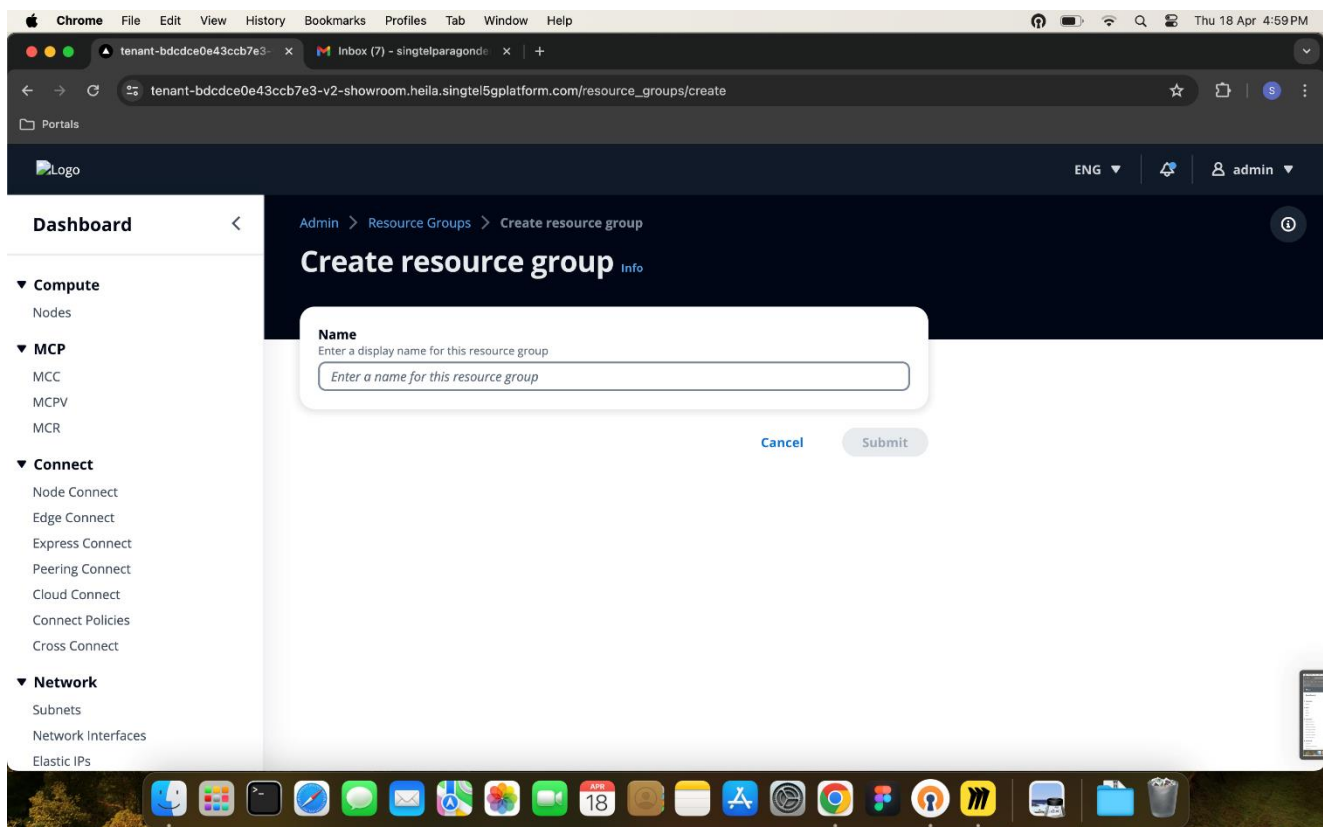


Figure: Resource Groups – Creating a Resource Group

You can start by providing your resource group name and then subsequently assign resources like network information (subnets, IPs, etc.) by editing the resource group from the main page.

3.0.3 User Count and Status for Assigned Roles

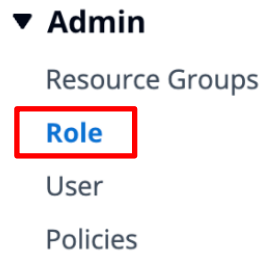
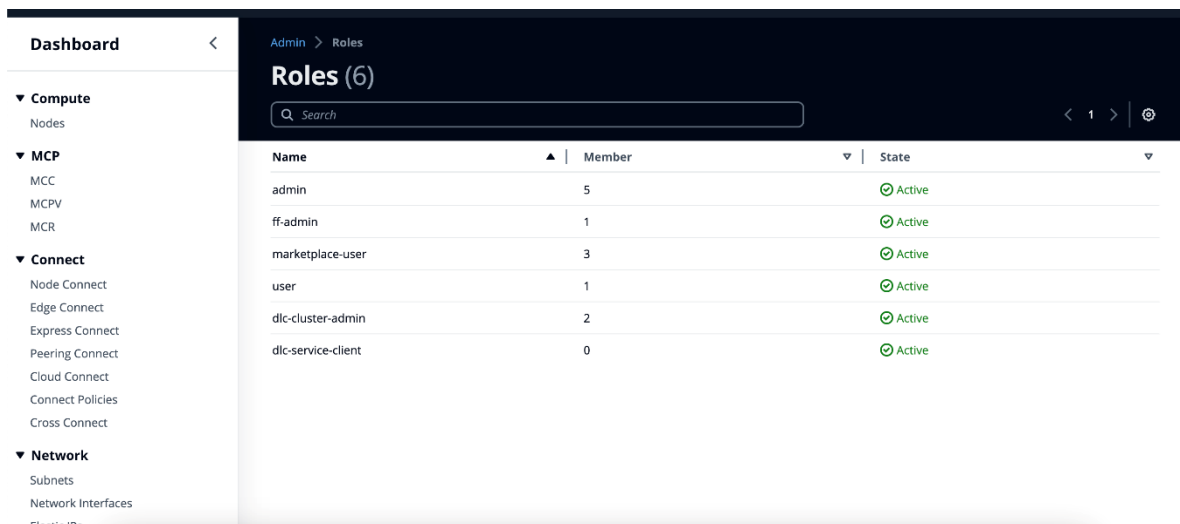


Figure: Admin – Roles

To get started, select the "Admin" tab, and then choose "Roles."



A screenshot of the 'Roles (6)' overview page. The page shows a table with columns: Name, Member, and State. The table lists six roles, all with a status of 'Active'.

Name	Member	State
admin	5	Active
ff-admin	1	Active
marketplace-user	3	Active
user	1	Active
dlc-cluster-admin	2	Active
dlc-service-client	0	Active

Figure: Admin – Roles Overview

Within this view, you'll find a comprehensive list of existing roles, displaying their respective names, total member counts, and the status of the roles (active/disabled).

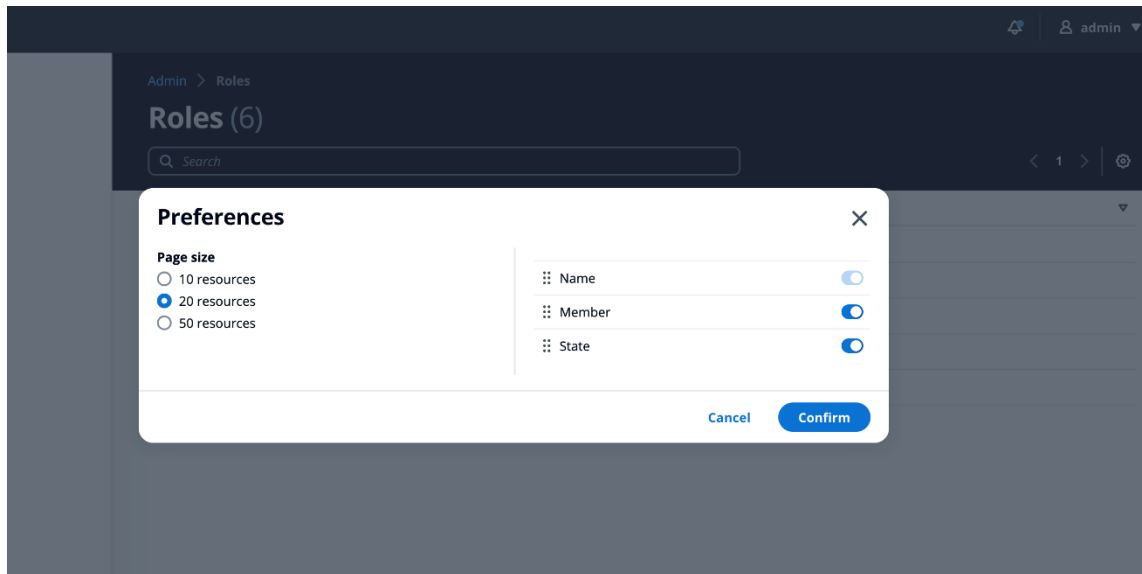


Figure: Admin – Roles – Preferences

Clicking on the settings icon within this dashboard will also allow the users to select the page size, and filters.

3.0.4 Creating a User

Do note that this action is only enabled for administrators.

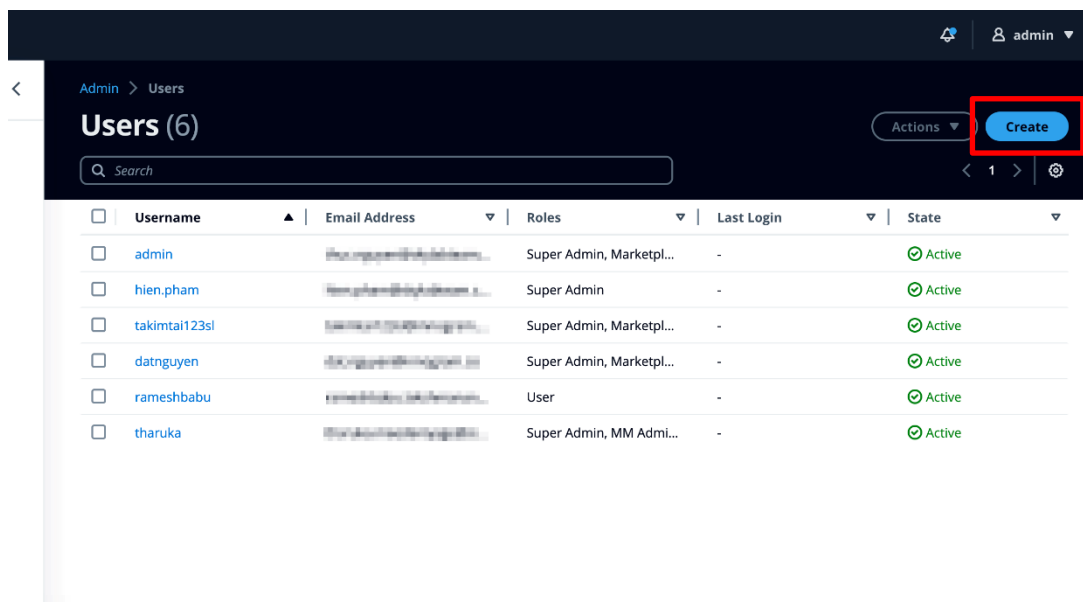


Figure: Admin – Creating a User

To begin the process of creating a user for MEC Manager, on the landing dashboard of the “Users” module, proceed to click on the “Create” button.

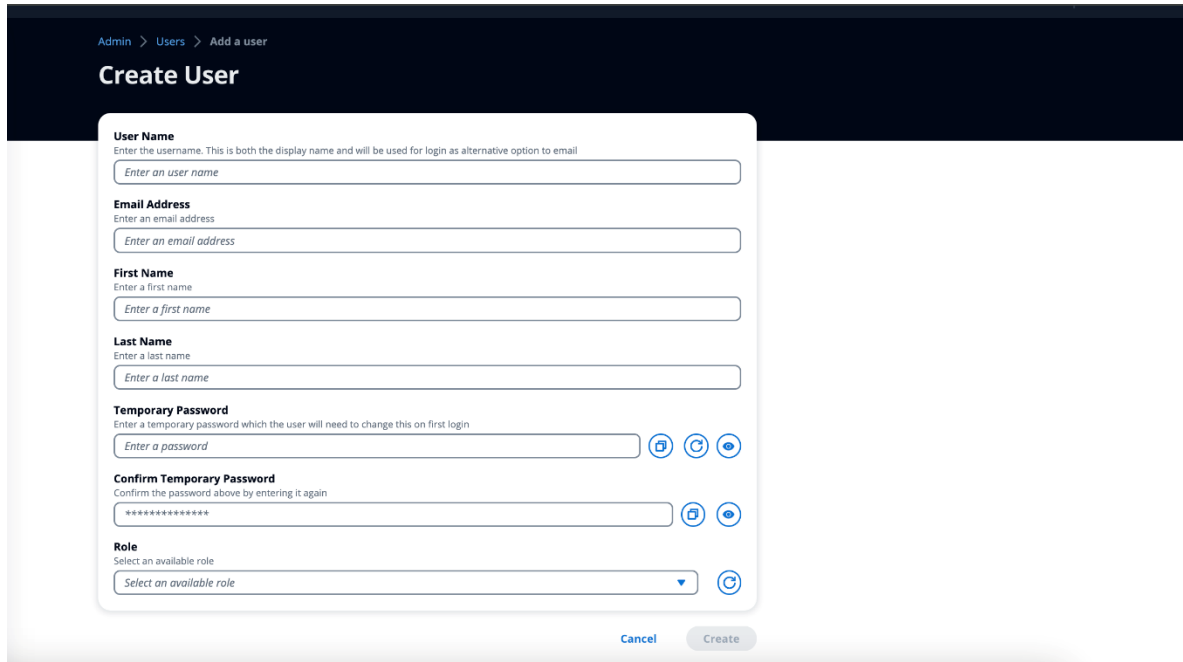


Figure: Admin – Steps to creating a User.

Once the user has landed on this view, proceed to follow the steps as listed below:

Step 1: Enter your username, which can be used as a backup for the email address you were given upon logging in.

Step 2: To access this user account, provide its email address.

Steps 3 and 4: Give the user's first and last name.

Step 5: Create a temporary password. The user will need to update it the first time they log in.

Step 6: Verify the temporary password.

Step 7: Based on the options provided from the dropdown menu, select the user's role.

3.0.5 Viewing User Details

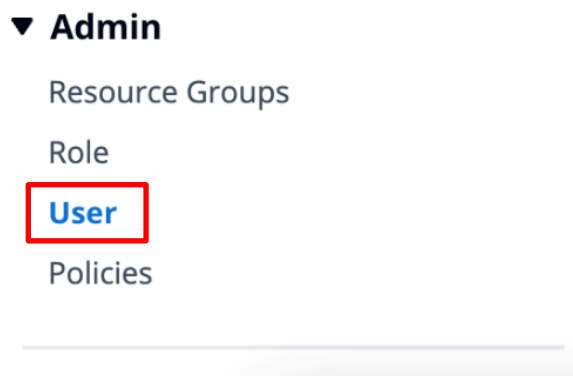


Figure: Admin – Users

To get started, select the "Admin" tab, and then choose "Users."

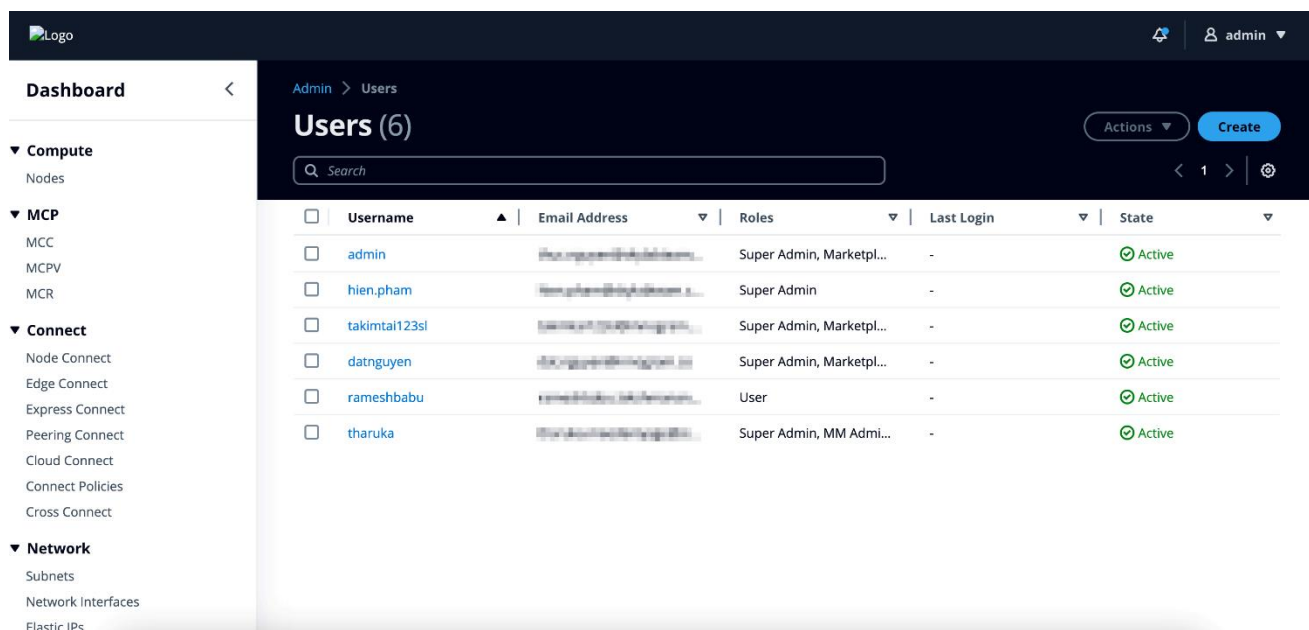


Figure: Admin – Users Dashboard

In the "Users" section of the admin panel, a user-friendly dashboard provides a table view featuring key details. This table includes the following columns:

- **Username:** Displays the user's name.
- **Email:** Shows the user's email address.
- **Role:** Indicates the role/roles assigned to the user.
- **Last Login:** Indicates a timestamp of the most recent login.
- **State:** Reflects the current status of the user account.

This streamlined presentation enables administrators to efficiently manage user information, facilitating quick access to essential details and actions.

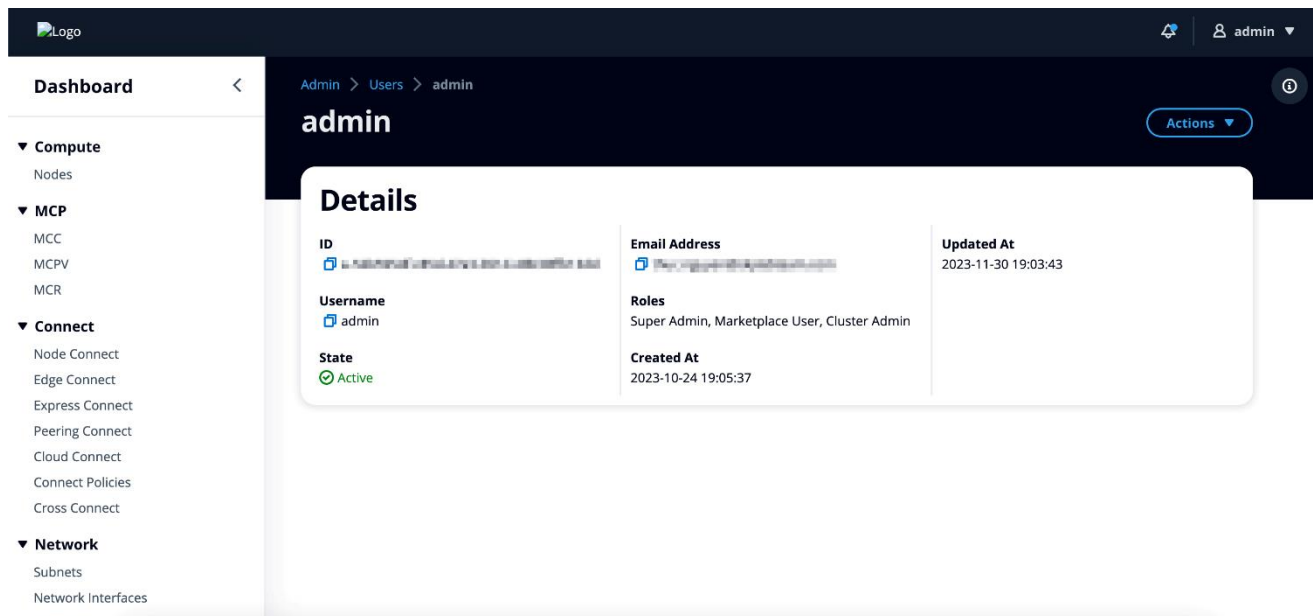


Figure: Users – View details

The following information will be visible: user ID, username, creation date, email address, state, and the types of roles associated with the user.

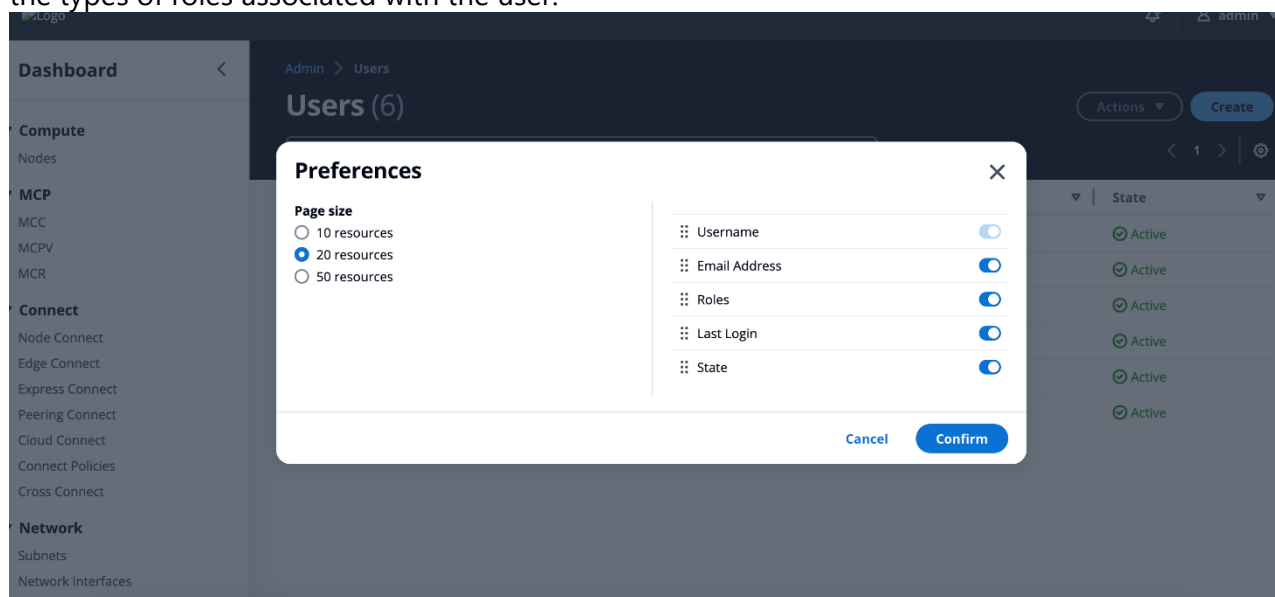


Figure: Admin – Users - Preferences

Clicking on the settings icon within this dashboard will also allow the users to select the page size, and filters.

3.1.0 Change of Passwords

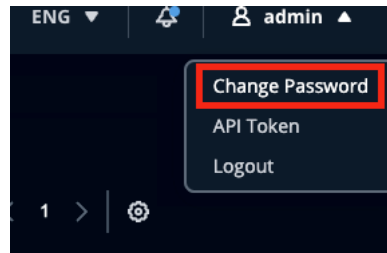
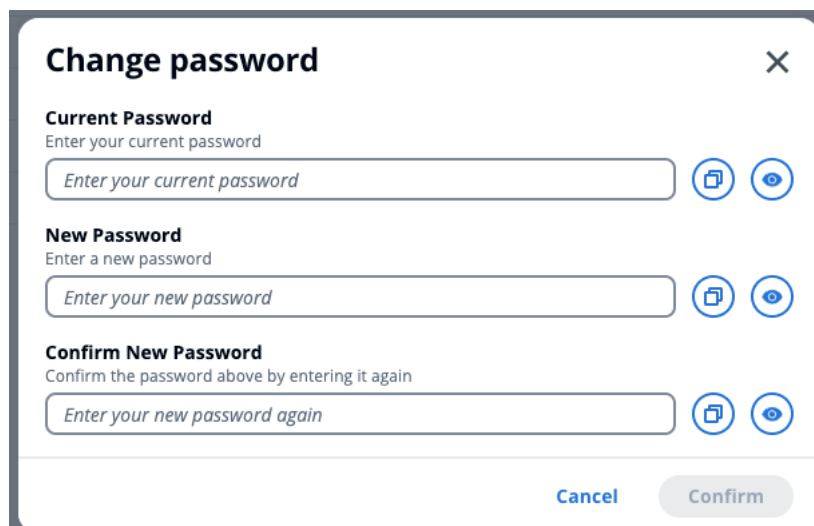


Figure: MEC Manager – Change Password

At the top right of the browser, click on your profile icon and then select “Change Password.”


 A screenshot of the 'Change password' form. The form has a title 'Change password' and a close button (X) in the top right corner. It contains three input fields:

- Current Password**: Labeled 'Enter your current password'.
- New Password**: Labeled 'Enter a new password'.
- Confirm New Password**: Labeled 'Confirm the password above by entering it again'.

 Each input field has a placeholder text 'Enter your current password', 'Enter your new password', and 'Enter your new password again' respectively. To the right of each input field are two icons: a clipboard icon and an eye icon. At the bottom right of the form are two buttons: 'Cancel' and 'Confirm'.

Figure: Change Password – Step 1 to 4

Step 1: Begin by providing your existing password.

Step 2: Select a new password that is strong and unique. A strong password typically includes a mix of uppercase and lowercase letters, numbers, and special characters. Avoid using easily guessable information such as names, birthdays, or frequently used words. Creating a complex password enhances the security of your account.

Step 3: Re-enter the newly chosen password exactly as you did in Step 2. This confirmation ensures that you have correctly inputted your new password.

Step 4: Once you have confirmed your new password, look for a button labelled “Confirm”. Click on this button to apply the changes. By doing so, you have updated your password securely.

4.0 Generating an API Token

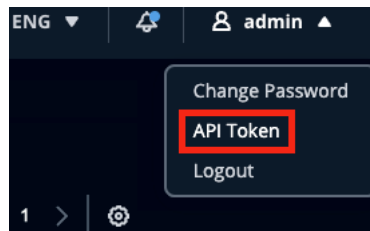


Figure: MEC Manager – API Token

At the top right of the browser, click on your profile icon and then select “API Token.”

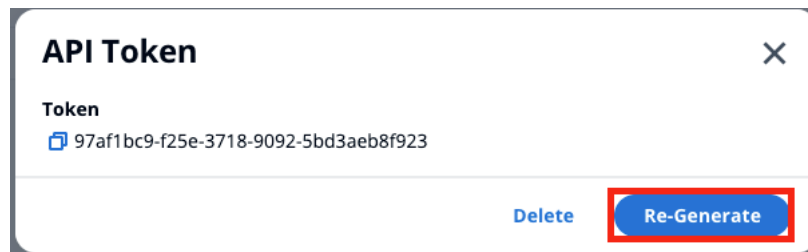


Figure: MEC Manager – Regenerate API Token

Upon landing on the “API Token” page, you may click on “Re-Generate” to regenerate the token.

5.0 Understanding Transactions Logs

- Edge Connect
- Express Connect
- Peering Connect
- Cloud Connect
- Connect Policies
- Cross Connect
- ▼ **Network**
 - Subnets
 - Network Interfaces
 - Elastic IPs
 - NAT Gateways
 - Network Policies
 - Routes
 - Zone Load Balancer
- ▼ **Object Storage**
 - Buckets
 - Bucket Users
- ▼ **Storage**
 - Block Volumes
 - Snapshots
 - Backups
 - Scheduler
- ▼ **Services**
 - SSH Keys
 - Transaction Logs**
 - Notifications
- ▼ **Admin**
 - Resource Groups
 - Role
 - User
 - Policies

Transactions (1832) Info							
ID	Type	Resource Type	Name	Resource ID	SKU	User	
0000002346	Resize	Node	-	ed8756c5-add0-46...	7568912e-8487-47f...	admin	
0000002345	Decommission	Node	AutoVM-150420240...	6aed3711-16ac-4d6...	7568912e-8487-47f...	admin	
0000002344	Provision	Node	AutoVM-150420240...	0e5ca0bd-ced5-46c...	c55c49a9-19f5-467...	admin	
0000002343	Resize	Node	AutoVM-150420240...	0e5ca0bd-ced5-46c...	c55c49a9-19f5-467...	admin	
0000002342	Decommission	Node	AutoVM-150420240...	2a34906d-5184-4a...	c55c49a9-19f5-467...	admin	
0000002341	Provision	Node	AutoVM-150420240...	6aed3711-16ac-4d6...	7568912e-8487-47f...	admin	
0000002340	Resize	Node	AutoVM-150420240...	6aed3711-16ac-4d6...	7568912e-8487-47f...	admin	
0000002339	Decommission	Node	AutoVM-150420240...	006a823a-25cc-42c...	7568912e-8487-47f...	admin	
0000002338	Provision	Node	AutoVM-150420240...	2a34906d-5184-4a...	c55c49a9-19f5-467...	admin	
0000002337	Resize	Node	AutoVM-150420240...	2a34906d-5184-4a...	c55c49a9-19f5-467...	admin	
0000002336	Decommission	Node	AutoVM-150420240...	91f83ba2-2b75-486...	c55c49a9-19f5-467...	admin	
0000002335	Provision	Node	AutoVM-150420240...	006a823a-25cc-42c...	7568912e-8487-47f...	admin	
0000002334	Resize	Node	AutoVM-150420240...	006a823a-25cc-42c...	7568912e-8487-47f...	admin	
0000002333	Decommission	Node	AutoVM-150420240...	27a4d6ab-0ee8-4a...	7568912e-8487-47f...	admin	
0000002332	Provision	Node	AutoVM-150420240...	91f83ba2-2b75-486...	c55c49a9-19f5-467...	admin	
0000002331	Resize	Node	AutoVM-150420240...	91f83ba2-2b75-486...	c55c49a9-19f5-467...	admin	
0000002330	Node Start	Node	AutoVM-150420240...	27a4d6ab-0ee8-4a...	-	admin	
0000002328	Decommission	Node	AutoVM-150420240...	ad319e3a-cdb6-4a3...	c55c49a9-19f5-467...	admin	
0000002327	Provision	Node	AutoVM-150420240...	27a4d6ab-0ee8-4a...	7568912e-8487-47f...	admin	
0000002326	Resize	Node	AutoVM-150420240...	27a4d6ab-0ee8-4a...	7568912e-8487-47f...	admin	

Figure: MEC Manager – Transaction Logs

- **Transaction ID:** Each operation or transaction is assigned a unique ID. This ID helps you keep track of specific activities, making it easier to reference and understand the context of each operation.
- **Type:** type of transaction, shown below
- **User:** username of user who initiated the transaction
- **Resource Type:** Type of resources, shown below
- **Resource Name:** Name of resource if available
- **Resource ID:** the UUID of the resource
- **Zone ID:** Zone of the resource (may be null)
- **Region ID:** Region of the resource (may be null)
- **SKU ID:** The SKU ID of the resource or produce
- **State:** State of the transaction, available states are Complete, Failed, or In Progress
- **Started:** Date and time that the transaction started
- **Completed:** The date and time the transaction completed, If the transaction is still in progress it will be null

Types of Transaction:

Provision, Decommission, Node Start, Node stop, Node Resize, Enable Node High Availability, Disable Node High Availability, Enable Node Auto Recovery, Disable Node Auto Recovery, Volume Attach, Volume Detach, Networking Interface Attach, Network Interface Detach, Restore, Application Purchase



Types of Resource Types:

Subnet, Node, Storage Block, Storage Snapshot, Storage Backup, Network Interface, Elastic IP, Network Route, NAT Gateway, Network Policy, Load Balancer, Node Connect, MCC (Multi-cloud Container Cluster), MCC Node, MCPV (Multi-cloud Container Persistent Volume), MCR (Multi-cloud Container Registry), MCR Account, MCC Connect, Cross Connect, Edge Connect, Peering Connect, Express Connect, Cloud Connect, Connect Policy, Application Purchase.

In summary, this dashboard provides you with a clear overview of all the transactions happening within the cloud infrastructure. You can see what kind of operations are taking place, which resources are involved, where they are located, and whether these operations were successful or not. This detailed information is crucial for managing and troubleshooting activities in a cloud environment effectively.

Click on the "Services" tab and select "Transaction Logs" to begin.

ID	Type	Resource Type	Name	Resource ID	SKU	User	Completed
0000002346	Resize	Node	-	ed8756c5-add0-46...	7568912e-8487-47f...	admin	-
0000002345	Decommission	Node	AutoVM-150420240...	6aed3711-16ac-4d6...	7568912e-8487-47f...	admin	2024-04-17 10:54:...
0000002344	Provision	Node	AutoVM-150420240...	0e5ca0bd-ced5-46c...	c55c49a9-19f5-467...	admin	2024-04-17 10:54:...
0000002343	Resize	Node	AutoVM-150420240...	0e5ca0bd-ced5-46c...	c55c49a9-19f5-467...	admin	2024-04-17 10:54:...
0000002342	Decommission	Node	AutoVM-150420240...	2a34906d-5184-4a...	c55c49a9-19f5-467...	admin	2024-04-17 10:45:...
0000002341	Provision	Node	AutoVM-150420240...	6aed3711-16ac-4d6...	7568912e-8487-47f...	admin	2024-04-17 10:45:...
0000002340	Resize	Node	AutoVM-150420240...	6aed3711-16ac-4d6...	7568912e-8487-47f...	admin	2024-04-17 10:45:...
0000002339	Decommission	Node	AutoVM-150420240...	006a823a-25cc-42c...	7568912e-8487-47f...	admin	2024-04-17 10:38:...
0000002338	Provision	Node	AutoVM-150420240...	2a34906d-5184-4a...	c55c49a9-19f5-467...	admin	2024-04-17 10:38:...
0000002337	Resize	Node	AutoVM-150420240...	2a34906d-5184-4a...	c55c49a9-19f5-467...	admin	2024-04-17 10:38:...
0000002336	Decommission	Node	AutoVM-150420240...	91f83ba2-2b75-486...	c55c49a9-19f5-467...	admin	2024-04-17 10:35:...
0000002335	Provision	Node	AutoVM-150420240...	006a823a-25cc-42c...	7568912e-8487-47f...	admin	2024-04-17 10:35:...
0000002334	Resize	Node	AutoVM-150420240...	006a823a-25cc-42c...	7568912e-8487-47f...	admin	2024-04-17 10:35:...
0000002333	Decommission	Node	AutoVM-150420240...	27a4d6ab-0ee8-4a...	7568912e-8487-47f...	admin	2024-04-17 09:53:...
0000002332	Provision	Node	AutoVM-150420240...	91f83ba2-2b75-486...	c55c49a9-19f5-467...	admin	2024-04-17 09:53:...
0000002331	Resize	Node	AutoVM-150420240...	91f83ba2-2b75-486...	c55c49a9-19f5-467...	admin	2024-04-17 09:53:...
0000002330	Node Start	Node	AutoVM-150420240...	27a4d6ab-0ee8-4a...	-	admin	2024-04-17 09:52:...
0000002328	Decommission	Node	AutoVM-150420240...	ad319e3a-cdb6-4a3...	c55c49a9-19f5-467...	admin	2024-04-16 21:24:...
0000002327	Provision	Node	AutoVM-150420240...	27a4d6ab-0ee8-4a...	7568912e-8487-47f...	admin	2024-04-16 21:24:...
0000002326	Resize	Node	AutoVM-150420240...	27a4d6ab-0ee8-4a...	7568912e-8487-47f...	admin	2024-04-16 21:24:...

Figure: MEC Manager – View Detail

To view more information about a specific transaction, click on the "Actions" tab and click "View Detail."



Transaction: 0000002345		✕	
ID	0000002345		
Type	Decommission		
User	admin		
Resource Type	Node		
Resource Name	AutoVM-15042024060555554		
Resource ID			
Zone	Zone 02		
Region	Region01		
Provider	FusionFlowCloud		
SKU ID			
State	Completed		
Started	2024-04-17 10:54:55		
Completed	2024-04-17 10:54:55		
		Close	

Figure: MEC Manager – Transaction

When you click on "View Detail," a pop-up window will appear displaying a summary of the transaction. It includes details such as Transaction ID, Transaction type, User ID, Provider, Region, zone, Resource type, Resource ID, SKU ID, Status, provisioning time, creation time, and update time.

6.0 Notifications

▼ MCP

MCC

MCPV

MCR

▼ Connect

Node Connect

Edge Connect

Express Connect

Peering Connect

Cloud Connect

Connect Policies

Cross Connect

▼ Network

Subnets

Network Interfaces

Elastic IPs

NAT Gateways

Network Policies

Routes

Zone Load Balancer

▼ Object Storage

Buckets

Bucket Users

▼ Storage

Block Volumes

Snapshots

Backups

Scheduler

▼ Services

SSH Keys

Transaction Logs

Notifications

▼ Admin

Resource Groups

Role

User

Policies

Services > Notifications

Notifications (2144) Info

Search

Filter by a date range

< 1 2 3 4 5 6 7 ... 188 >

Actions

Severity	Title	Message	Code	Received At
▼ Low	Node resized successfully	Node AutoVM-1504202406055554 resized ... Transaction: PM-NO50_0000002346_04 Resource: ed8756c5-ad00-465d-a9b5-c3762...	002-NO12_0000	2024-04-17 10:57:16
▼ Low	Node resized successfully	Node AutoVM-1504202406055554 resized ... Transaction: PM-NO50_0000002343_04 Resource: 0e5ca0bd-ced5-46cb-bb10-3cf1b4...	002-NO12_0000	2024-04-17 10:54:56
▼ Low	Node resized successfully	Node AutoVM-1504202406055554 resized ... Transaction: PM-NO50_0000002340_04 Resource: 6aed3711-16ac-4d63-81bf-edfda9...	002-NO12_0000	2024-04-17 10:45:11
▼ Low	Node resized successfully	Node AutoVM-1504202406055554 resized ... Transaction: PM-NO50_0000002337_04 Resource: 2a34906d-5184-4a57-8d30-fd65cf...	002-NO12_0000	2024-04-17 10:38:37
▼ Low	Resource Group deleted successfully	test group deleted successfully Transaction: null Resource: 94a01153-86db-46eb-98fa-27ca21...	002-RSG11_0000	2024-04-17 10:36:56
▼ Low	Resource Group created successfully	test group created successfully Transaction: null Resource: 94a01153-86db-46eb-98fa-27ca21...	002-RSG10_0000	2024-04-17 10:36:19
▼ Low	Node resized successfully	Node AutoVM-1504202406055554 resized ... Transaction: PM-NO50_0000002334_04 Resource: 006a823a-25cc-42c3-ab17-480157...	002-NO12_0000	2024-04-17 10:35:23
▼ Low	Node resized successfully	Node AutoVM-1504202406055554 resized ... Transaction: PM-NO50_0000002331_04 Resource: 91f83ba2-2b75-4864-bb14-b7d69...	002-NO12_0000	2024-04-17 09:53:39
▼ Low	Node command sent	Start command sent to 27a4d6ab-0ee8-4aa... Transaction: PM-NO14_0000002330_02 Resource: 27a4d6ab-0ee8-4aa5-bc7d-6d6ec...	002-NO60_0000	2024-04-17 09:52:04
▼ Low	Node resized successfully	Node AutoVM-1504202406055554 resized ... Transaction: PM-NO50_0000002326_04 Resource: 27a4d6ab-0ee8-4aa5-bc7d-6d6ec...	002-NO12_0000	2024-04-16 21:24:26
▼ Low	Node command sent	Stop command sent to AutoVM-15042024060... Transaction: PM-NO15_0000002325_02 Resource: ad319e3a-cdb6-4a36-ad9b-ac197e...	002-NO61_0000	2024-04-16 20:03:20
▼ Low	Node command sent	Start command sent to ad319e3a-cdb6-4a3... Transaction: PM-NO14_0000002324_02 Resource: ad319e3a-cdb6-4a36-ad9b-ac197...	002-NO60_0000	2024-04-16 19:56:43

Figure: MEC Manager – Notifications

In this dashboard, you will find a table view presenting the notifications in an organized manner. Each notification is categorized by its severity level, which indicates its importance. The severity level helps you quickly understand the urgency of the notification.

For each notification, these are the key pieces of information displayed:

- **Severity:** This indicates how critical the notification is, ranging from low importance to high urgency.
- **Title:** The title provides a brief summary of the notification content, giving you an idea of what it is about
- **Message:** This section contains the detailed information or description related to the notification.
- **Code:** The code associated with the notification, which can be useful for technical reference or troubleshooting.
- **Received At:** This shows the date and time when the notification was received, helping you understand the timeline of events.

In addition to viewing the notifications, these are the actions that you can perform:



- **View:** Clicking on this option allows you to see the complete details of the notification, providing you with more context about the event or message.
- **Mark as Read:** This action lets you acknowledge that you have seen the notification, making it easier to keep track of which ones you have addressed.
- **Remove:** You can choose to remove the notification from the dashboard, decluttering your view and focusing on the remaining ones.

7.0 Node Management

This section is a central hub for overseeing various tasks related to nodes within your system. Here is a brief description of the key functionalities it offers:

- **Create:** This feature allows you to initiate the process of adding new nodes to your system. You can define their specifications and configurations, enabling the expansion of your infrastructure as needed.
- **View:** The "View" option provides you with an organized overview of all existing nodes. You can easily access details about each node, including their status, resources, and associated information.
- **Start:** When you need to activate a node that is currently in a stopped state, the "Start" function enables you to initiate its operation, making it available for processing tasks.
- **Stop:** Conversely, the "Stop" function allows you to halt the operation of a node. This is useful when you want to temporarily suspend a node's activity without the need for removal.

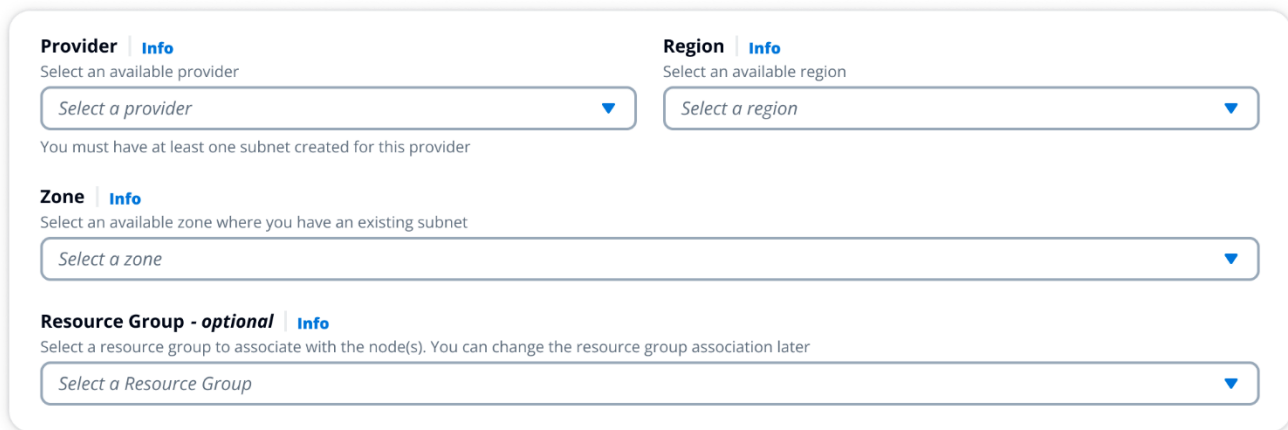
The Node Management section empowers you to efficiently create, view, start, and stop nodes, ensuring that your infrastructure remains scalable and responsive to your operational needs.

Nodes provide virtualized compute resources that can be provisioned for your use across different providers, regions, and zones available in the platform. A "node" is the term for these virtualized compute resources. Provisioning a node requires a Subnet, Node Size, and an Operating System Image.

7.0.1 Creating Nodes – Virtual Machines

A subnet must already be onboarded to your instance for a node to function.

1. Select Compute > Nodes.
2. Press the Create button.
3. Decide on an accessible Provider. 4. Choose a Region and Zone from the drop-down menus. Based on the subnets you have specified; providers will be loaded.
4. Choose a Resource Group Optionally

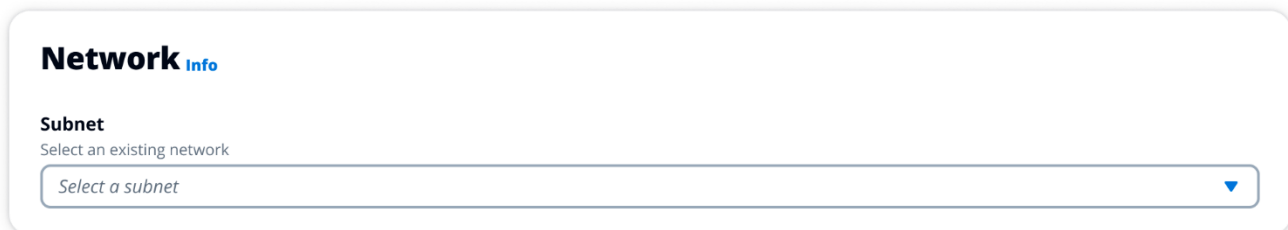


The screenshot shows a form with four sections for node creation:

- Provider** | Info: Select an available provider. A dropdown menu shows "Select a provider". Below it, a note states: "You must have at least one subnet created for this provider".
- Region** | Info: Select an available region. A dropdown menu shows "Select a region".
- Zone** | Info: Select an available zone where you have an existing subnet. A dropdown menu shows "Select a zone".
- Resource Group - optional** | Info: Select a resource group to associate with the node(s). You can change the resource group association later. A dropdown menu shows "Select a Resource Group".

Figure: Creating Nodes – Step 1 to 4

5. Choose a Subnet using the drop-down menu.
6. You will be able to give this node a public IP address if the provider you have chosen is a public cloud provider.



The screenshot shows the **Network** section of the form:

- Subnet**: Select an existing network. A dropdown menu shows "Select a subnet".

Figure: Creating Nodes – Step 5

7. From the list, choose a Node Size that is accessible. The availability of nodes in the chosen zone is used to filter node sizes. By selecting display all node sizes, you may broaden the display and access more details as well as a search feature.
8. You can specify a figure for the Number of nodes if you wish to provision more than one node—up to ten nodes. Every node will use the same node size, reservation, and HA settings and will be part of the same subnet.
9. From the drop-down menu, choose the Reservation Type.
10. Toggle the Enable High Availability switch, if desired.
11. Select the Enable Auto Recovery toggle, if desired.

Node Settings Info

Node Size
[View all node sizes](#)

Select a node size from the available list. Some node sizes are restricted to certain operating system types

NodeSize01 (4 vCPU, 8GB Memory, Nvidia A2 (x2))

Number of nodes

Enter how many nodes you want to provision. The node size and reservation type will be the same for each node but you can customize other individual node settings

1

Reservation Type

Select an available reservation type

Select a reservation type

High Availability Options

Enabling the below toggles will enable your nodes for high availability to protect from zone or node outages. Additional costs may apply for nodes with this state enabled

☐ Enable High Availability
☐ Enable Auto Recovery

Figure: Creating Nodes – Step 7 to 11

12. You will see each node as a tab, depending on how many nodes there are. The following configuration parameters can be set up separately for every node.
13. Optionally: Turn on the Create from backup image checkbox if you want to build a node from an already-existing backup. If the backup image satisfies the aforementioned parameters, you can then choose it.
14. Choose an available Operating System Image from the drop-down menu if you are not starting from a backup. Additionally, you can enlarge this view by selecting View all images, which will include a search feature and additional information.
15. Give the node a hostname.
16. Put in your username here. Using RDP or SSH, this will be used to access the node.
17. Choose an Access Key if it is a Linux-type operating system. This is the public SSH key that will be used to get access to the node. If you do not already have an access key, you can create one.
18. If Windows is the operating system type, provide a password that satisfies the pre-set requirements for complexity.
19. At the very least, you must choose the Volume Type for the root volume. Depending on the operating system image used, the root volume size is predetermined and cannot be changed.
20. Optionally - You can click the Add Volume button to add each additional Volume that you would like to add to this node. Choose the Volume Type and enter the Name and Size. These volumes are going to be made and linked to this node.

Image Info

☐ Create from existing backup image

Image

Select an image from the available list

[View all images](#)

OS-Image-01 (Server 2019 Core Base, Window) ▼



Access Info

Hostname

Enter the initial hostname for this node that will be set at the time of creation

Enter a hostname

Username

Enter a username

Enter a username

Access Key

Select an access key

Select an access key ▼

Create new access key

Storage Info

Name

Enter the volume name

Size (GB)

50

Volume Type

Select a volume type ▼

Enter the volume name

Enter the volume size

Select a volume type ▼

Remove

Enter the volume name

Enter the volume size

Select a volume type ▼

Remove

Add Volume

Figure: Creating Nodes – Steps 12 to 20

21. Select Next.
22. Before proceeding, you can have a look at an overview of what will be made. Click Create if you are satisfied with the details. On the list view page, you can see the creation's status.

7.0.2 Starting a Node – Virtual Machines

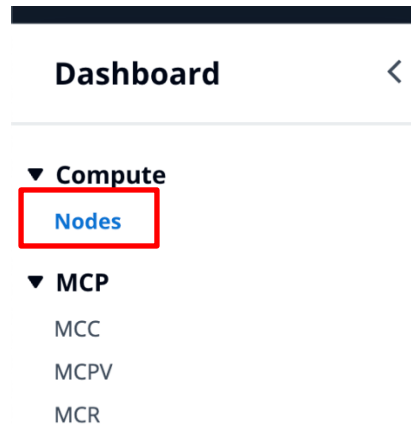


Figure: Starting a Node – Step 1

Step 1: Click on the “Compute” tab on the top of the dashboard, then click on “Nodes”.

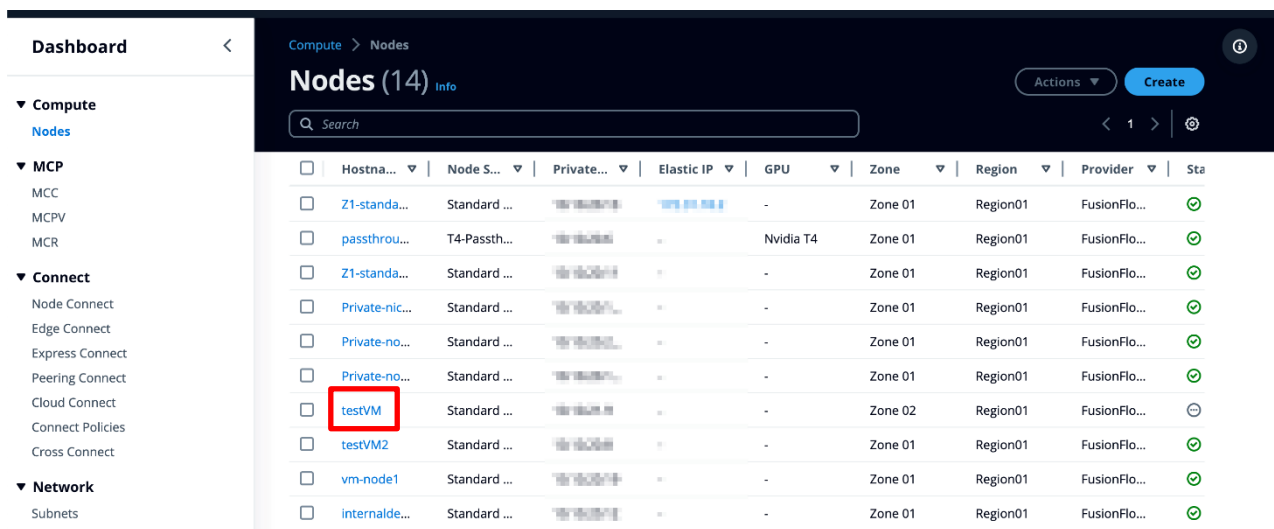


Figure: Starting a Node – Step 2

Step 2: Choose the node that you wish to start from the list of nodes in the dashboard, click on the node. *Keep in mind that only a non-active node can be started*

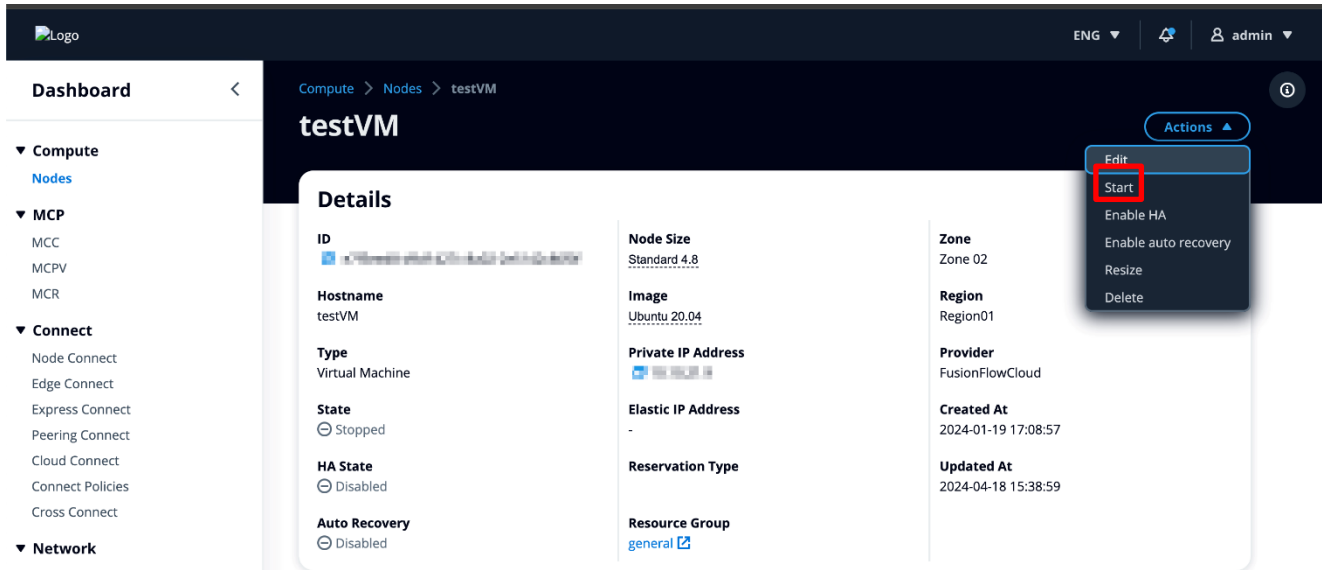


Figure: Starting a Node – Step 3

Step 3: Click on the “Start” button and you will see the selected node have successfully started back on the “Nodes” dashboard with the state displaying “Running”.

7.0.3 Stopping a Node – Virtual Machines

****IMPORTANT****

Stopping a node may result in data loss.

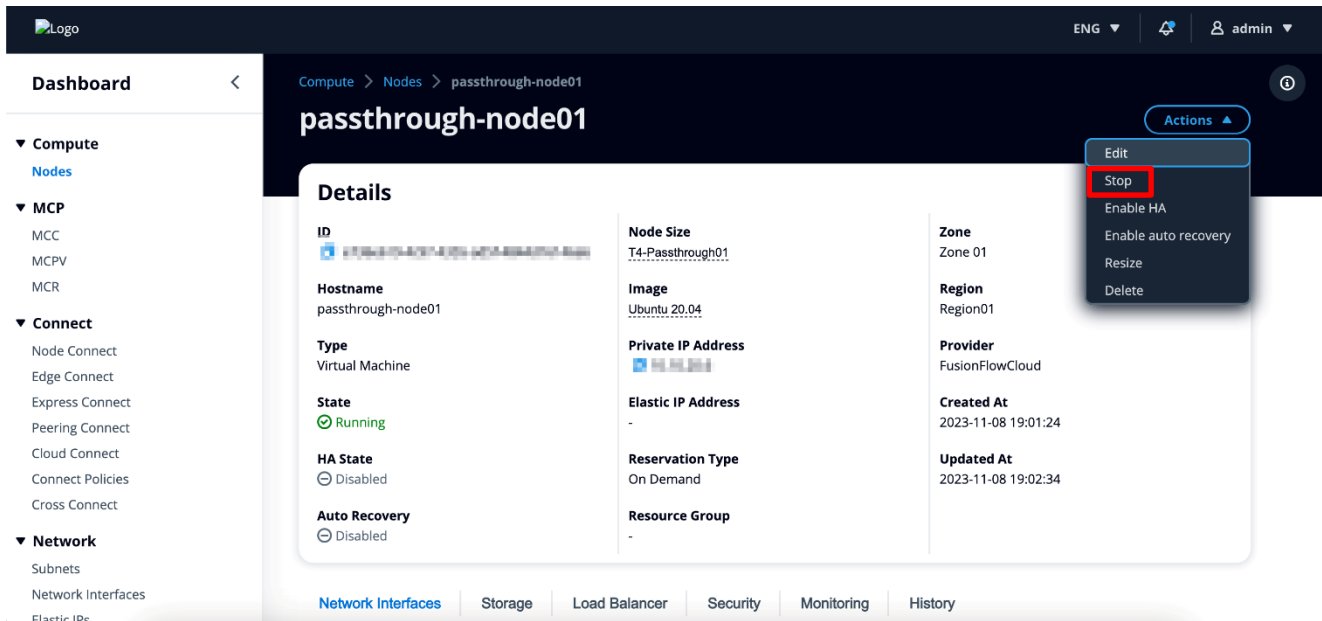


Figure: Stopping a Node

Choose the node that you wish to start from the list of nodes in the dashboard, click “Stop” and you will see the state displayed as “Stopped”.

When a node is successfully stopped, you will observe its status change to "Stopped," prominently highlighted within the main dashboard for nodes. This visual cue serves as a clear indicator that the node has transitioned to a stopped state.

7.0.4 Enabling/Disabling High Availability on a Node

A node that has the high availability flag enabled will be protected against hardware failure at the zone level. In the event of an unplanned outage, the system will try to immediately transfer your node to available hardware so that it may continue operating with the least amount of downtime.

As long as a node is not in the creating, updating, removing, failed, or error states, users can enable HA on it whenever users want. As soon as the success message is received, the setting will go into effect.

This can be accessed via an action drop-down menu on already-existing nodes or as a toggle during the node creation process.

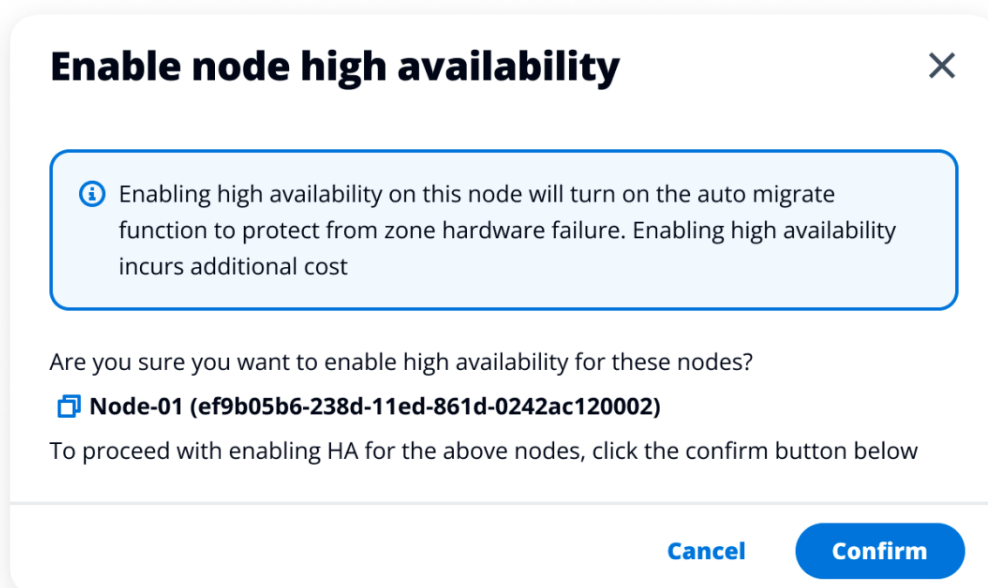


Figure: Enabling Node HA

If users enable this, a transaction indicating that HA has been enabled for this node will be generated.

As long as a node is not in the creating, updating, removing, failing, or error states, users can also disable HA on it whenever users like. As soon as the success message is received, the setting will go into effect.

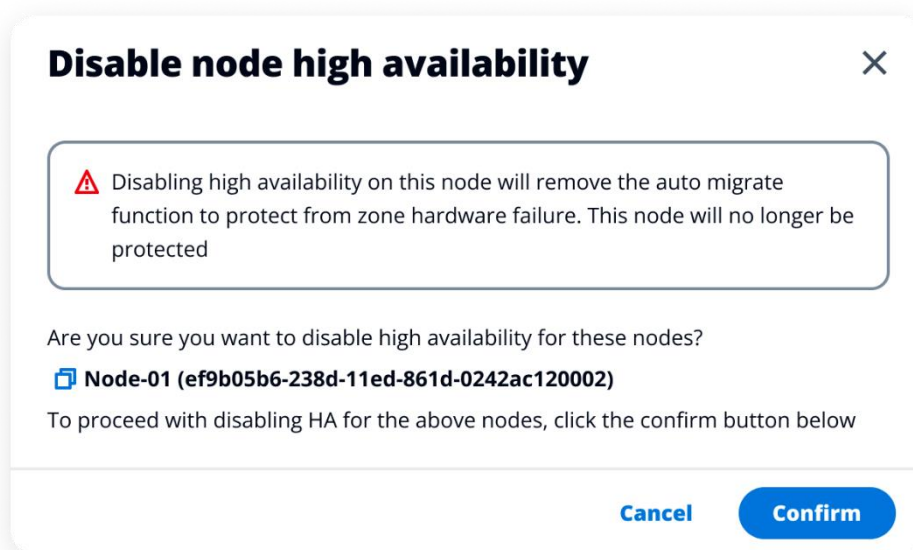


Figure: Disabling Node HA

When this is disabled, a transaction indicating that HA has been disabled for this node will be generated.

7.0.5 Provisioning Status on Nodes

Nodes (13) Info										
Search										
☐	Hostname	Node Size	Private IP	Elastic IP	GPU	Zone	Region	Provider	State	
☐	Z1-standard-1	Standard 2.4	10.10.20.13	172.31.18.2	-	Zone 01	Region01	FusionFlowCloud		Running
☐	passthrough-node...	T4-Passthrough01	10.10.20.6	-	Nvidia T4	Zone 01	Region01	FusionFlowCloud		Running
☐	Z1-standard-3	Standard 4.8	10.10.20.11	-	-	Zone 01	Region01	FusionFlowCloud		Running
☐	Private-nic-demo01	Standard 4.8	10.10.20.147	-	-	Zone 01	Region01	FusionFlowCloud		Running
☐	Private-node-subn...	Standard 4.8	10.10.20.228	-	-	Zone 01	Region01	FusionFlowCloud		Running
☐	Private-node-Subn...	Standard 4.8	10.10.20.146	-	-	Zone 01	Region01	FusionFlowCloud		Running
☐	testVM	Standard 4.8	10.10.21.9	-	-	Zone 02	Region01	FusionFlowCloud		Stopped
☐	testVM2	Standard 2.4	10.10.20.8	-	-	Zone 01	Region01	FusionFlowCloud		Running
☐	vm-node1	Standard 2.4	10.10.20.19	-	-	Zone 01	Region01	FusionFlowCloud		Stopped
☐	internaldemo01	Standard 2.4	10.10.20.12	-	-	Zone 01	Region01	FusionFlowCloud		Running
☐	TestVM	Standard 4.8	10.10.20.4	172.31.18.6	-	Zone 01	Region01	FusionFlowCloud		Running
☐	AutoVM-15042024...	Standard 4.8	10.10.21.29	-	-	Zone 02	Region01	FusionFlowCloud		Running
☐	test-VM	Standard 1.1	-	-	-	Zone 01	Region01	FusionFlowCloud		Provisioning

Figure: Nodes – Provisioning Status

A node will first appear as though it is provisioning, as shown and then it will be operational, as indicated by the notification icon located in the upper right corner.

Compute > Nodes

Nodes (13) info

Search

	Hostname	Node Size	Private IP	Elastic IP	GPU	Zone	Region	Provider
☐	Z1-standard-1	Standard 2.4	10.10.20.13	172.31.18.2	-	Zone 01	Region01	FusionFlowCloud
☐	passthrough-node...	T4-Passthrough01	10.10.20.6	-	Nvidia T4	Zone 01	Region01	FusionFlowCloud
☐	Z1-standard-3	Standard 4.8	10.10.20.11	-	-	Zone 01	Region01	FusionFlowCloud
☐	Private-nic-demo01	Standard 4.8	10.10.20.147	-	-	Zone 01	Region01	FusionFlowCloud
☐	Private-node-subn...	Standard 4.8	10.10.20.228	-	-	Zone 01	Region01	FusionFlowCloud
☐	Private-node-Subn...	Standard 4.8	10.10.20.146	-	-	Zone 01	Region01	FusionFlowCloud
☐	testVM	Standard 4.8	10.10.21.9	-	-	Zone 02	Region01	FusionFlowCloud
☐	testVM2	Standard 2.4	10.10.20.8	-	-	Zone 01	Region01	FusionFlowCloud
☐	vm-node1	Standard 2.4	10.10.20.19	-	-	Zone 01	Region01	FusionFlowCloud
☐	internaldemo01	Standard 2.4				Region01	Region01	FusionFlowCloud
☐	TestVM	Standard 4.8				Region01	Region01	FusionFlowCloud
☐	AutoVM-15042624...	Standard 4.8				Region01	Region01	FusionFlowCloud
☐	test-VM	Standard 1.1				Region01	Region01	FusionFlowCloud

Node created successfully

Today, 16:58:22

test-VM was created successfully

Transaction: [DLC-VM10_0000002349_02](#)

Resource: [b567f683-9973-4dc1-9f8f-37c2ef46f108](#)

Close

Node created successfully

Today, 16:58:22

SSH Key deletion successfully

Today, 11:54:20

SSH Key created successfully

Today, 11:41:12

Node resized successfully

Today, 10:57:16

Node resized successfully

Today, 10:54:56

Node resized successfully

Today, 10:45:11

Node resized successfully

Today, 10:38:37

Resource Group deleted successfully

Today, 10:36:56

Resource Group created successfully

Today, 10:36:19

Node resized successfully

Today, 10:35:23

Node resized successfully

Today, 09:53:39

Node command sent

Today, 09:52:04

Node resized successfully

Yesterday, 21:24:26

Node command sent

Yesterday, 19:56:43

Node command sent

Yesterday, 19:46:07

Figure: Nodes – Indication of successful provisioning

8.0 Snapshot and Backup Lifecycle Management

This section covers a set of essential activities related to creating, maintaining, and utilizing snapshots and backups within your system. This involves:

- **Creating Snapshots:** Initiating the creation of snapshots to capture the current state of data or system configurations. Snapshots provide a point-in-time representation that can be used for recovery or reference.
- **Restoring Snapshots:** The process of reverting to a previous state by restoring snapshots. This is useful for recovering from errors or unwanted changes.
- **Deleting Snapshots:** Managing storage resources by removing obsolete snapshots, ensuring efficient use of storage capacity.
- **Creating Backups:** Establishing backup routines, including creating backup copies of data and system configurations. Backups are typically scheduled to occur regularly and serve as a safety net against data loss or system failures.
- **Scheduling Backups:** Configuring automated backup schedules to ensure that data is regularly saved, reducing the risk of data loss in case of unexpected events.
- **Managing Backups:** Overseeing the management of backups, which includes tracking backup versions, organizing backups for easy retrieval, and ensuring data integrity.
- **Restoring Backups:** The process of recovering data or system configurations from backups in the event of data corruption, hardware failures, or other issues.

Snapshot and Backup Lifecycle Management is critical for data protection, disaster recovery, and system stability. It ensures that you can efficiently create, maintain, and utilize snapshots and backups to safeguard your valuable information and maintain operational continuity.

8.0.1 Creating a Snapshot

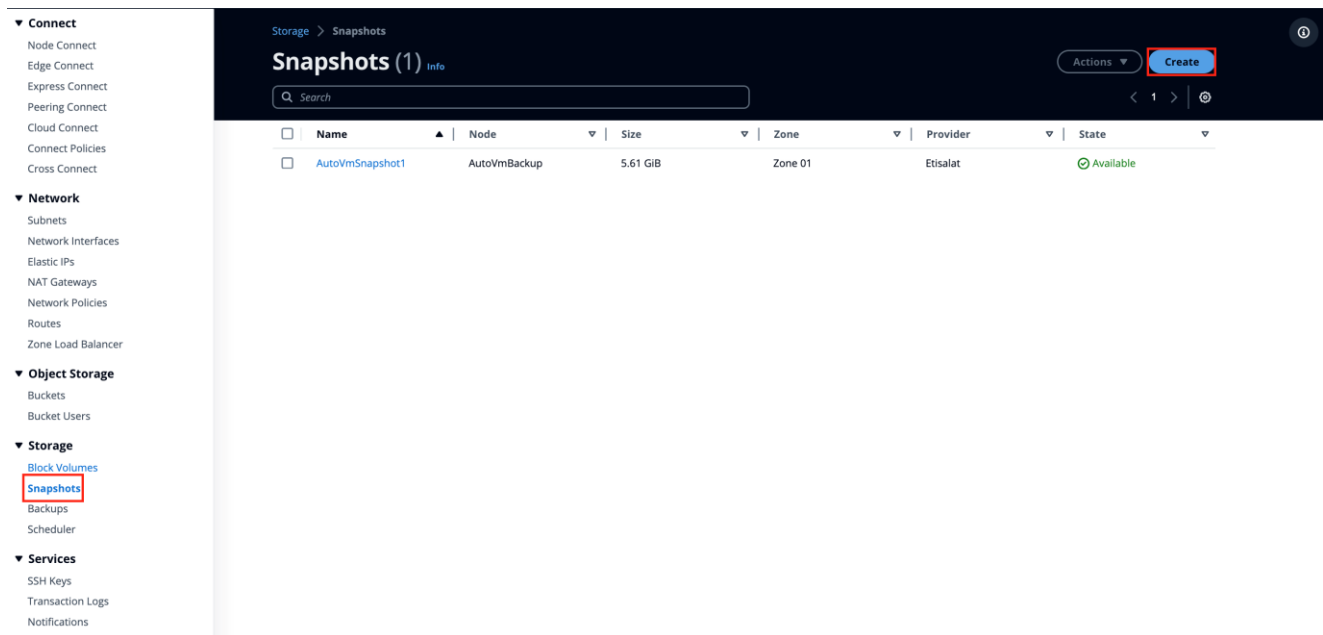
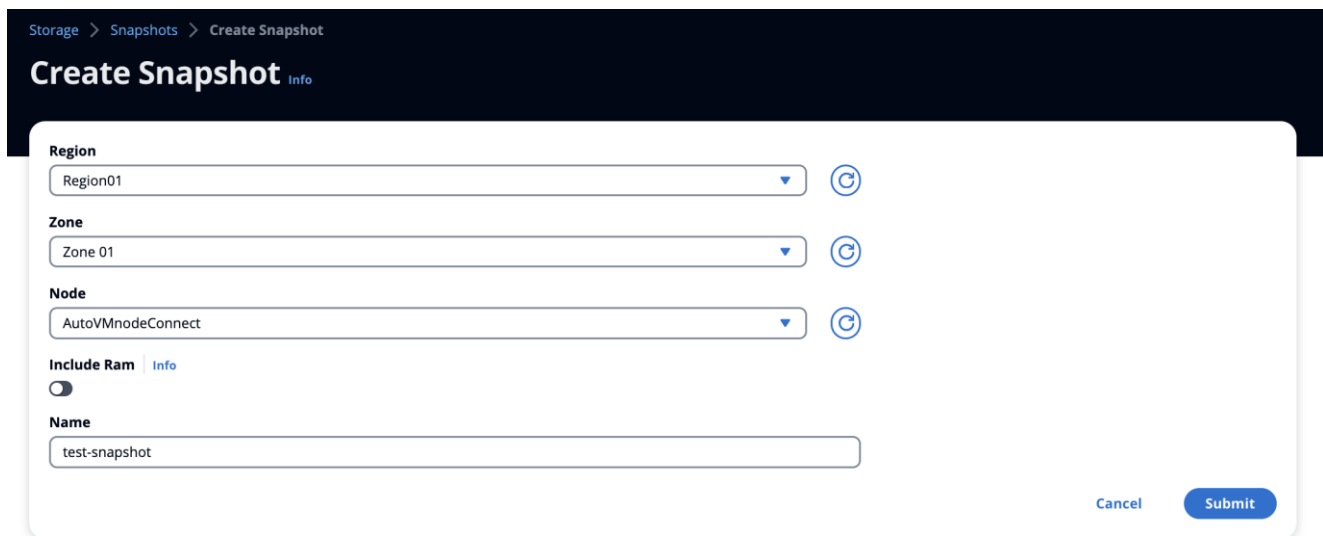


Figure: Creating a Snapshot – Step 1

Step 1: To begin, click on the “Storage” tab and click on “Snapshots”.

At this page, you will be able to view a table of all the available snapshots created and manage them.



The screenshot shows the 'Create Snapshot' form. The breadcrumb trail at the top is 'Storage > Snapshots > Create Snapshot'. The form has the following fields and options:

- Region:** A dropdown menu with 'Region01' selected.
- Zone:** A dropdown menu with 'Zone 01' selected.
- Node:** A dropdown menu with 'AutoVMnodeConnect' selected.
- Include Ram:** A toggle switch that is currently turned off.
- Name:** A text input field containing 'test-snapshot'.

At the bottom right of the form are two buttons: 'Cancel' and 'Submit'.

Figure: Creating a Snapshot – Step 3 to 8

Step 3: Choose an available region.

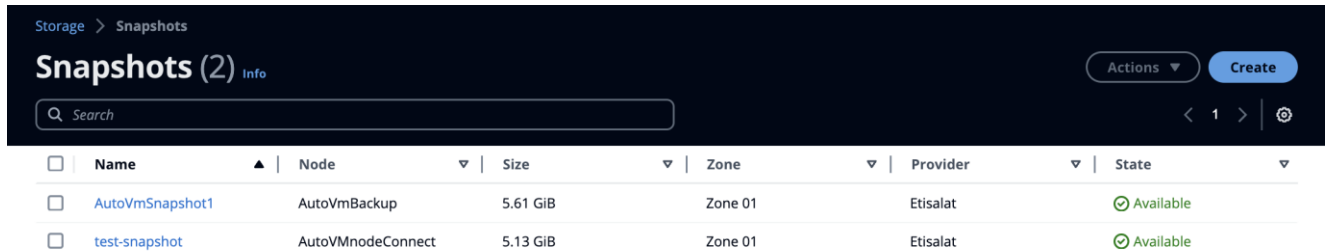
Step 4: Choose an available zone from the region.

Step 5: Select the node you would like to take a snapshot of.

Step 6: State if you would like to include RAM.

Step 7: Provide a name for the snapshot.

Step 8: Ensure all the information are correctly filled in and selected, click on the “Create” button once this is verified.

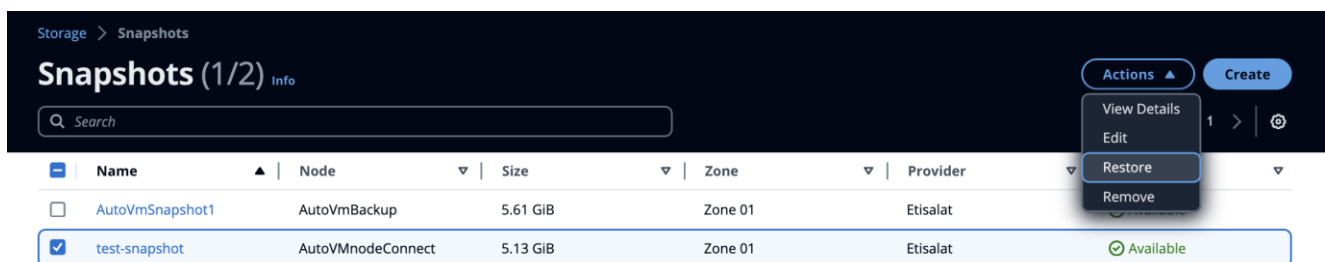


	Name	Node	Size	Zone	Provider	State
☐	AutoVmSnapshot1	AutoVmBackup	5.61 GiB	Zone 01	Etisalat	Available
☐	test-snapshot	AutoVMnodeConnect	5.13 GiB	Zone 01	Etisalat	Available

Figure: Snapshots – Successful creation

The created snapshot will then appear on the “Snapshot” dashboard with the state progressing from “In Progress” to “Available.”

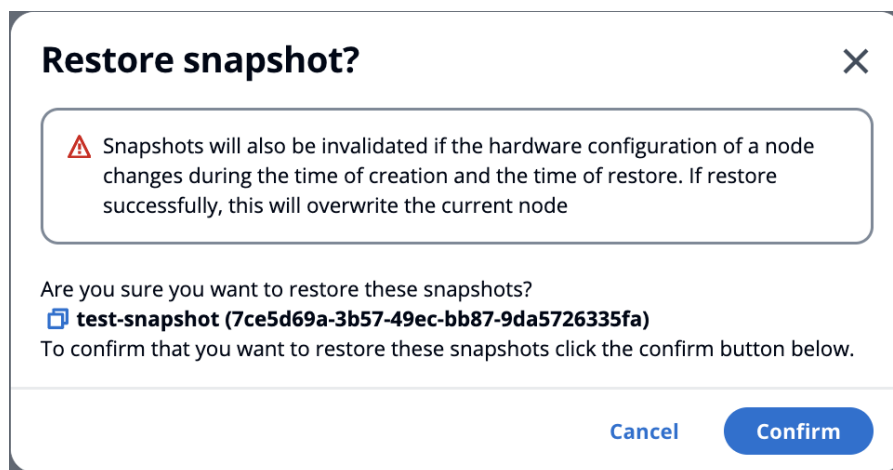
8.0.2 Restoring a Snapshot



	Name	Node	Size	Zone	Provider	State
☐	AutoVmSnapshot1	AutoVmBackup	5.61 GiB	Zone 01	Etisalat	Available
☒	test-snapshot	AutoVMnodeConnect	5.13 GiB	Zone 01	Etisalat	Available

Figure: Restoring a Snapshot – Step 1

Step 1: Click on the snapshot you want to restore under the “Actions” column.



Restore snapshot?

⚠ Snapshots will also be invalidated if the hardware configuration of a node changes during the time of creation and the time of restore. If restore successfully, this will overwrite the current node

Are you sure you want to restore these snapshots?

☒ **test-snapshot (7ce5d69a-3b57-49ec-bb87-9da5726335fa)**

To confirm that you want to restore these snapshots click the confirm button below.

Cancel
Confirm

Figure: Restoring a Snapshot – Step 2

Step 2: After the popup box displaying the accurate node information appears, simply click "Confirm" to initiate the restoration process. You can also view this event in the "Transactions Logs" tab located within the "Services" section.

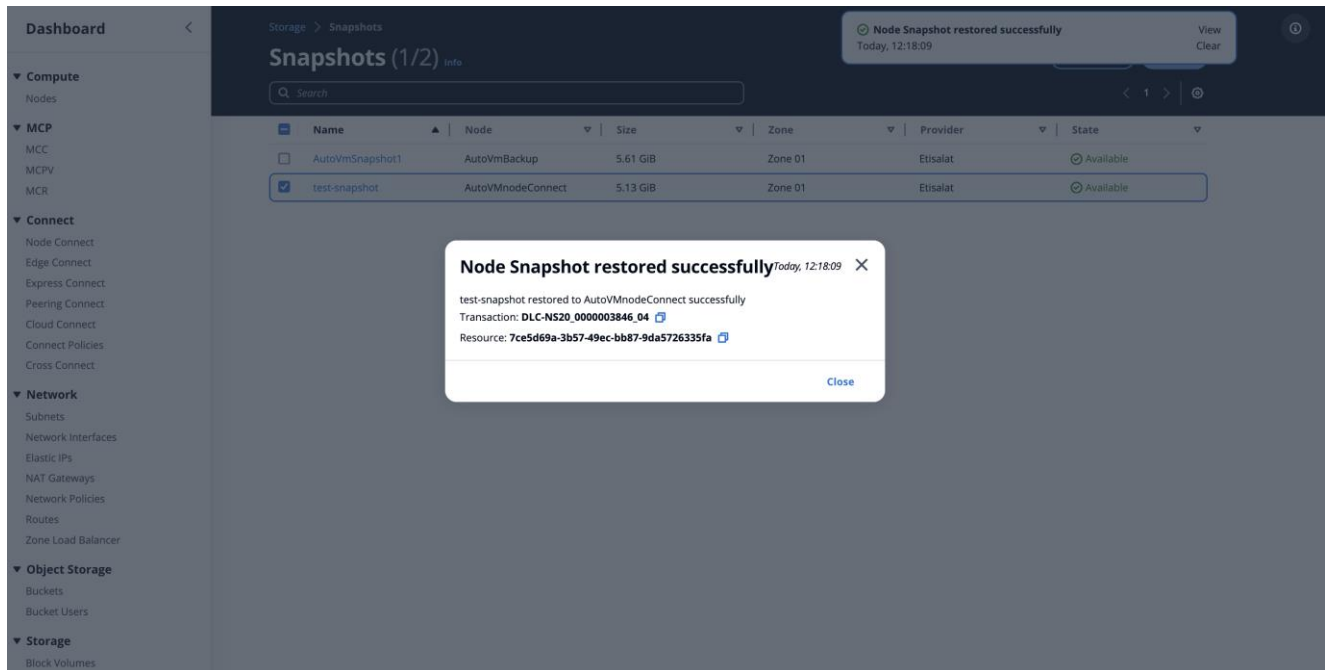


Figure: Snapshots – Successful restoration of a snapshot

Step 3: Following confirmation, watch for the notification indicating the successful restoration of the snapshot.

8.0.3 Deleting a Snapshot

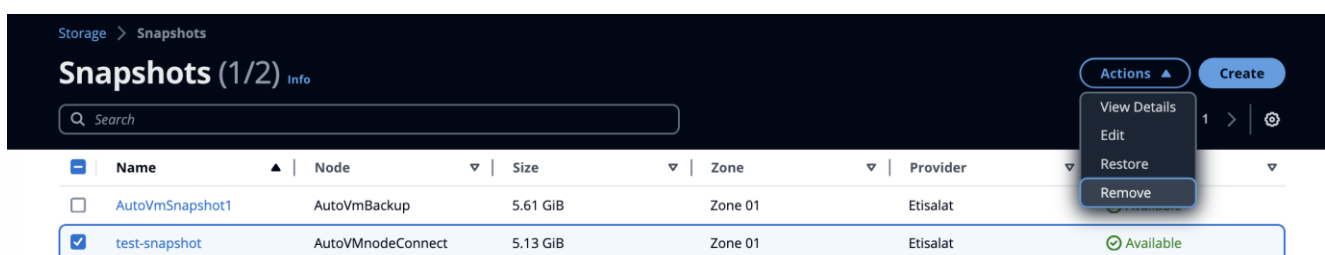


Figure: Snapshots – Deleting a snapshot

Step 1: Click on the snapshot you want to delete under the "Actions" column.

****IMPORTANT****

It is important to proceed with caution when deleting records, data, or rules.

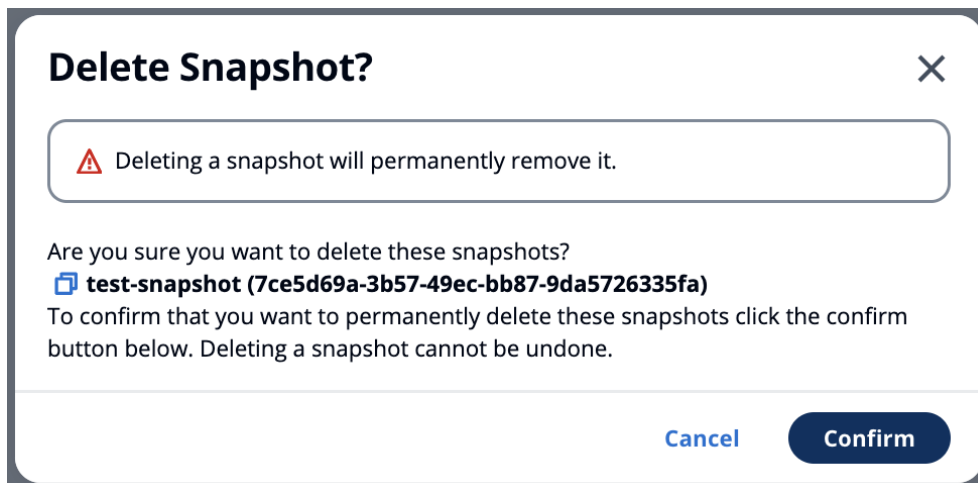


Figure: Snapshot – Pop up box for confirming deletion.

Step 2: After the popup box displaying the accurate node information appears, simply click "Confirm" to initiate the deletion process. You can also view this event in the "Transactions Logs" tab located within the "Services" section.

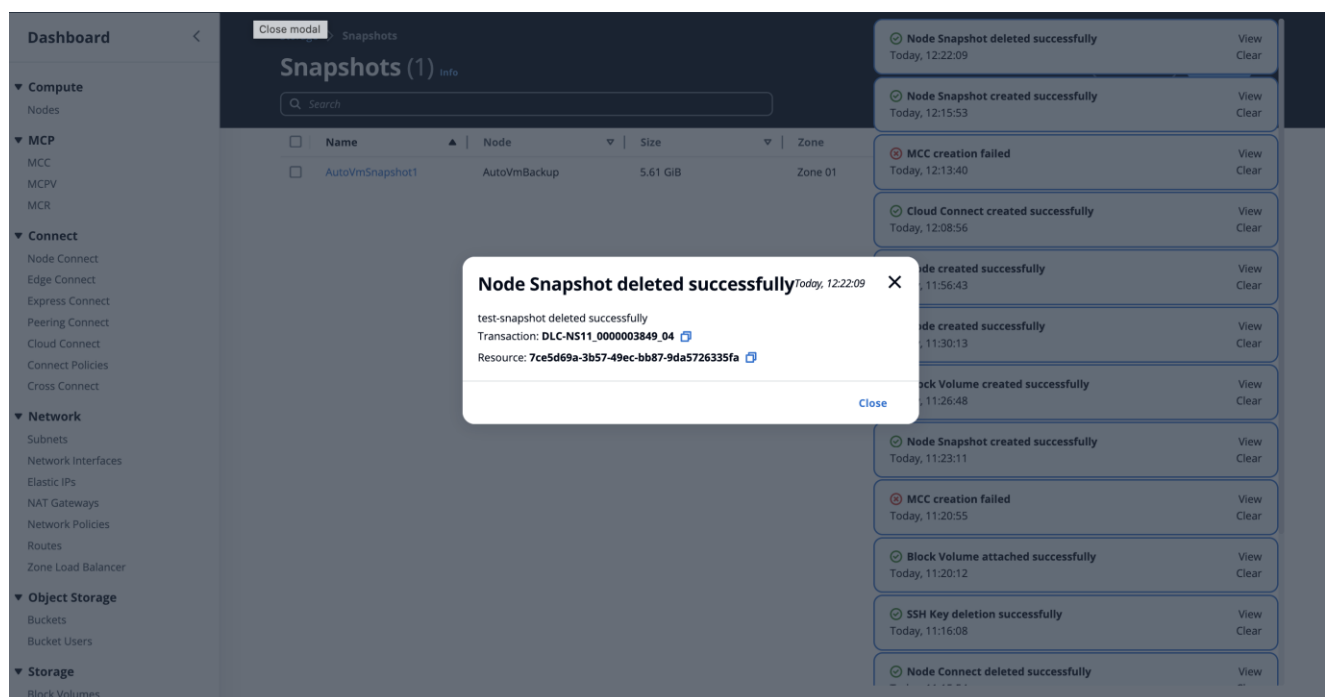


Figure: Snapshot – Pop up for successful deletion of snapshot.

8.0.4 Creating a Backup

▼ Connect

Node Connect
Edge Connect
Express Connect
Peering Connect
Cloud Connect
Connect Policies
Cross Connect

▼ Network

Subnets
Network Interfaces
Elastic IPs
NAT Gateways
Network Policies
Routes
Zone Load Balancer

▼ Object Storage

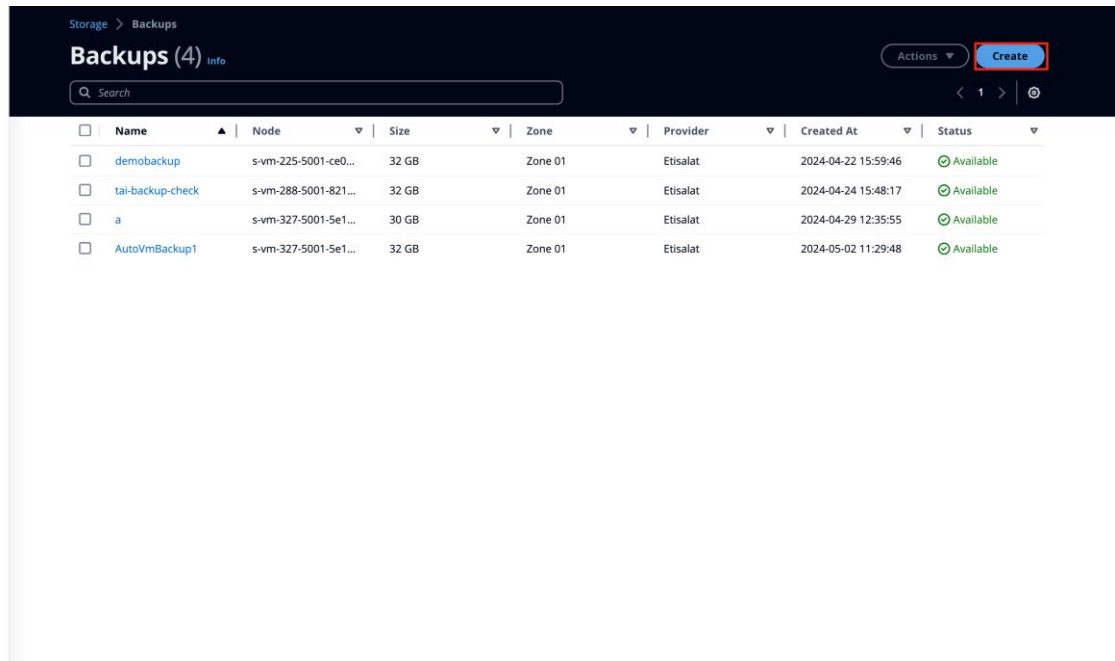
Buckets
Bucket Users

▼ Storage

Block Volumes
Snapshots
Backups
Scheduler

▼ Services

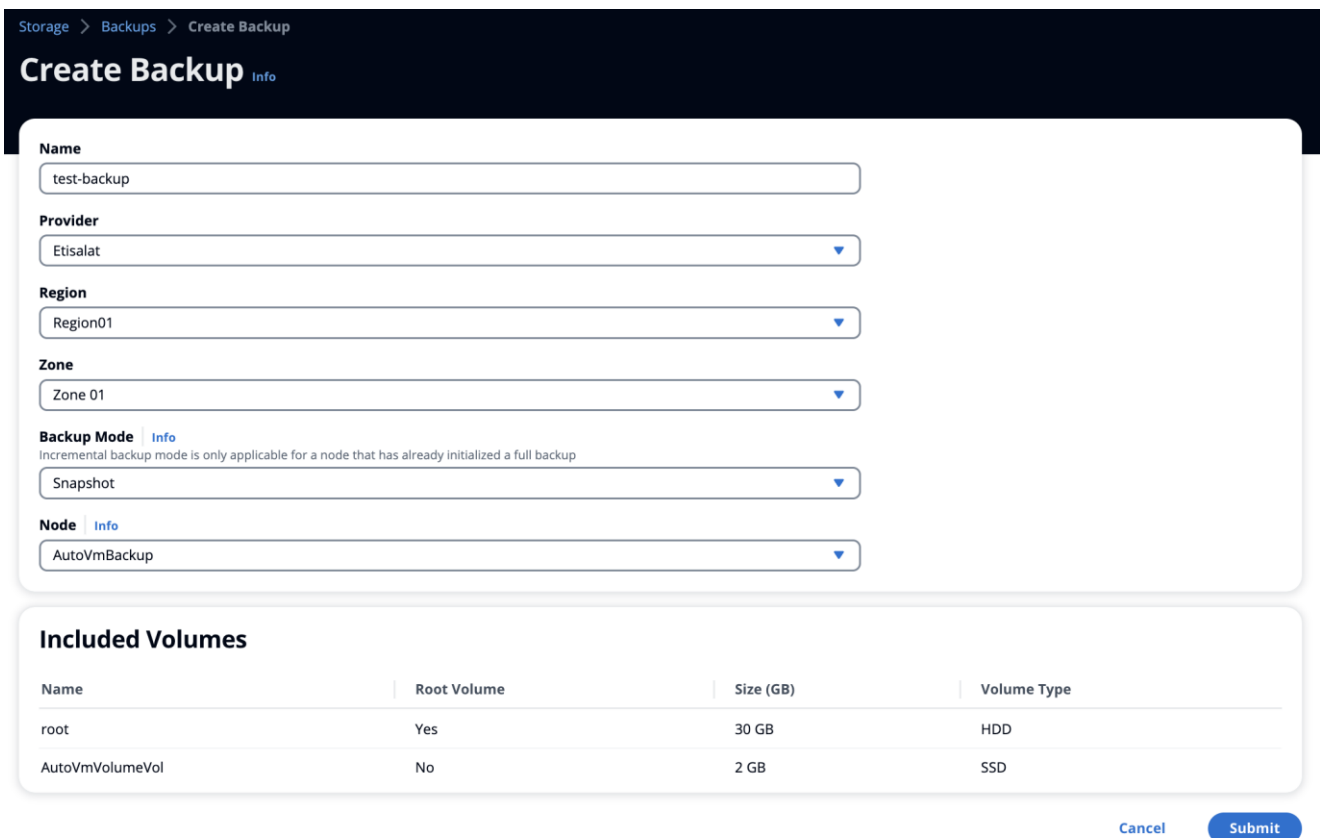
SSH Keys
Transaction Logs
Notifications



☐	Name	Node	Size	Zone	Provider	Created At	Status
☐	demobackup	s-vm-225-5001-ce0...	32 GB	Zone 01	Etisalat	2024-04-22 15:59:46	Available
☐	tai-backup-check	s-vm-288-5001-821...	32 GB	Zone 01	Etisalat	2024-04-24 15:48:17	Available
☐	a	s-vm-327-5001-5e1...	30 GB	Zone 01	Etisalat	2024-04-29 12:35:55	Available
☐	AutoVmBackup1	s-vm-327-5001-5e1...	32 GB	Zone 01	Etisalat	2024-05-02 11:29:48	Available

Figure: Creating a Backup – Step 1

Step 1: Go to Backups under Storage. To start creating a backup for a particular node, click Create.



Create Backup Info

Name
test-backup

Provider
Etisalat

Region
Region01

Zone
Zone 01

Backup Mode Info
Incremental backup mode is only applicable for a node that has already initialized a full backup
Snapshot

Node Info
AutoVmBackup

Included Volumes

Name	Root Volume	Size (GB)	Volume Type
root	Yes	30 GB	HDD
AutoVmVolumeVol	No	2 GB	SSD

Cancel Submit

Figure: Backup – Form inputs for creation

Step 2: Complete the necessary information in the details.

- Some details to complete are as follows: -Node backup provider.
- The node's current region and zone.
- -The desired backup mode, either Safe or Snapshot.

While safe mode stops the node to perform a slower but more consistent backup, snapshot has the lowest operation downtime and keeps the node running.

Click submit once you've chosen the node you wish to backup.

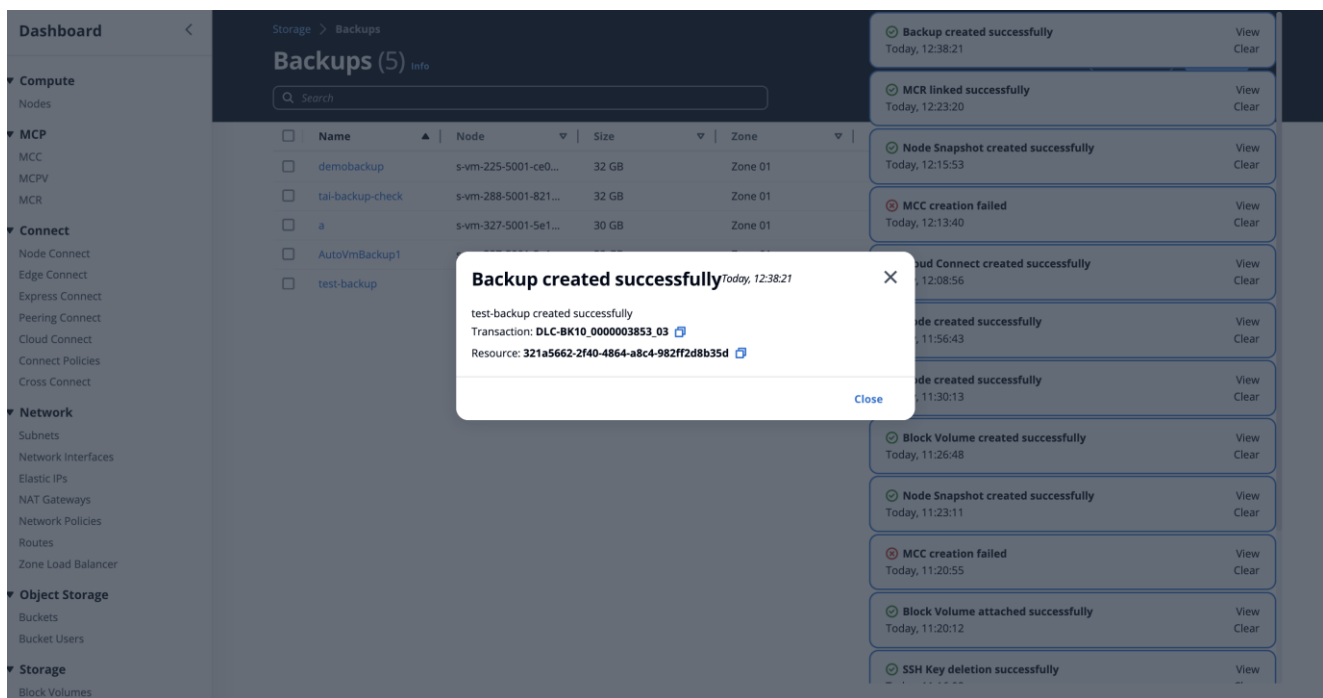


Figure: Backups - Backup successfully created.

Step 3: Confirm that the backup has been created when the backup node shows up on the backup tab and when the notification appears after submitting.

8.0.5 Restoring a Backup

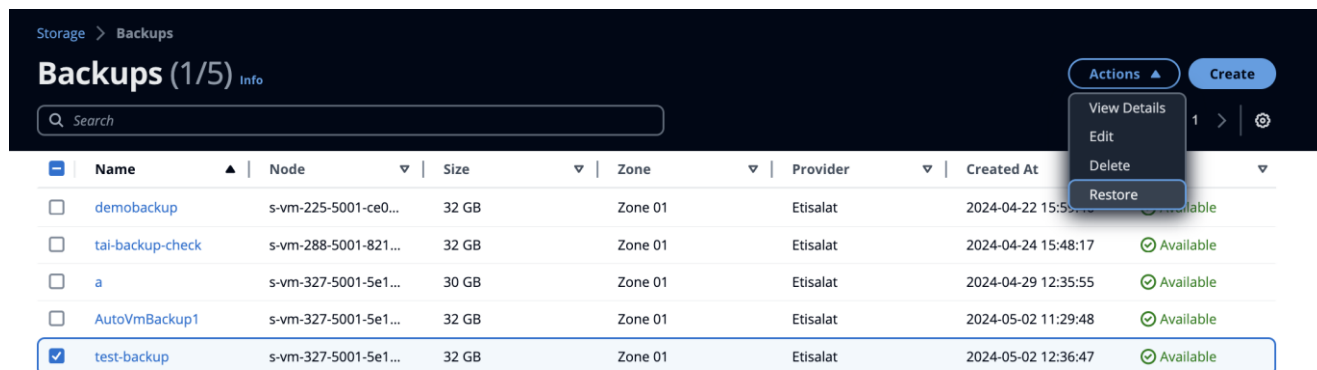


Figure: Restoring a Backup – Step 1

Step 1: At the main page, click on the “Storage” section and within the section, click on the “Backups” tab to proceed.

Step 2: At the “Backup” dashboard, select the backup from the table, click on “Restore” under the “Action” column.

Restore Backup

⚠

Restoring this backup will create a new volumes based on those included in the backup. You will need to specify a new name for these volumes which will be automatically attached to the restored node. The volumes that will be created are listed in the table below.

Volume ID	Volume Name	Volume Size	New Volume Name
09dac4b4-90b1-4253-bf99-ea9b92a8908f	AutoVmVolumeVol	2.0 GB	AutoVmVolumeVo

Any non-root volumes currently attached to the node can either be detached for use later or destroyed. Please select what you would like to happen using the selection box. Please note if you detach the volumes then you will still be billed for these volumes in addition to the new volumes created by the restore

☒ Detach any additional non-root volumes currently attached to the node
 ☐ Delete any additional non-root volumes currently attached to the node

Cancel

Confirm

Figure: Restore Backup – Pop up box.

Step 3: Provide a new name for the volume.

Step 4: This step will vary based on your preferences to detach any non-root volumes that are currently attached to the node for later use or destroyed.

Step 5: Click on the “Restore” button once you have filled and selected the respective fields.

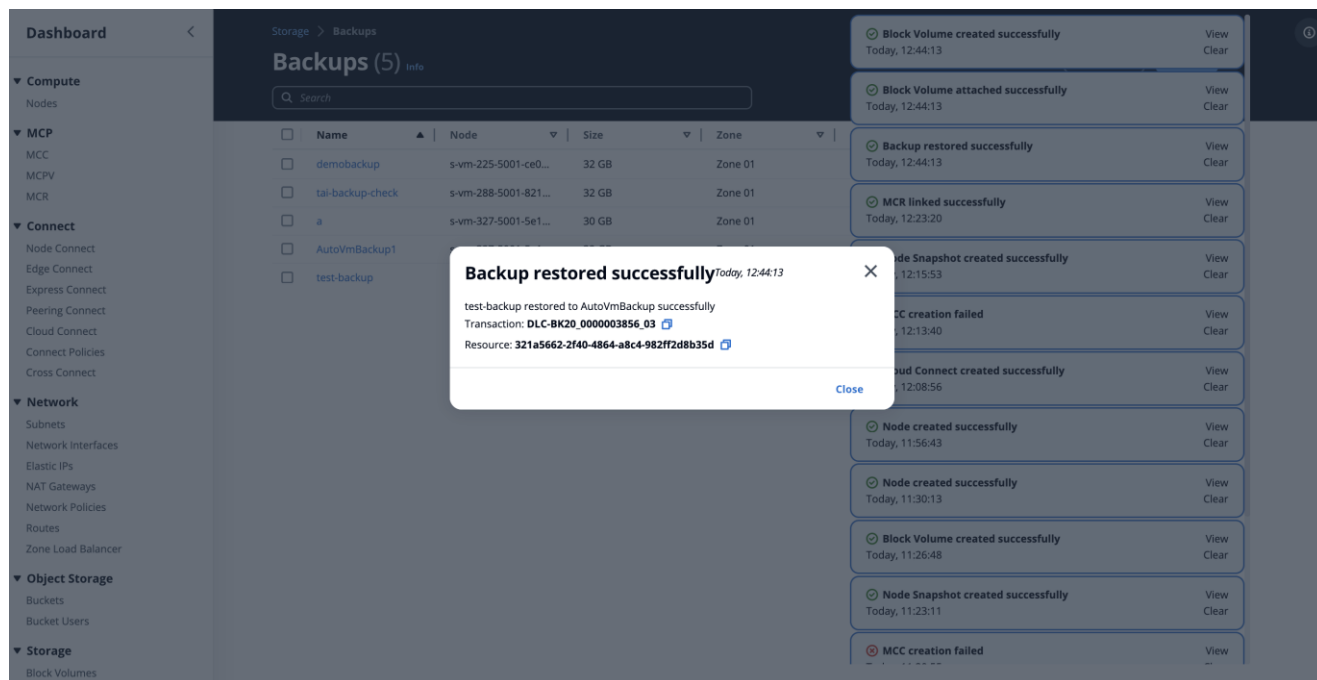


Figure: Backups – Successful restoration of backup.

8.0.6 Deleting a Backup

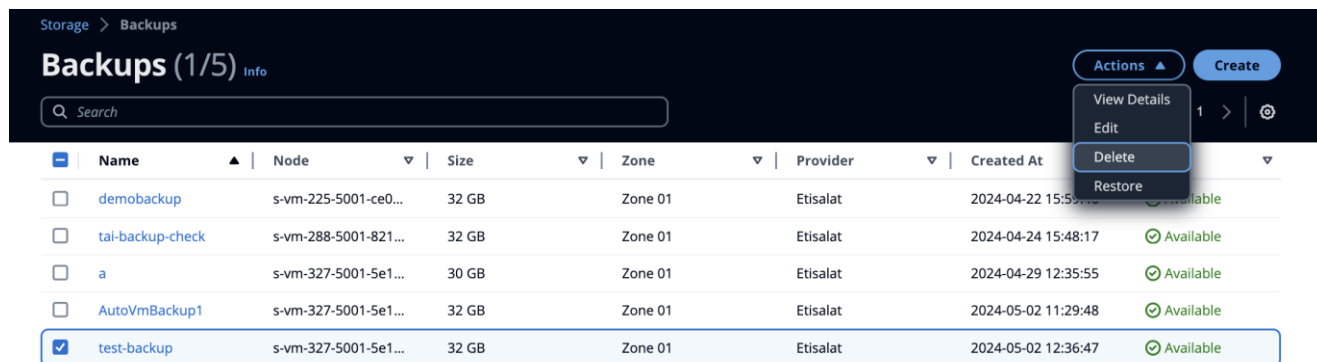


Figure: Backups – Deleting a backup.

Step 1: Click on the snapshot and select Actions then Delete to remove a backup.

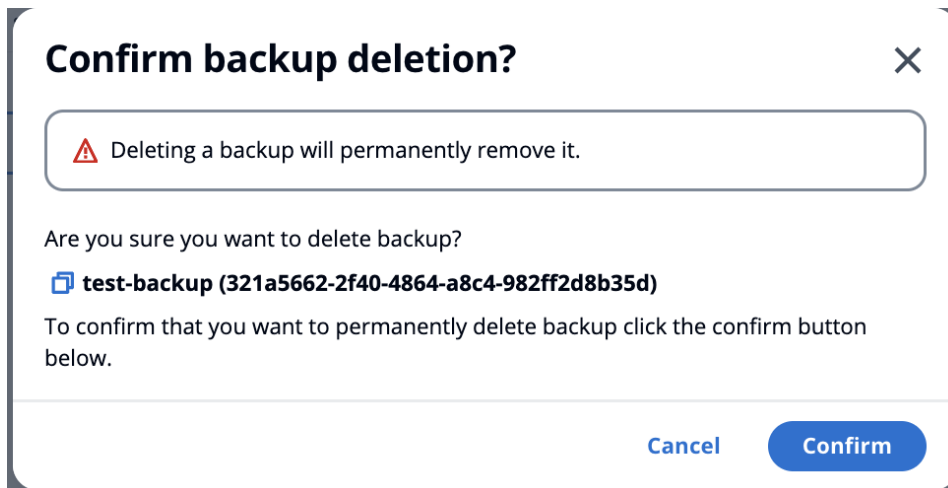


Figure: Backups – Confirmation for backup deletion

Step 2: To confirm the deletion of the backup, a pop-up window would display. Select "Confirm".

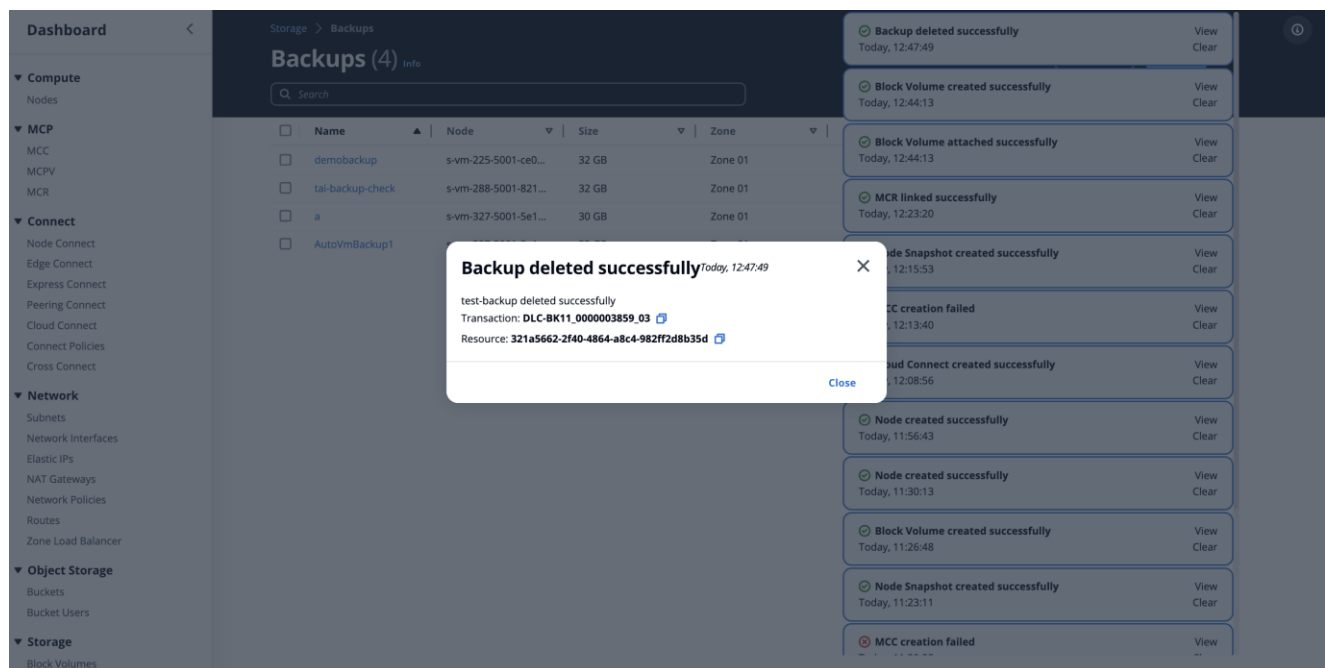


Figure: Backups – Successful deletion of backup.

Step 3: Users should receive a notification indicating that the backup has been deleted after they acknowledge the deletion. To verify that the removed node backup is not shown in the backup dashboard view, which would indicate that the deletion was successful.

9.0 Managing your Schedules

A schedule enables automated backups or snapshots at designated intervals and times. You can select multiple nodes within a single schedule, but they must share the same type (backup or snapshot) and belong to the same zone for effective scheduling.

9.0.1 Scheduling a Backup

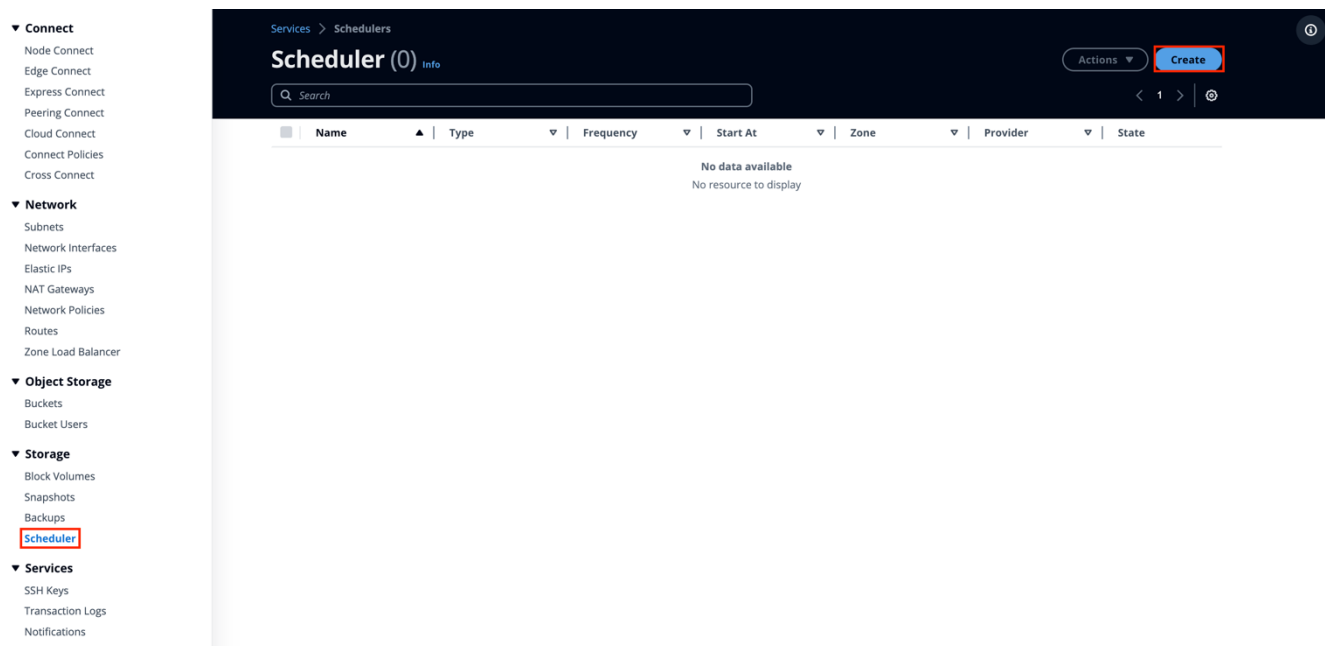


Figure: Scheduling a Backup – Step 1

Step 1: Go to the "Services" section and choose "Scheduler."

Step 2: Click on "Create."

Services > Schedulers > Create Schedule

Create Schedule Info

Provider
 Etisalat

Region
 Region01

Zone
 Zone 01

Type
 Backup

Mode
 Snapshot

Frequency
 Monthly

Start Time
 12:50

Select

1 X

Node Select
 Select one or multiple nodes within the selected zone

AutoVMnodeConnect X

Name
 test-scheduler

[Cancel](#)
[Submit](#)

Figure: Scheduling a Backup – Form Inputs

Step 2: Fill out the schedule creation page with the information below:

1. Indicate the node's Zone and Region.
2. Select the Type from the list of options, choosing between backup and snapshot. Make sure to choose the backup method as well if backup is selected.
3. Establish the frequency, or the number of times per day that the backup should run.
4. Establish the Start time and plan for the task to begin no later than one hour from the selected time.

After finishing, click "Submit" to continue.

Services > Schedulers

Scheduler (1) Info

Search

Scheduler created successfully
Today, 12:51:37 [View](#) [Clear](#)

☐	Name	Type	Frequency	Start At	Zone	Provider	State
☐	test-scheduler	Backup	Monthly	12:50	Zone 01	Etisalat	Enabled

Figure: Scheduler – Created successfully.

Step 3: Verify that the scheduler is established when the backup appears as indicated and when the notification pops up after submitting.

9.0.2 Disabling a Scheduler

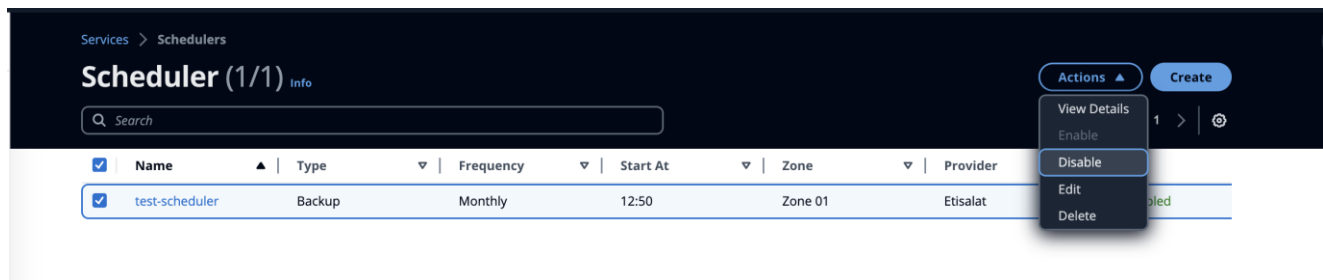


Figure: Scheduler – Disabling a Scheduler

Step 1: Choose the schedule from the scheduler tab, then select Actions then Disable.

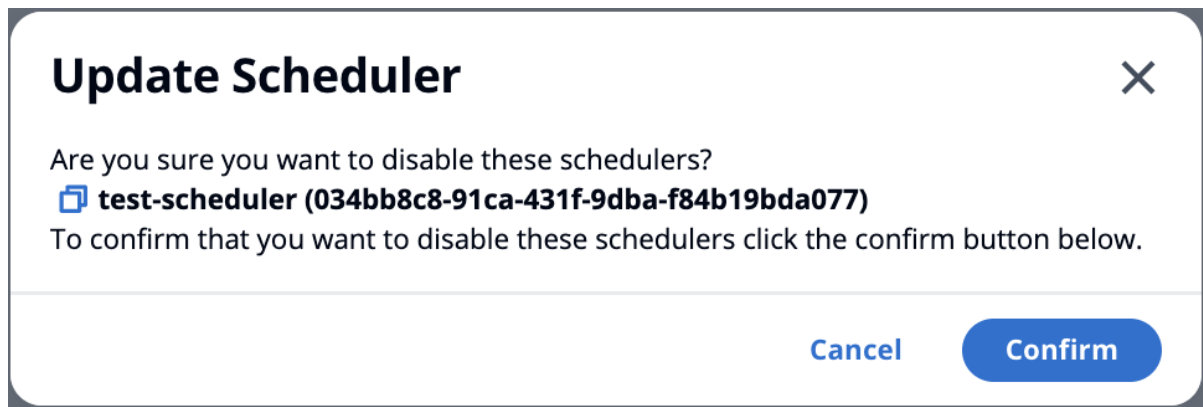


Figure: Scheduler – Confirmation for disabling scheduler.

Step 2: You'll be prompted with a popup window to confirm that you wish to turn off the scheduler. Select "Confirm".

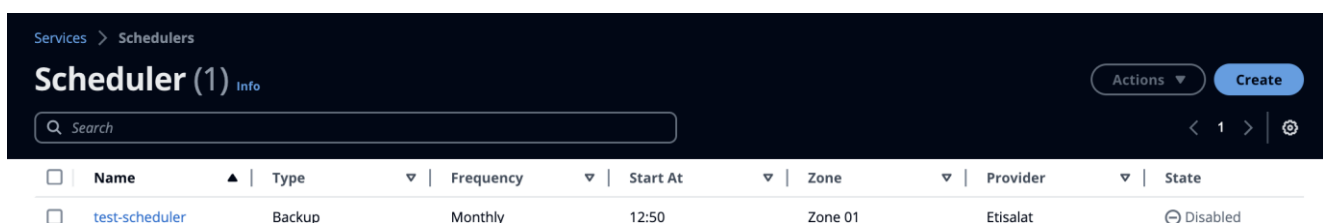


Figure: Scheduler – State reflects as Disabled.

Step 3: The schedule has been successfully disabled when, after some time, the scheduler state indicates that it is disabled.

9.0.3 Enabling a Scheduler

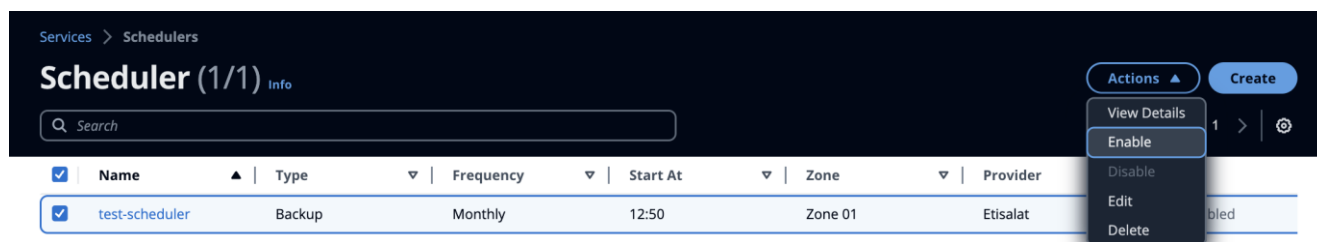


Figure: Scheduler – Enabling a Scheduler

Step 1: Choose the schedule from the scheduler tab, then select Actions then Enable.

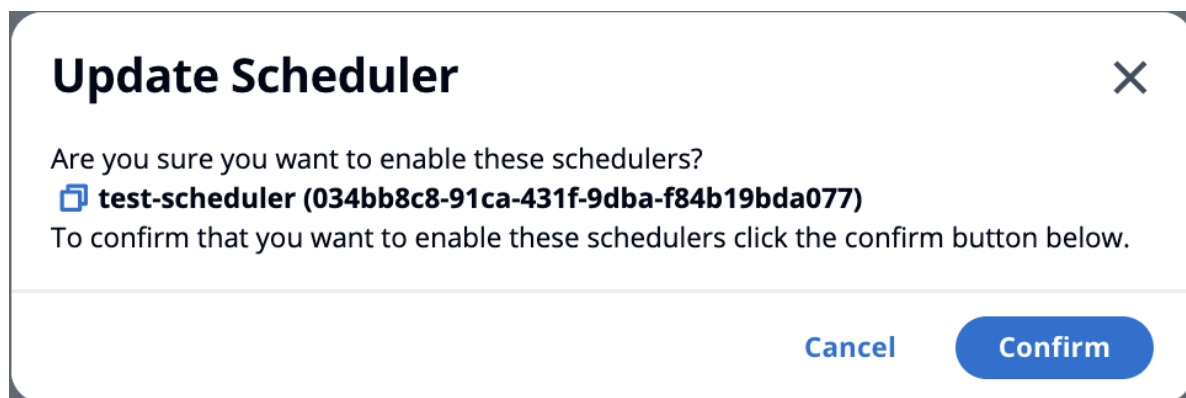


Figure: Scheduler – Confirmation for enabling.

Step 2: You'll be prompted to confirm if you wish to enable the scheduler using a popup box. Select "Confirm".

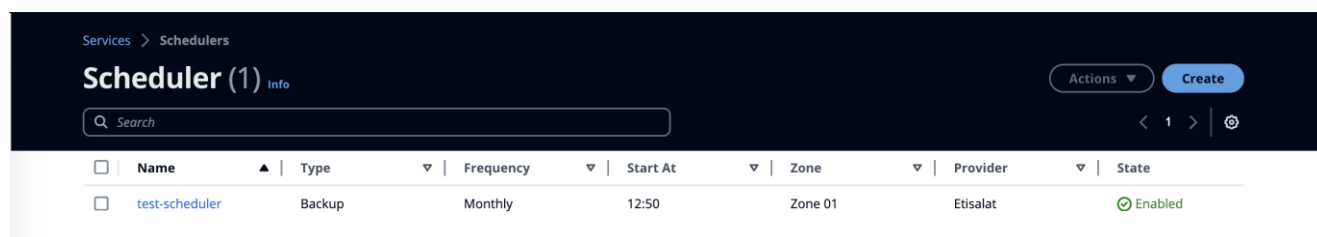


Figure: Scheduler – State displayed as Enabled.

Step 3: The schedule has been properly activated when, after a while, the scheduler state shows that it is enabled.

9.0.4 Deleting a Scheduler

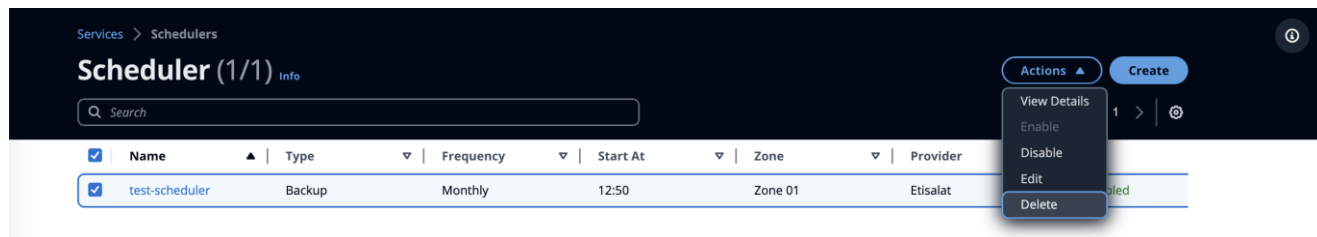


Figure: Scheduler – Deleting a Scheduler

Step 1: Choose the schedule from the scheduler tab, then select Actions then Delete.

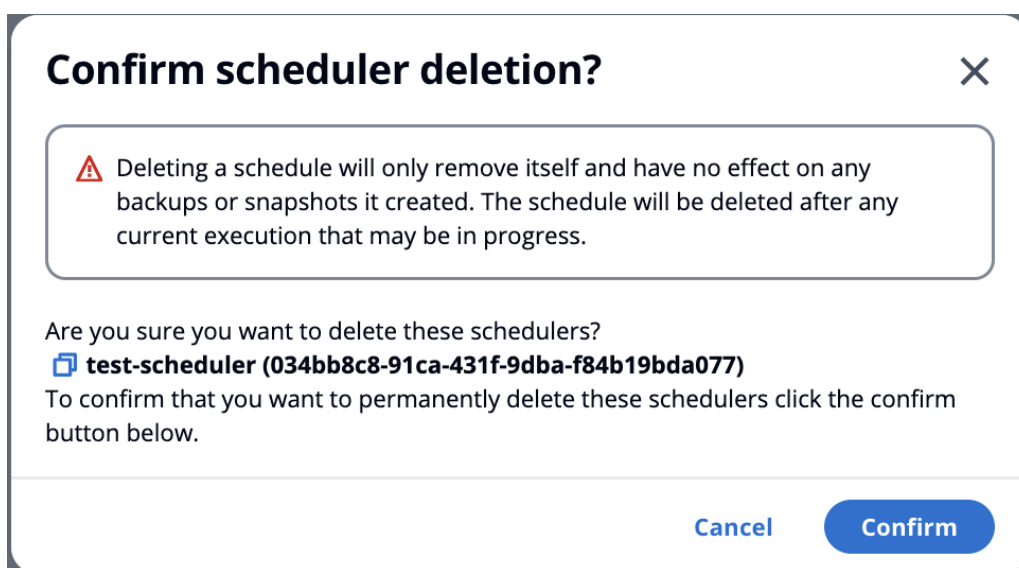


Figure: Scheduler – Confirmation for Deletion

Step 2: You'll be prompted to confirm that you wish to remove the scheduler using a popup box. Select "Confirm".

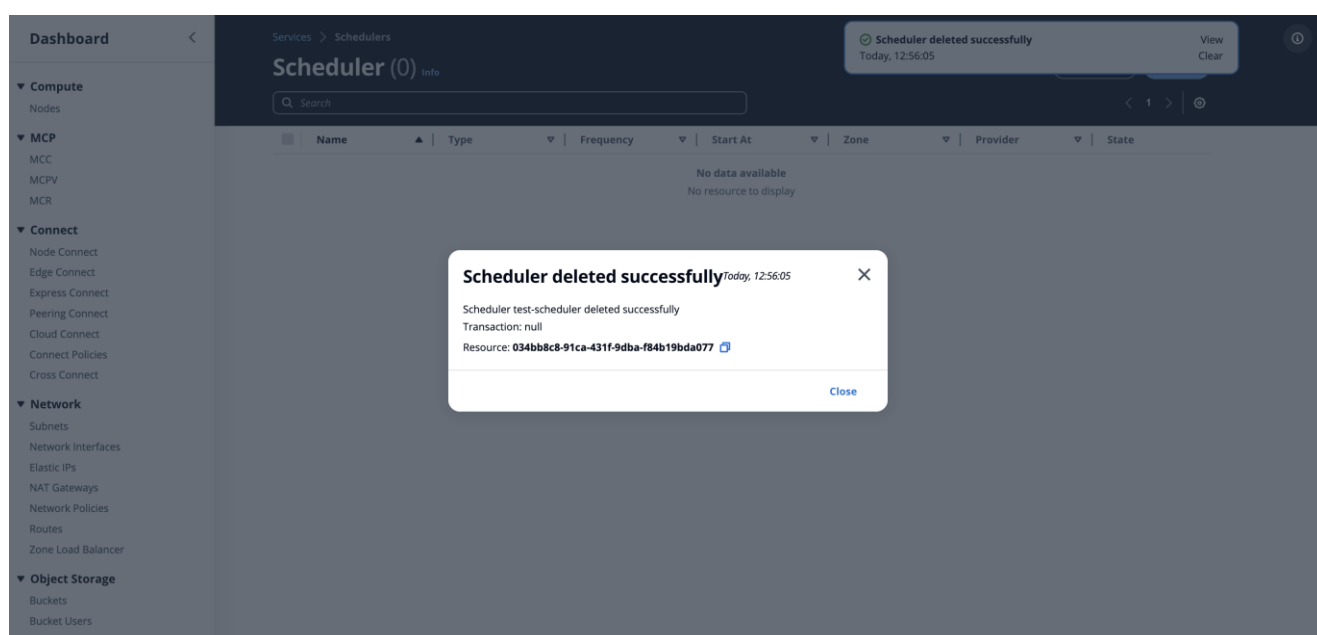


Figure: Scheduler – Successful deletion of Scheduler

Step 3: You ought to see a notification stating that the scheduler has been erased after validating the deletion. Verify that the deleted schedule is not shown in the scheduler dashboard view, which would indicate that the deletion was successful.

10.0 Network Management

10.0.1 Creating a Subnet

There are not any requirements to establish a subnet, but users must do it before users may establish a node.

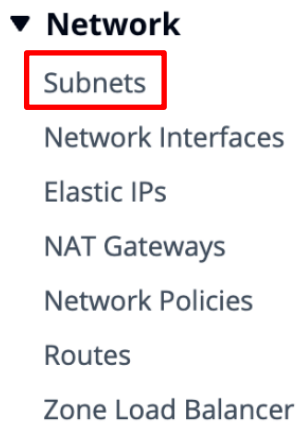
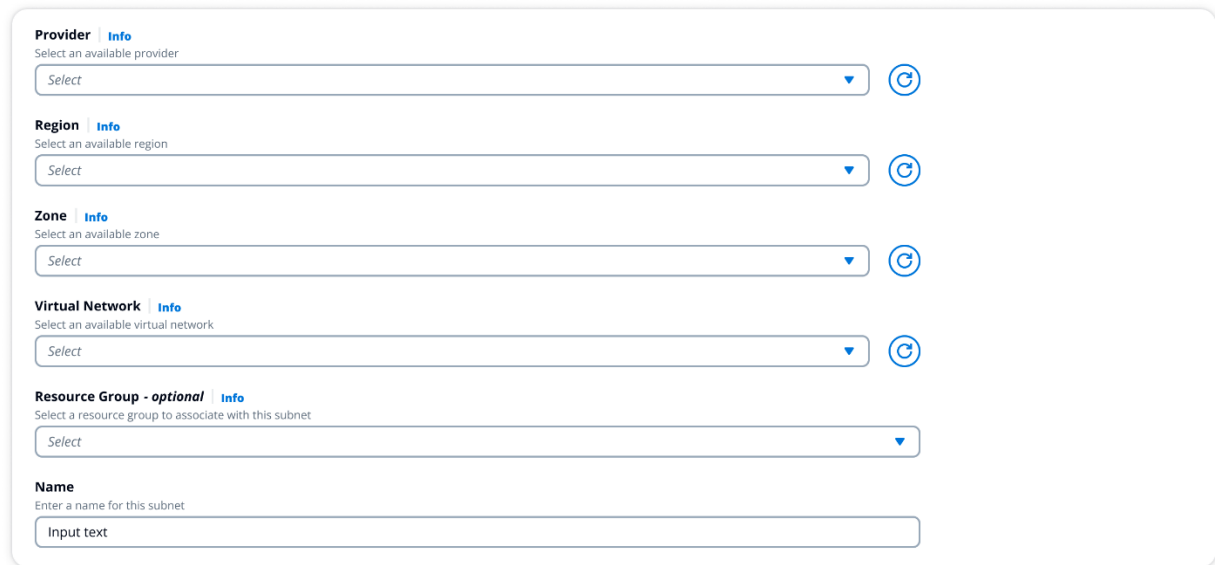


Figure: Creating Subnets

1. Choose Network > Subnets.
2. Select Create.
3. From the drop-down menu, choose a provider. This will load the regions that are available.
4. To load the available zones, select the Region.
5. Choose the Zone in which you want to construct your subnet.
6. Choose a Virtual Network from the drop-down menu if one is available. This CIDR will be used to assign a subnet.
7. Choose a Resource Group to link this subnet to if you would like.
8. Press the Create button. Once this is finished, you will receive a notification and the subnet will be ready for usage.



Provider [Info](#)
Select an available provider
Select

Region [Info](#)
Select an available region
Select

Zone [Info](#)
Select an available zone
Select

Virtual Network [Info](#)
Select an available virtual network
Select

Resource Group - optional [Info](#)
Select a resource group to associate with this subnet
Select

Name
Enter a name for this subnet
Input text

Figure: Creating a Subnet – Steps 2 to 8

10.0.2 Allocating/Releasing Elastic IPs

▼ Network

Subnets

Network Interfaces

Elastic IPs

NAT Gateways

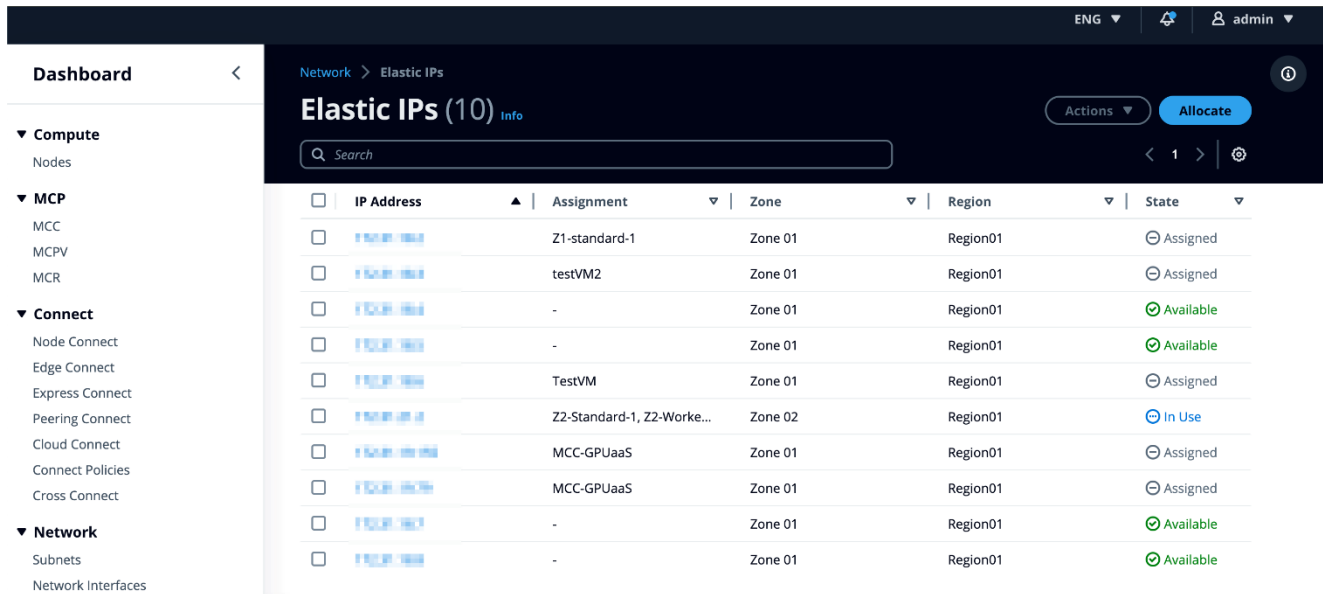
Network Policies

Routes

Zone Load Balancer

Figure: Selecting Elastic IPs

A static IPv4 address that is available in designated zones, each Elastic IP is mostly utilized to grant access to your resources from other networks. Depending on the service providing, these elastic IPs can be public (facing the internet) or limited to certain networks.

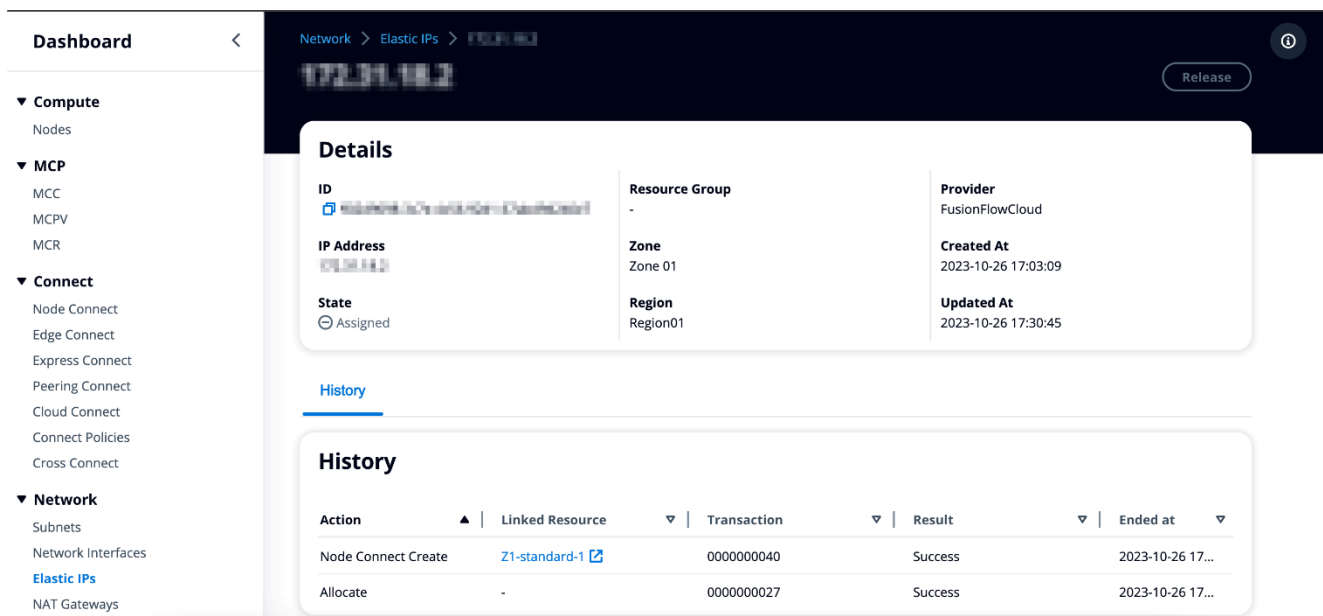


IP Address	Assignment	Zone	Region	State
172.31.10.1	Z1-standard-1	Zone 01	Region01	Assigned
172.31.10.2	testVM2	Zone 01	Region01	Assigned
172.31.10.3	-	Zone 01	Region01	Available
172.31.10.4	-	Zone 01	Region01	Available
172.31.10.5	TestVM	Zone 01	Region01	Assigned
172.31.10.6	Z2-Standard-1, Z2-Worke...	Zone 02	Region01	In Use
172.31.10.7	MCC-GPUaaS	Zone 01	Region01	Assigned
172.31.10.8	MCC-GPUaaS	Zone 01	Region01	Assigned
172.31.10.9	-	Zone 01	Region01	Available
172.31.10.10	-	Zone 01	Region01	Available

Figure: Elastic IP - Dashboard

An Elastic IP must be assigned before users can use it for anything. The instance will be allocated and made usable using an accessible IP address. As soon as it is assigned, users will have it assigned to them until it is relinquished.

Only resources from the same provider and within the same zone may be utilized with elastic IPs. Elastic IPs cannot be transferred between zones.



Details	
ID [ID]	Resource Group -
IP Address [IP Address]	Zone Zone 01
State Assigned	Region Region01
Provider FusionFlowCloud	
Created At 2023-10-26 17:03:09	
Updated At 2023-10-26 17:30:45	

History				
Action	Linked Resource	Transaction	Result	Ended at
Node Connect Create	Z1-standard-1	0000000040	Success	2023-10-26 17...
Allocate	-	0000000027	Success	2023-10-26 17...

Figure: Elastic IP – Selected Record

Once the instance is up and running, users can request the allocation of an Elastic IP at any time. By doing this, the IP provided will become exclusively theirs to use.

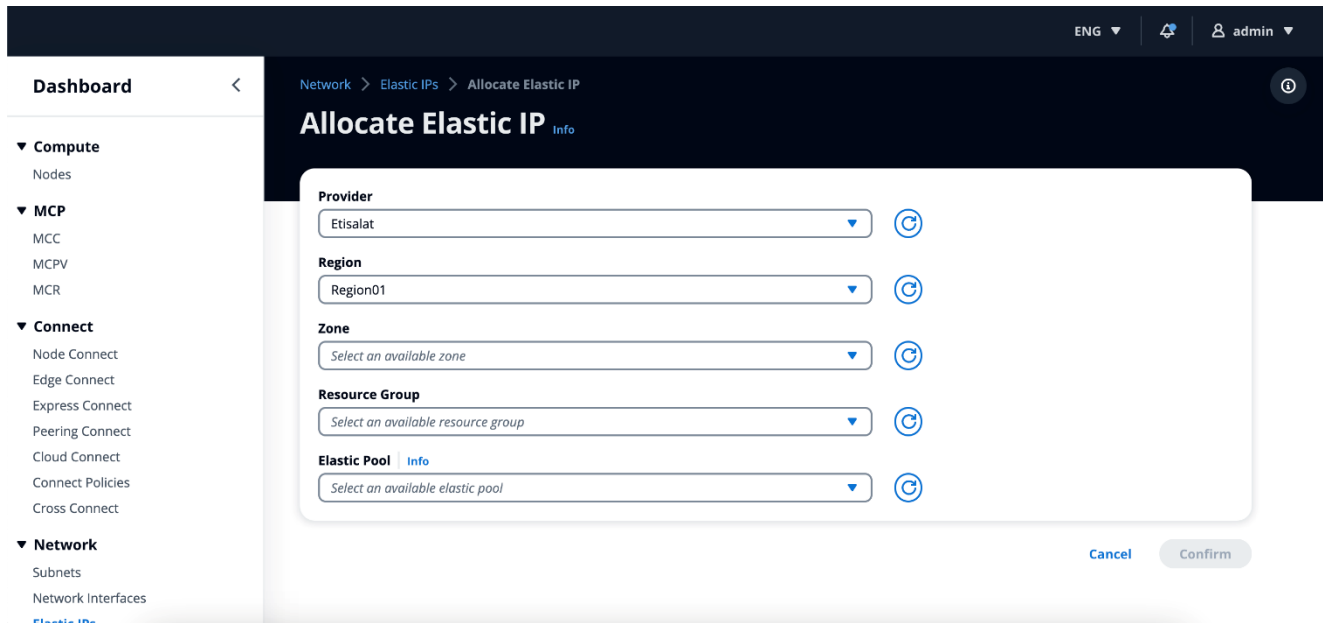


Figure: Elastic IP - Allocating Elastic IP

1. Open the Network > Elastic IPs menu.
2. Select "Allocate"
3. Decide on a Zone and Region.
4. Choose a Provider that is available. What resources can be used with this IP will be determined by this. For instance, the node provider and elastic IP must match.
5. Select Allocate. You will be assigned an IP as long as there are IPs available within the Zone and Provider. In the list view, it ought to appear as Available.

You can use an Elastic IP with **Node Connect** or **App Connect** after it is in the Available state.

You will not be able to utilize an elastic IP after you release it; the IP will be returned to the service provider. An Elastic IP can only be released while it is in the Available state.

Open the Network > Elastic IPs menu.

Click Actions, then Release after selecting the Elastic IP you want to release.

After it is returned, you will not be able to use the IP anymore.

10.0.3 Creating NAT Gateways

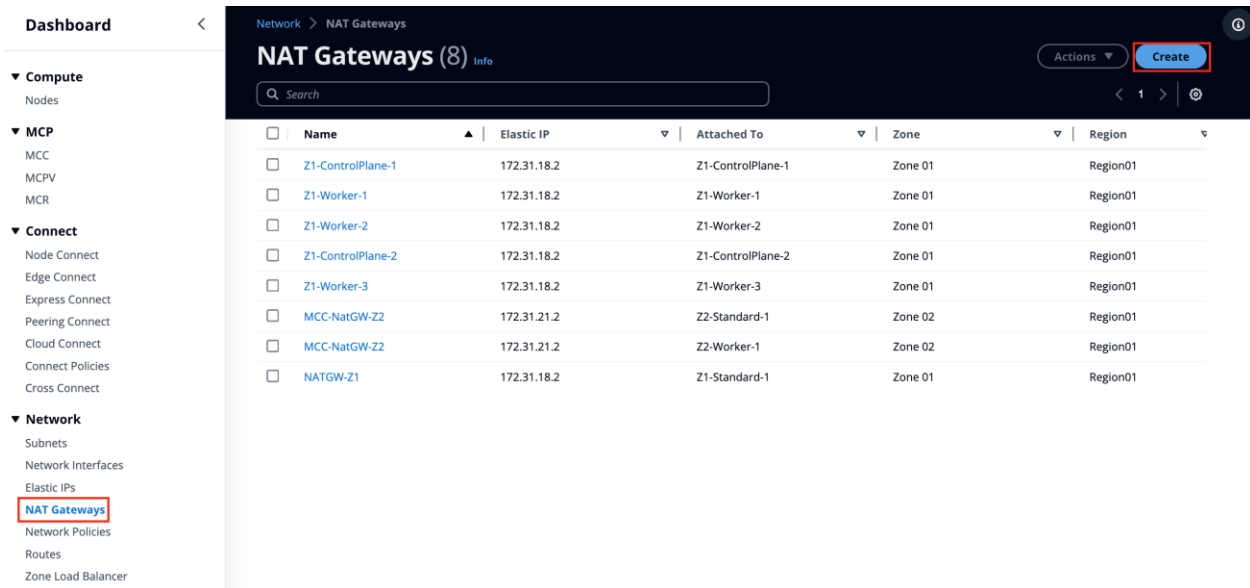
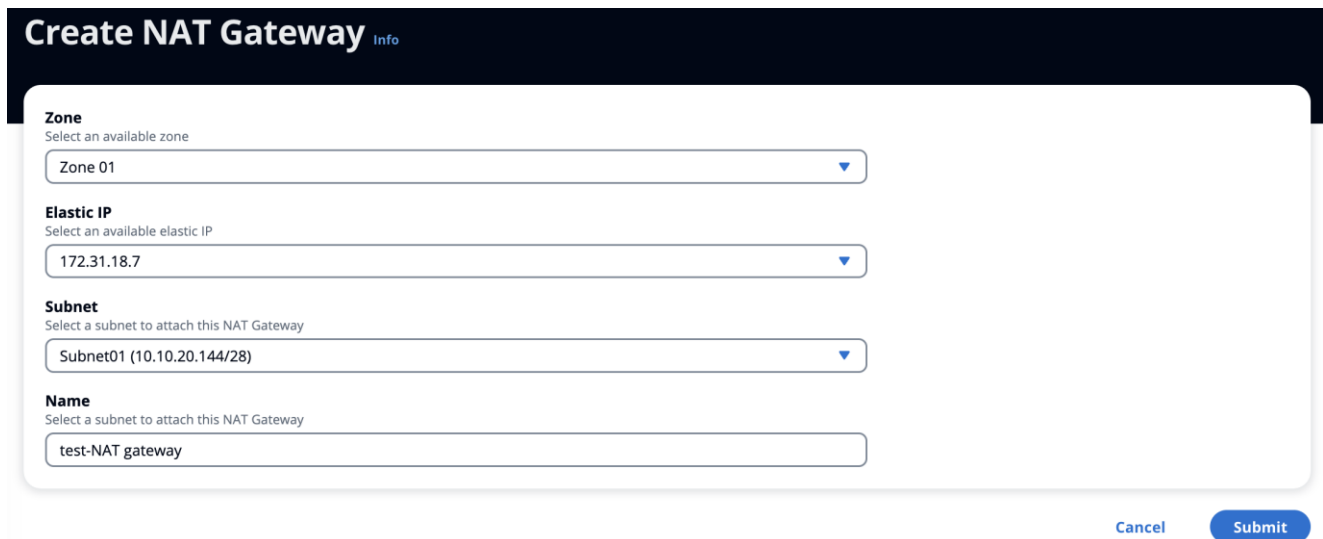


Figure: Creating a NAT Gateway – Step 1

Step 1: Click on the “Network” tab, select “NAT Gateways” to begin.

Step 2: Once you have landed on the “NAT Gateways” page, click on the “Create” button.



The screenshot shows the 'Create NAT Gateway' form. It has four main sections: Zone, Elastic IP, Subnet, and Name. Each section has a dropdown menu for selection. The 'Zone' dropdown is set to 'Zone 01'. The 'Elastic IP' dropdown is set to '172.31.18.7'. The 'Subnet' dropdown is set to 'Subnet01 (10.10.20.144/28)'. The 'Name' field is a text input with 'test-NAT gateway'. At the bottom right, there are 'Cancel' and 'Submit' buttons.

Figure: Selecting your network settings from the available options – Step 3 to 5

Step 3: Select your zone from the available options in the dropdown list under the “Zone” field.

Step 4: Select your elastic IP from the available options in the dropdown list under the “Elastic IP” field.

Step 5: Select the correct subnet in the “Subnet” field.

Step 6: Provide a name to associate with the NAT gateway in the “Name” field.

Step 7: Once you have verified all the relevant information are correct, proceed to click on the “Submit” button.

Dashboard

▼ Compute

Nodes

▼ MCP

MCC

MCPV

MCR

▼ Connect

Node Connect

Edge Connect

Express Connect

Peering Connect

Cloud Connect

Connect Policies

Cross Connect

▼ Network

Subnets

Network Interfaces

Elastic IPs

NAT Gateways

Network Policies

Routes

Zone Load Balancer

Network > NAT Gateways

NAT Gateway created successfully

Today, 16:08:06

View

Clear

NAT Gateways (9)

Search

0

0

0

0

2

0

☐	Name	Elastic IP	Attached To	Zone	Region
☐	Z1-ControlPlane-1	172.31.18.2	Z1-ControlPlane-1	Zone 01	Region01
☐	Z1-Worker-1	172.31.18.2	Z1-Worker-1	Zone 01	Region01
☐	Z1-Worker-2	172.31.18.2	Z1-Worker-2	Zone 01	Region01
☐	Z1-ControlPlane-2	172.31.18.2	Z1-ControlPlane-2	Zone 01	Region01
☐	Z1-Worker-3	172.31.18.2	Z1-Worker-3	Zone 01	Region01
☐	MCC-NatGW-Z2	172.31.21.2	Z2-Standard-1	Zone 02	Region01
☐	MCC-NatGW-Z2	172.31.21.2	Z2-Worker-1	Zone 02	Region01
☐	NATGW-Z1	172.31.18.2	Z1-Standard-1	Zone 01	Region01
☐	test-NAT gateway	172.31.18.7	Subnet01	Zone 01	Region01

Figure: NAT Gateway – Created successfully.

10.0.4 Creating Network Policies

Network policies provide you the power to designate and establish guidelines for any network or elastic IP that you control. This enables you to specifically control outgoing and incoming rules for every Elastic IP address and subnet that you have.

These policies let you implicitly allow or prohibit protocols, ports, and IP ranges in either direction of communication, giving you the ability to create precise traffic limitations (rules) between your networks. You can safeguard your environments with a "three tier architecture" by utilizing these policies.

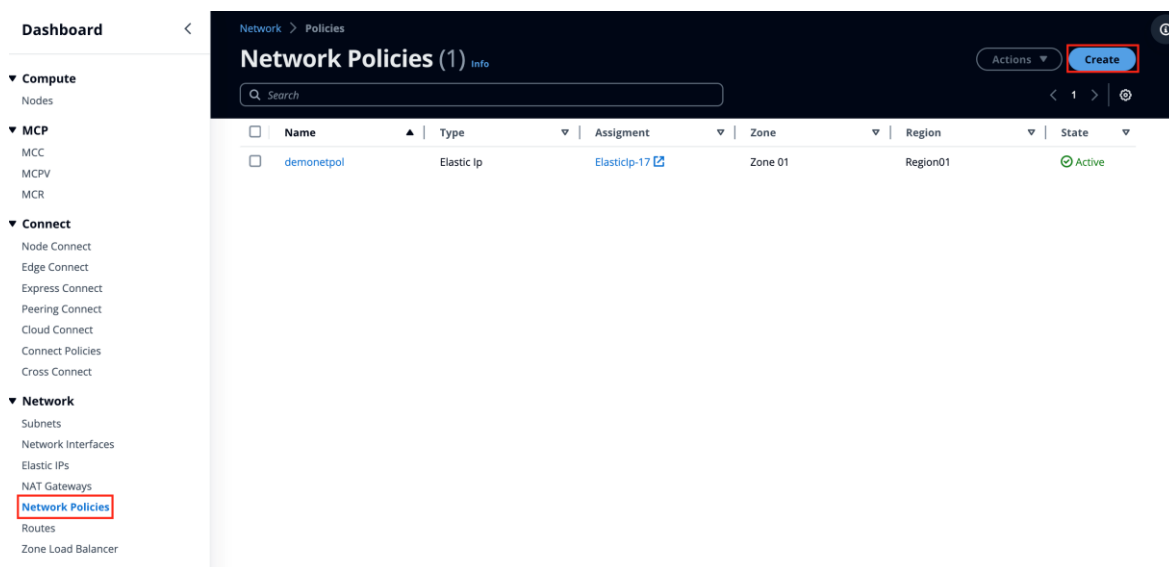
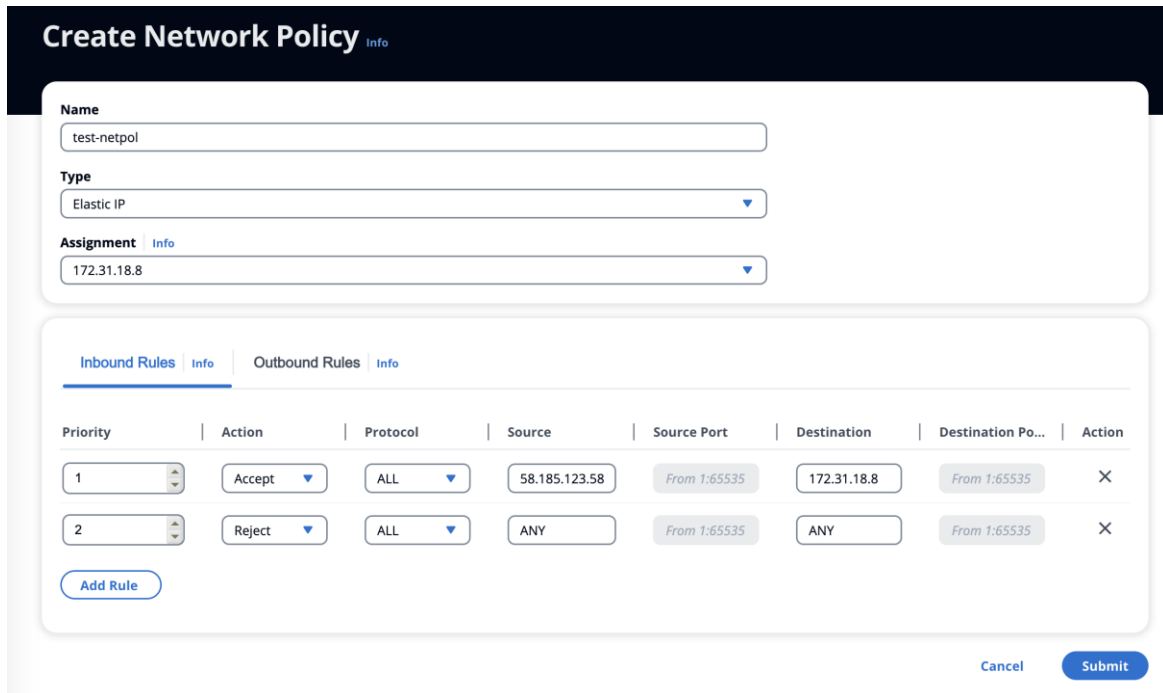


Figure: Creating a Network Policy

Step 1: Click on the "Network" tab, select "Network Policies" to begin.

Step 2: Once you have landed on the "Network Policies" dashboard, click on the "Create" button.



Create Network Policy Info

Name
test-netpol

Type
Elastic IP

Assignment Info
172.31.18.8

Inbound Rules Info | **Outbound Rules** Info

Priority	Action	Protocol	Source	Source Port	Destination	Destination Po...	Action
1	Accept	ALL	58.185.123.58	From 1:65535	172.31.18.8	From 1:65535	×
2	Reject	ALL	ANY	From 1:65535	ANY	From 1:65535	×

[Add Rule](#)

[Cancel](#) [Submit](#)

Figure: Create Network Policy - Form Inputs

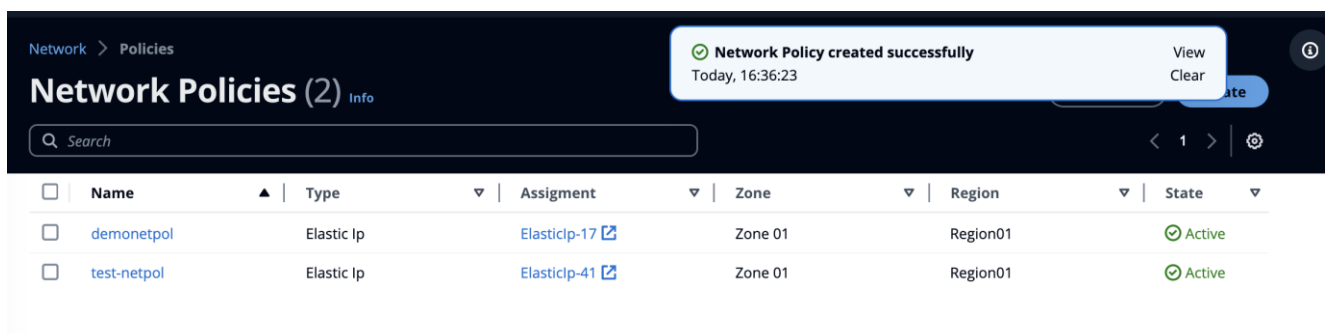
Step 3: Provide a name for the network policy under the “Name” field.

Step 4: Select the type (Subnet OR Elastic IP) for this network policy to be assigned to under the “Type” field.

Step 5: Select the Elastic IP OR Subnet as defined in Step 4 that the network will be assigned to under the “Assignment” field.

Step 6: Define your inbound/outbound rules, accordingly, enter a valid IP, CIDR, port.

Step 7: Once you have verified all the relevant information are correct, proceed to click on the “Create” button.



Network Policies (2) Info

[Search](#)

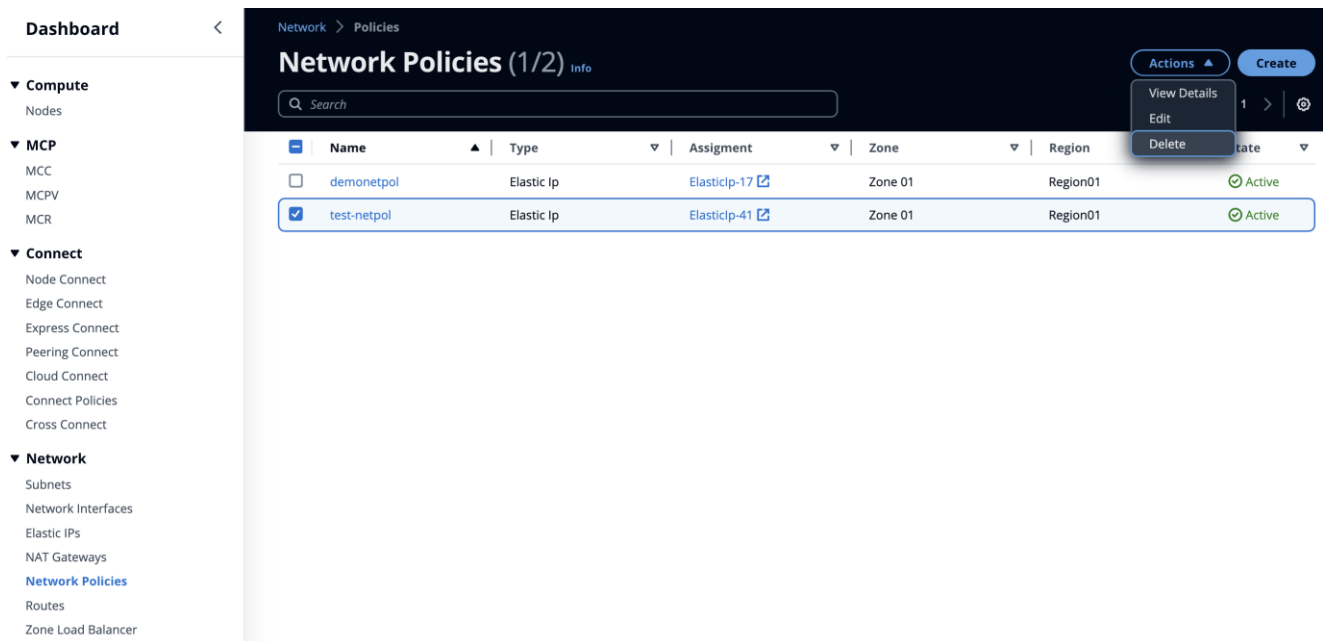
Network Policy created successfully
Today, 16:36:23 [View](#) [Clear](#)

	Name	Type	Assignment	Zone	Region	State
☐	demonetpol	Elastic Ip	ElasticIp-17	Zone 01	Region01	Active
☐	test-netpol	Elastic Ip	ElasticIp-41	Zone 01	Region01	Active

Figure: Network Policy – Created successfully.

Step 8: Once you have successfully created the policy, it will be displayed in the “Network Policies” dashboard.

10.0.5 Deleting Network Policies

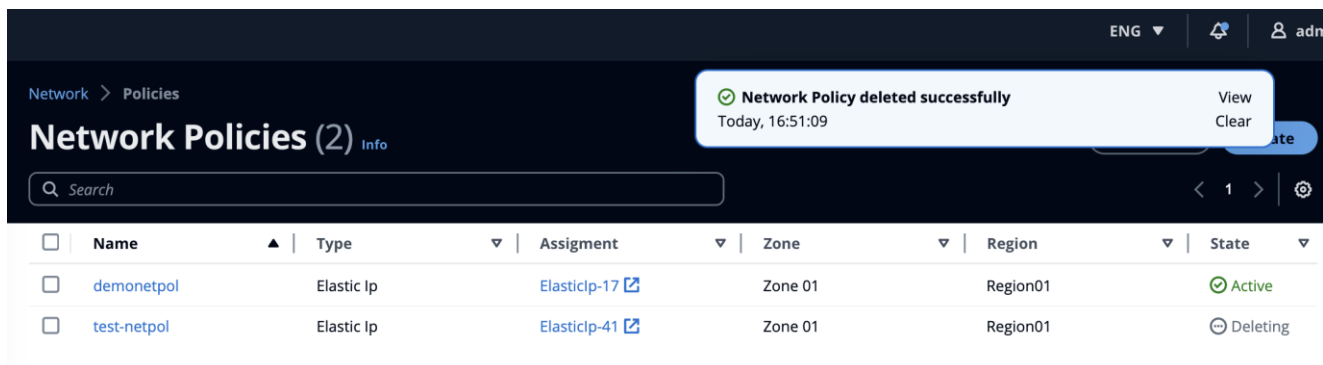


The screenshot shows the 'Network Policies' dashboard. On the left, a sidebar lists various network components under 'Compute', 'MCP', 'Connect', and 'Network'. The 'Network Policies' link is highlighted. The main panel shows a table of network policies. The 'test-netpol' policy is selected, and the 'Delete' action is being performed from the 'Actions' menu.

Name	Type	Assignment	Zone	Region	State
demonetpol	Elastic Ip	ElasticIp-17	Zone 01	Region01	Active
test-netpol	Elastic Ip	ElasticIp-41	Zone 01	Region01	Active

Figure: Network Policies – Delete

Step 1: To begin the deletion process, make sure the network policy is selected, then click Actions then Delete. When the pop-up window displays, confirm the deletion.



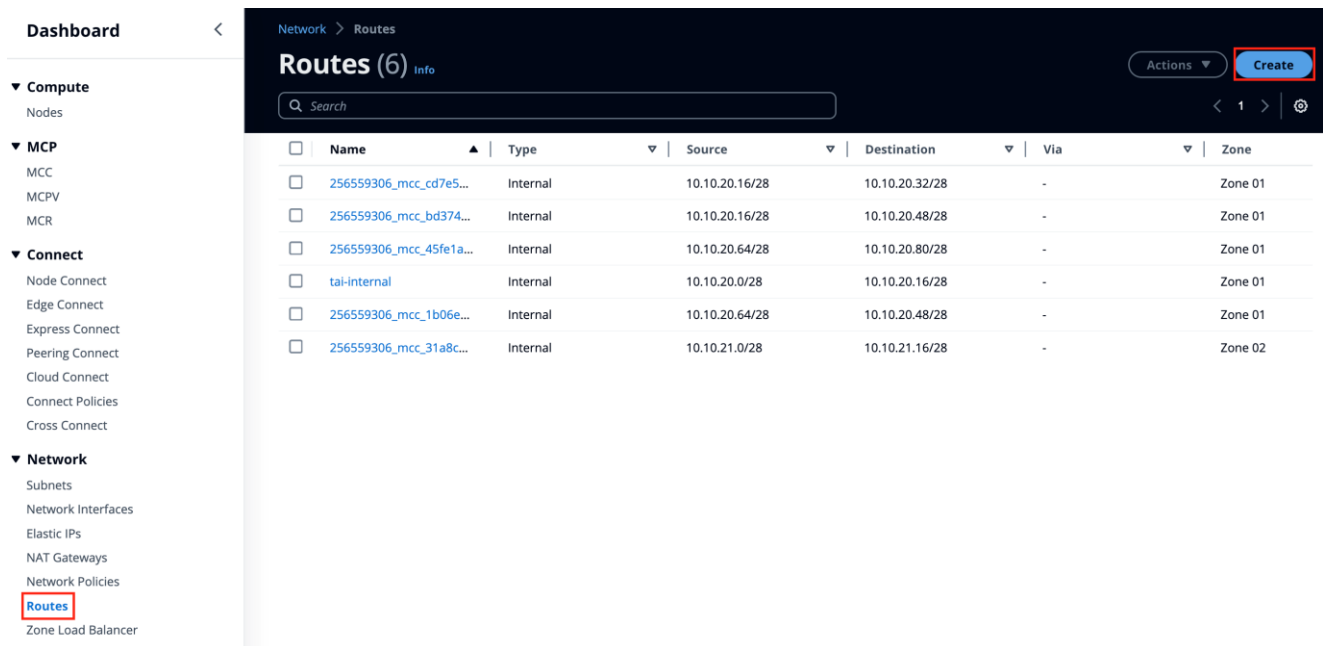
The screenshot shows the 'Network Policies' dashboard after the deletion of the 'test-netpol' policy. A notification banner at the top right indicates 'Network Policy deleted successfully' with the timestamp 'Today, 16:51:09'. The table now shows only the 'demonetpol' policy.

Name	Type	Assignment	Zone	Region	State
demonetpol	Elastic Ip	ElasticIp-17	Zone 01	Region01	Active
test-netpol	Elastic Ip	ElasticIp-41	Zone 01	Region01	Deleting

Figure: Network Policies – Successful deletion

Step 2: Check to see if the network policy is absent from the dashboard and that the notification banner which appears indicating its deletion has been executed successfully.

10.0.6 Creating Routes



Dashboard < Network > Routes

Routes (6) Info

Search

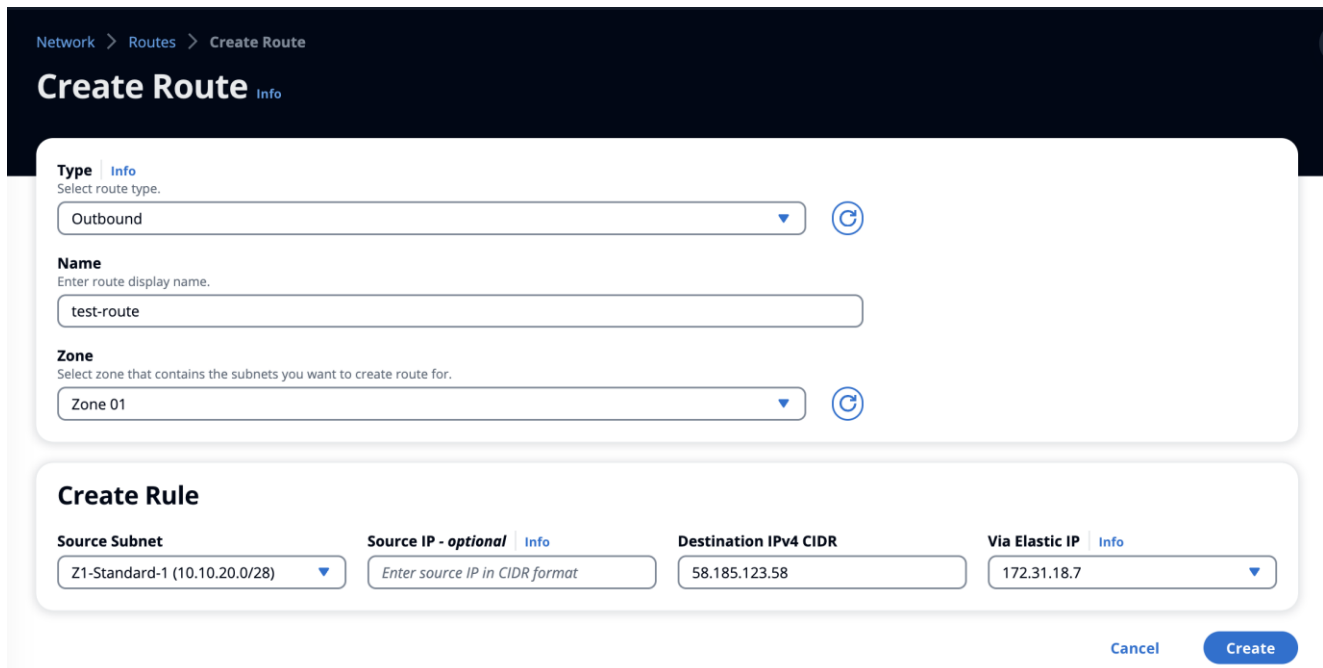
Actions Create

Name	Type	Source	Destination	Via	Zone
256559306_mcc_cd7e5...	Internal	10.10.20.16/28	10.10.20.32/28	-	Zone 01
256559306_mcc_bd374...	Internal	10.10.20.16/28	10.10.20.48/28	-	Zone 01
256559306_mcc_45fe1a...	Internal	10.10.20.64/28	10.10.20.80/28	-	Zone 01
tal-internal	Internal	10.10.20.0/28	10.10.20.16/28	-	Zone 01
256559306_mcc_1b06e...	Internal	10.10.20.64/28	10.10.20.48/28	-	Zone 01
256559306_mcc_31a8c...	Internal	10.10.21.0/28	10.10.21.16/28	-	Zone 02

Figure: Creating a Route – Step 1

Step 1: Click on the “Network” tab, select “Routes” to begin.

Step 2: Once you have landed on the “Routes” dashboard, click on the “Create” button.



Network > Routes > Create Route

Create Route

Type Info

Select route type.

Outbound

Name

Enter route display name.

test-route

Zone

Select zone that contains the subnets you want to create route for.

Zone 01

Create Rule

Source Subnet Z1-Standard-1 (10.10.20.0/28)

Source IP - optional Info Enter source IP in CIDR format

Destination IPv4 CIDR 58.185.123.58

Via Elastic IP Info 172.31.18.7

Cancel Create

Figure: Creating a Route – Form inputs

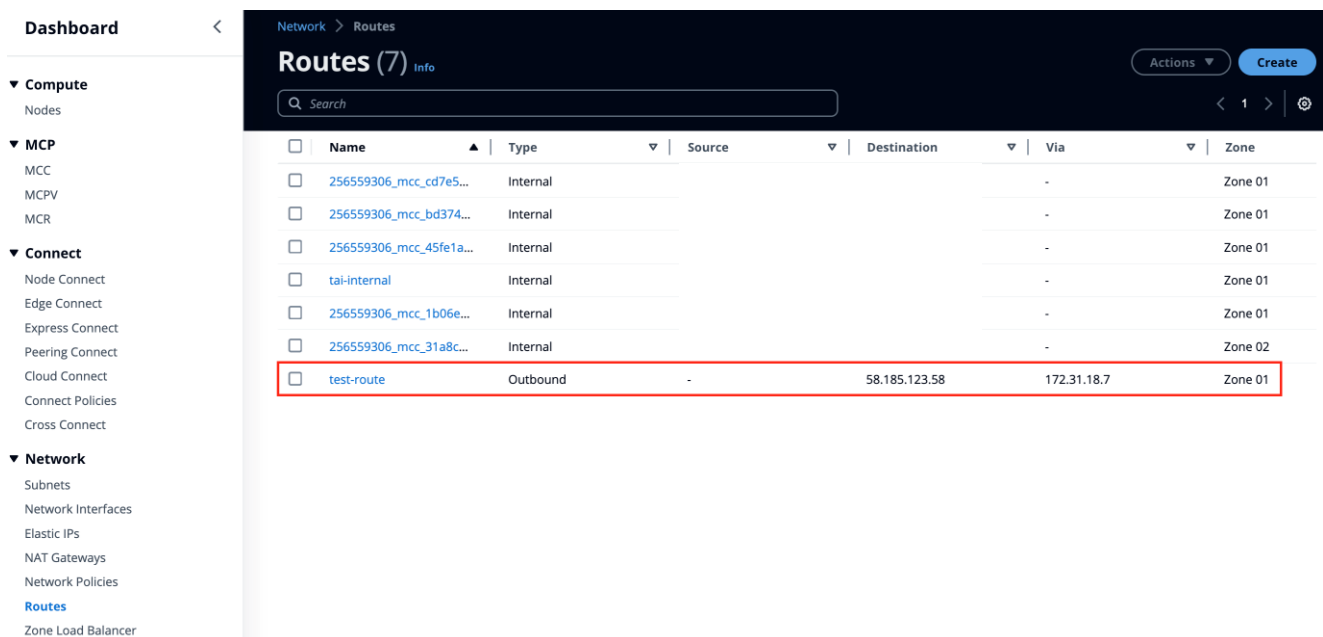
Step 3: Select the type of route (Outbound OR Internal) under the “Type” field.

Step 4: Provide a name for the route under the "Name" field.

Step 5: Select the zone that this route will be created under the "Zone" field.

Step 6: In the "Create Rule" section, you need to select the source subnet, provide the specific source IP address, specify the destination IP address, and indicate the Elastic IP through which the traffic will be routed.

Step 7: Once you have verified all the relevant information are correct, proceed to click on the "Add" button.



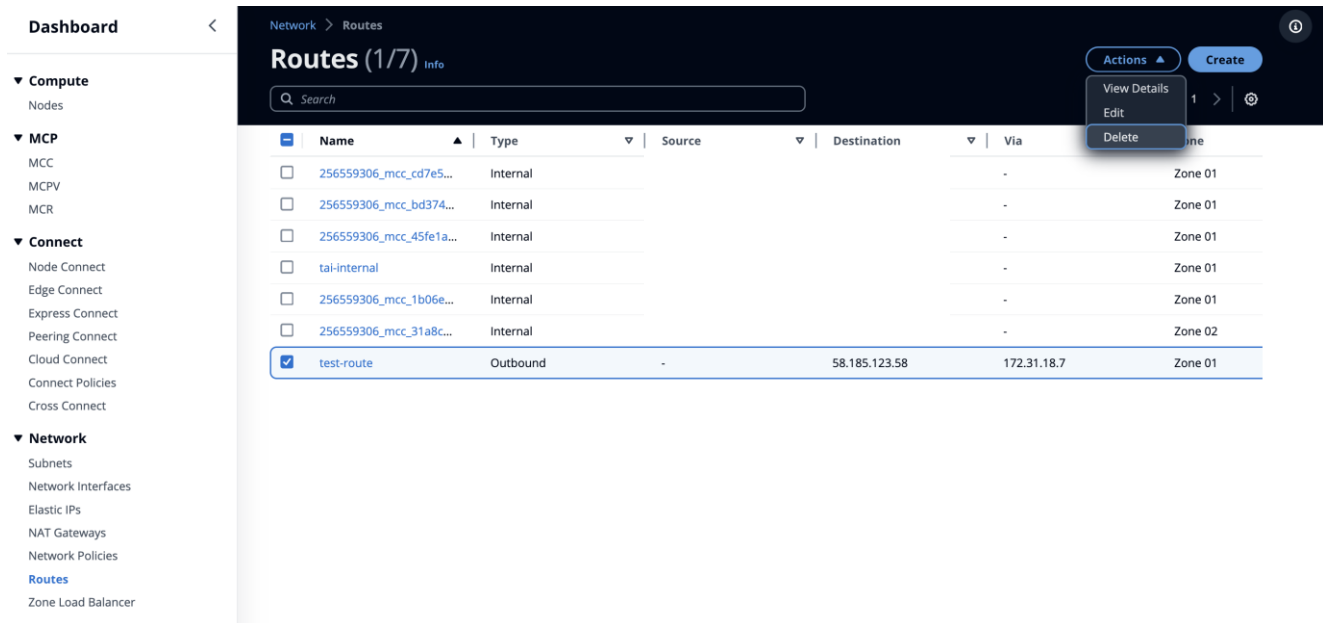
The screenshot displays the 'Routes (7)' dashboard in the Singtel Cloud Manager. The left sidebar shows the navigation menu with categories: Compute (Nodes), MCP (MCC, MCPV, MCR), Connect (Node Connect, Edge Connect, Express Connect, Peering Connect, Cloud Connect, Connect Policies, Cross Connect), and Network (Subnets, Network Interfaces, Elastic IPs, NAT Gateways, Network Policies, Routes, Zone Load Balancer). The main panel shows a table of routes. The 'test-route' is highlighted with a red box.

Name	Type	Source	Destination	Via	Zone
256559306_mcc_cd7e5...	Internal	-	-	-	Zone 01
256559306_mcc_bd374...	Internal	-	-	-	Zone 01
256559306_mcc_45fe1a...	Internal	-	-	-	Zone 01
tai-internal	Internal	-	-	-	Zone 01
256559306_mcc_1b06e...	Internal	-	-	-	Zone 01
256559306_mcc_31a8c...	Internal	-	-	-	Zone 02
test-route	Outbound	58.185.123.58	172.31.18.7	-	Zone 01

Figure: Validation of the Route

Step 8: Once you have successfully created the route, it will be displayed in the "Routes" dashboard.

10.0.7 Deleting a Route



Dashboard

- ▼ **Compute**
 - Nodes
- ▼ **MCP**
 - MCC
 - MCPV
 - MCR
- ▼ **Connect**
 - Node Connect
 - Edge Connect
 - Express Connect
 - Peering Connect
 - Cloud Connect
 - Connect Policies
 - Cross Connect
- ▼ **Network**
 - Subnets
 - Network Interfaces
 - Elastic IPs
 - NAT Gateways
 - Network Policies
 - Routes**
 - Zone Load Balancer

Network > Routes

Routes (1/7) Info

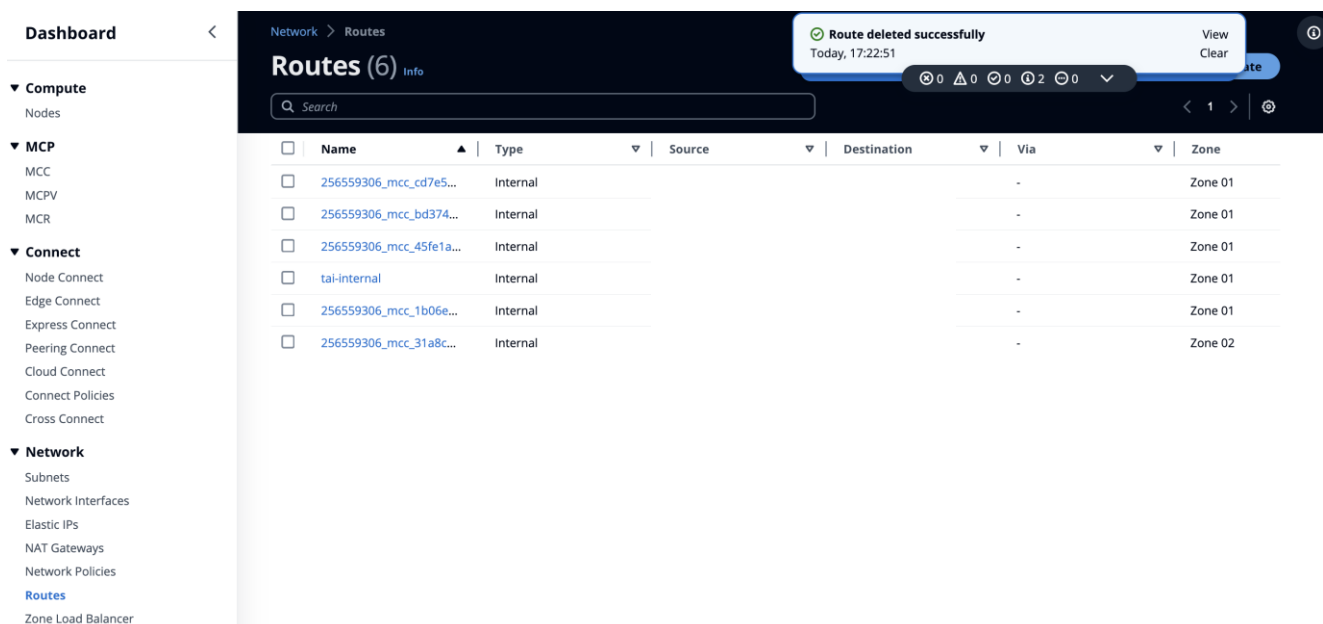
Search

Actions View Details Edit Delete Create

Name	Type	Source	Destination	Via	Zone
☐ 256559306_mcc_cd7e5...	Internal			-	Zone 01
☐ 256559306_mcc_bd374...	Internal			-	Zone 01
☐ 256559306_mcc_45fe1a...	Internal			-	Zone 01
☐ tai-internal	Internal			-	Zone 01
☐ 256559306_mcc_1b06e...	Internal			-	Zone 01
☐ 256559306_mcc_31a8c...	Internal			-	Zone 02
☒ test-route	Outbound	-	58.185.123.58	172.31.18.7	Zone 01

Figure: Routes – Delete

Step 1: To begin the deletion process, make sure the route is selected, then click Actions then Delete. When the pop-up window displays, confirm the deletion.



Dashboard

- ▼ **Compute**
 - Nodes
- ▼ **MCP**
 - MCC
 - MCPV
 - MCR
- ▼ **Connect**
 - Node Connect
 - Edge Connect
 - Express Connect
 - Peering Connect
 - Cloud Connect
 - Connect Policies
 - Cross Connect
- ▼ **Network**
 - Subnets
 - Network Interfaces
 - Elastic IPs
 - NAT Gateways
 - Network Policies
 - Routes**
 - Zone Load Balancer

Network > Routes

Routes (6) Info

Search

Route deleted successfully Today, 17:22:51 View Clear

Name	Type	Source	Destination	Via	Zone
☐ 256559306_mcc_cd7e5...	Internal			-	Zone 01
☐ 256559306_mcc_bd374...	Internal			-	Zone 01
☐ 256559306_mcc_45fe1a...	Internal			-	Zone 01
☐ tai-internal	Internal			-	Zone 01
☐ 256559306_mcc_1b06e...	Internal			-	Zone 01
☐ 256559306_mcc_31a8c...	Internal			-	Zone 02

Figure: Routes – Successful deletion

Step 2: Confirm that the route is not shown in the dashboard, along with the notification banner confirming that the route has been deleted as shown above.

11.0 Node Connect

Node Connect provides inbound connectivity from an external network to a node's network interface through an allocated elastic IP in the same zone. It is a 1:1 mapping of elastic IP to the network interface private IP.

The elastic IP used in a node connect will be dedicated to this connection and not available for other node connects while the current assignment is still active. You can create and delete node connects at any time to allow a single elastic IP to be dynamically assigned to different network interface (limited to one network interface at a time).

11.0.1 Creating a Node Connect

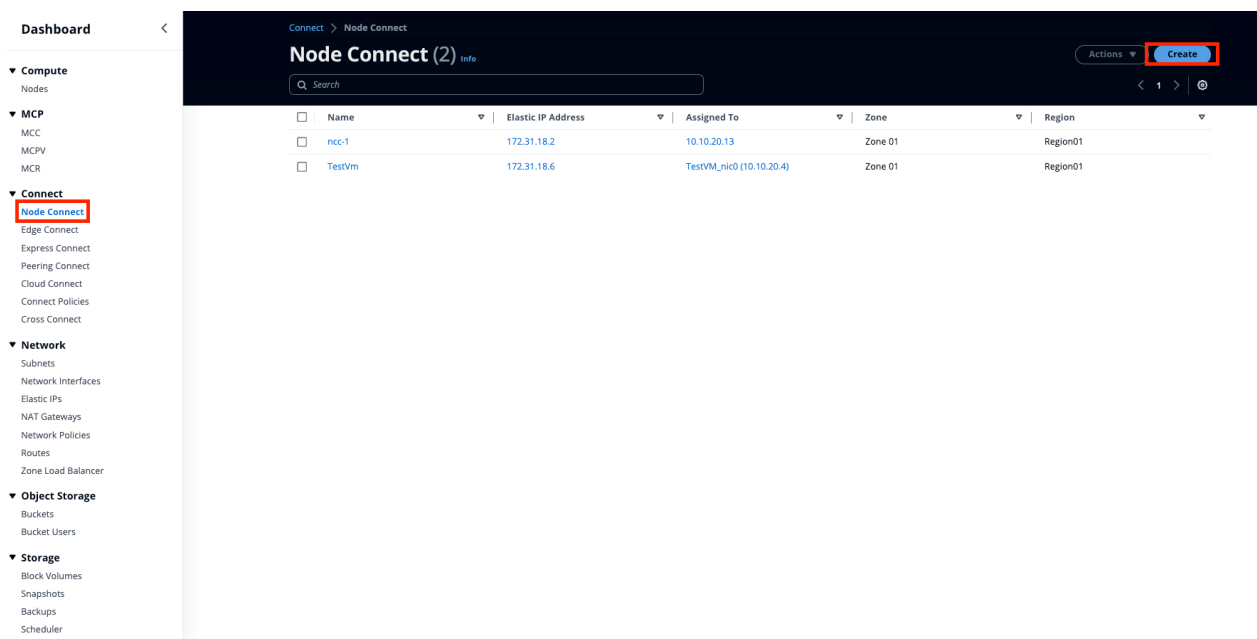


Figure: Node Connect – Create

Step 1: Navigate to Connect, then click on "Node Connect".

Step 2: Click on the "Create" button.

Connect > Node Connect > Create Node Connect

Create Node Connect Info

Name
Enter the display name for this Node Connect
test-vm

Zone
Select an private cloud zone
Zone 01

Node
Select a node to allocate an elastic IP address
testVM2

Network Interface
Select a network interface of the node for this node connect
bb35090c-a05a-41c8-8ba2-c3242b80a666

Elastic IP
Select an elastic IP to provide external connectivity to the target node
172.31.18.3

Cancel Submit

Figure: Node Connect – Form Input for Creating

Step 3: Enter the preferred display name.

Step 4: To load the available node, choose a zone.

Step 5: Choose a Node to load the network interface that is attached. 6. Decide on a Network Interface.

Step 6: Click "Submit" after choosing an Elastic IP.

Dashboard <

Connect > Node Connect

Node Connect created successfully Today, 13:01:43 View Clear

Node Connect (3) Info

Search

Name	Elastic IP Address	Assigned To	Zone	Region
nc-1	172.31.18.2	10.10.20.13	Zone 01	Region01
TestVM	172.31.18.6	TestVM, nc1 (10.10.20.4)	Zone 01	Region01
test-vm	172.31.18.3	10.10.20.8	Zone 01	Region01

Node Connect created successfully Today, 13:01:43 X

test-vm created successfully
Transaction: FF-NC10_0000002402_04
Resource: 8705403a-397c-4860-a5d1-b8b5c6e223f

Close

Figure: Node Connect – Successful creation.

12.0 Creating SSH Keys

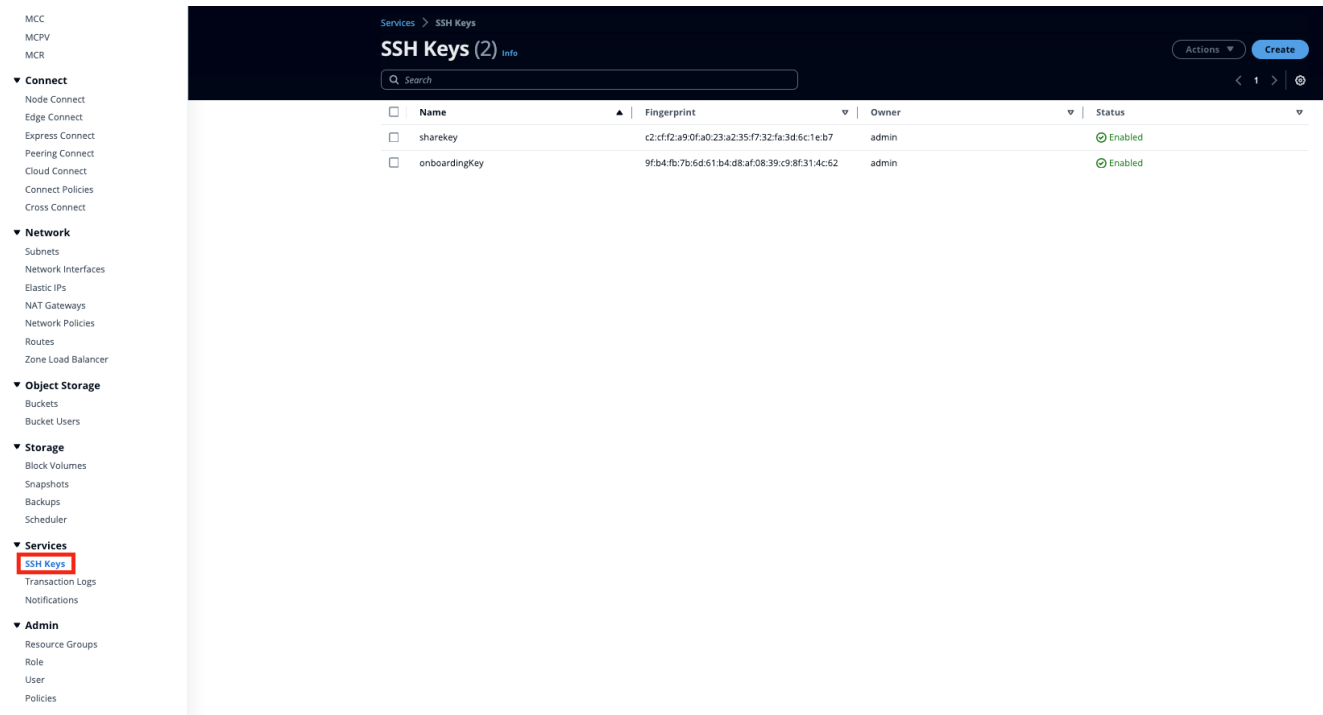


Figure: SSH Keys - Creating SSH Keys

Step 1: Go to “Services” tab, click on “SSH Keys” to begin.



Figure: SSH Keys - Generating SSH Keys

Step 2: Provide a name to associate the SSH Key with.

Step 3: Paste the public key in the textbox provided.

Step 5: Click on the “Create” button.

Once the steps are completed, you should be able to see your SSH key displayed on the dashboard.

Services > SSH Keys

SSH Keys (1/3) Info

Search

Actions Create

	Name	Fingerprint	Owner	Status
☐	sharekey	c2:cf:f2:a9:0f:a0:23:a2:35:f7:32:fa:3d:6c:1e:b7	admin	Enabled
☐	onboardingKey	9f:b4:fb:7b:6d:61:b4:d8:af:08:39:c9:8f:31:4c:62	admin	Enabled
☒	Testkey	92:1d:cc:35:29:08:9b:d8:e6:89:55:b3:48:cb:a1:...	admin	Enabled

Confirm SSH key deletion?

To confirm that you want to permanently delete these SSH keys click the confirm button below.

Are you sure you want to delete these SSH keys?

Testkey (f29bf2ea-fa10-4e0b-a1bd-6f848dfd565c)

Deleting an SSH key does not affect any running node that may have used this SSH key during creation. You will still be able to connect with the key pair you deleted if the node is still running.

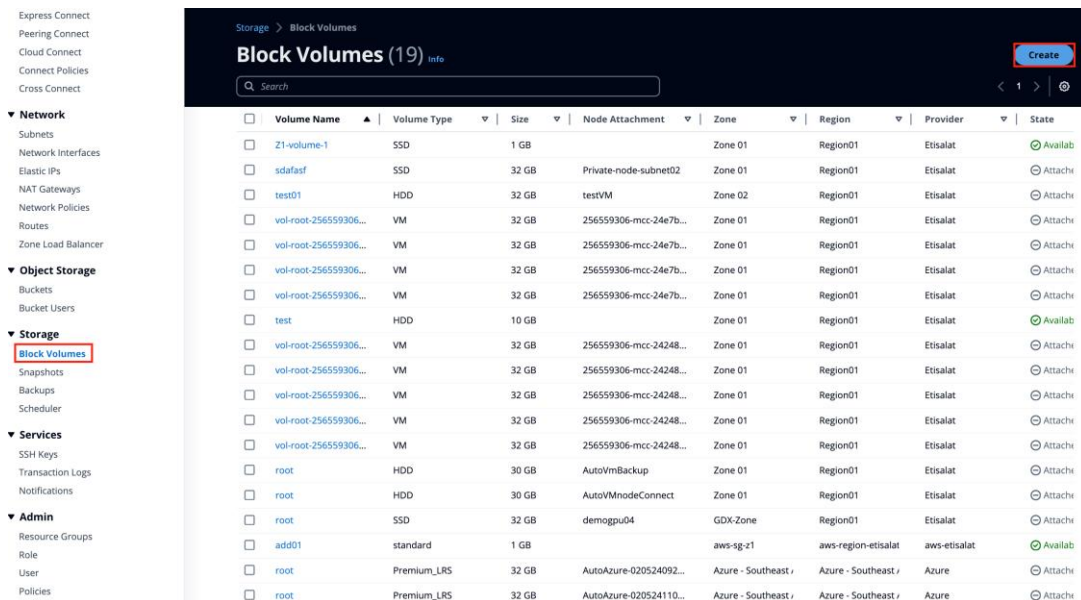
Cancel Confirm

Figure: SSH Keys - Deleting a SSH Key

Select the key you wish to remove, then pick Delete from the Actions menu in the upper right corner. Click on the "Confirm" button to execute the deletion.

13.0 Volume Management

Block volumes can be created from available shared storage offered by Platform Manager to attach directly to available nodes. These volumes are block storage type volumes which are provided by the storage clusters in each region and available zones. When creating a new volume, you can either specify the node you want the volume attached to, or simply create an unattached volume to be attached later on. If attaching at the time of creation, the volume and the node must be within the same zone. You can request any sized volume in GB; however, some larger sizes may not be available. Any volumes you attach will be presented in the node OS as an unformatted block device or volume. You will need to mount and format the volume as per your own requirements. Root volumes are created and attached as part of node provisioning and cannot be attached or detached. From the volumes page you can only manage additional volumes.



Volume Name	Volume Type	Size	Node Attachment	Zone	Region	Provider	State
Z1-volume-1	SSD	1 GB		Zone 01	Region01	Etisalat	Available
sdataf	SSD	32 GB	Private-node-subnet02	Zone 01	Region01	Etisalat	Attached
test01	HDD	32 GB	testVM	Zone 02	Region01	Etisalat	Attached
vol-root-256559306...	VM	32 GB	256559306-mcc-24e7b...	Zone 01	Region01	Etisalat	Attached
vol-root-256559306...	VM	32 GB	256559306-mcc-24e7b...	Zone 01	Region01	Etisalat	Attached
vol-root-256559306...	VM	32 GB	256559306-mcc-24e7b...	Zone 01	Region01	Etisalat	Attached
vol-root-256559306...	VM	32 GB	256559306-mcc-24e7b...	Zone 01	Region01	Etisalat	Attached
test	HDD	10 GB		Zone 01	Region01	Etisalat	Available
vol-root-256559306...	VM	32 GB	256559306-mcc-24248...	Zone 01	Region01	Etisalat	Attached
vol-root-256559306...	VM	32 GB	256559306-mcc-24248...	Zone 01	Region01	Etisalat	Attached
vol-root-256559306...	VM	32 GB	256559306-mcc-24248...	Zone 01	Region01	Etisalat	Attached
vol-root-256559306...	VM	32 GB	256559306-mcc-24248...	Zone 01	Region01	Etisalat	Attached
vol-root-256559306...	VM	32 GB	256559306-mcc-24248...	Zone 01	Region01	Etisalat	Attached
root	HDD	30 GB	AutoVmBackup	Zone 01	Region01	Etisalat	Attached
root	HDD	30 GB	AutoVmnodeConnect	Zone 01	Region01	Etisalat	Attached
root	SSD	32 GB	demogpu04	GDX-Zone	Region01	Etisalat	Attached
add01	standard	1 GB		aws-sg-z1	aws-region-etisalat	aws-etisalat	Available
root	Premium_LRS	32 GB	AutoAzure-020524092...	Azure - Southeast	Azure - Southeast	Azure	Attached
root	Premium_LRS	32 GB	AutoAzure-020524110...	Azure - Southeast	Azure - Southeast	Azure	Attached

Figure: Storage – Block Volumes

To get started, navigate to the "Storage" tab, and select "Block Volumes."

Storage > Block Volumes > Create Block Volume

Create Block Volume Info

Provider

Region

Zone

Volume Type Info

Volume Size

Node Attachment - optional Info

Name

Cancel

Submit

Figure: Block Volumes - Creating a Block Volume

Step 2: Complete the essential information in the details.

This includes:

1. Azure/AWS cloud as the preferred provider.
2. The volume type, which is decided by the service provider.
3. the zone from where the block volume will be located.
4. the block volume's location.
5. the volume size, which specifies the volume's overall size in gigabytes.
6. (Optional) Attaching a node from an available node above a zone and region of choice.
During the creation process, this volume will automatically attach; it won't be formatted or mounted.

Finally, click on "Submit".

Storage > Block Volumes

Block Volumes (22) Info

Search

Create

☐	Volume Name	Volume Type	Size	Node Attachment	Zone	Region	Provider	State
☐	AutoVmVolumeVol	SSD	2 GB	AutoVmBackup	Zone 01	Region01		⊖ Attached
☐	vm-test -block	VM	10 GB		Zone 01	Region01		⊕ Available

Figure: Block Volumes – Successful creation

If the volume is created and not assigned to a node, it should be reflected on the block volume dashboard with the state 'Available'.

13.0.1 Attaching a Volume

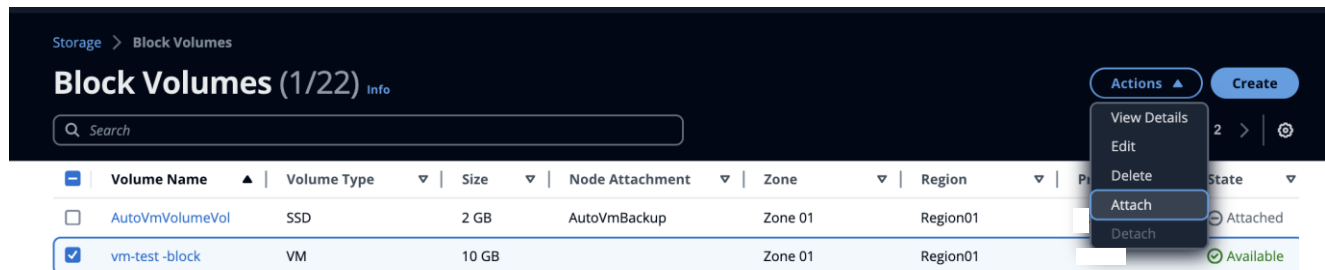


Figure: Block Volumes - Attaching a Volume

Step 1: To attach the block volume, click on the checkbox on the volume then click on Attach under Actions.

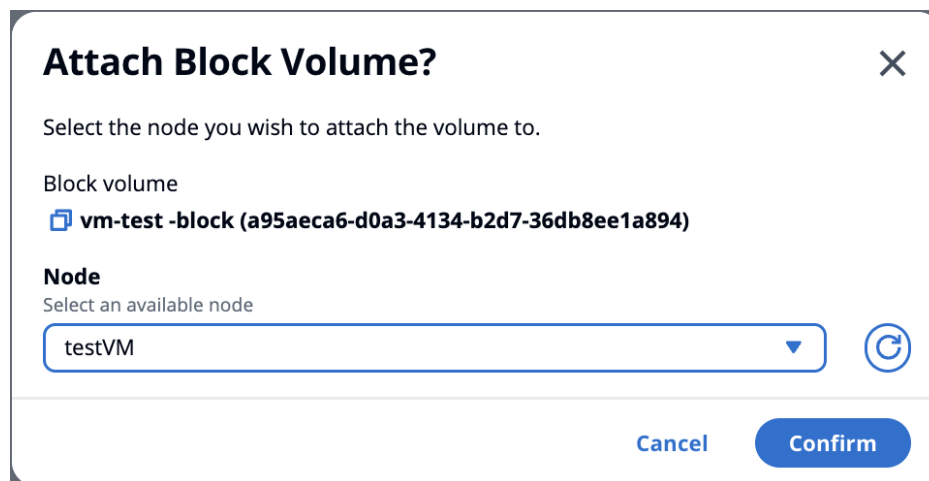


Figure: Attaching a Block Volume – Confirmation

Step 2: A pop up will appear to let you select on a node that you would need to attach to. Click on a node from the dropdown menu to select and confirm.

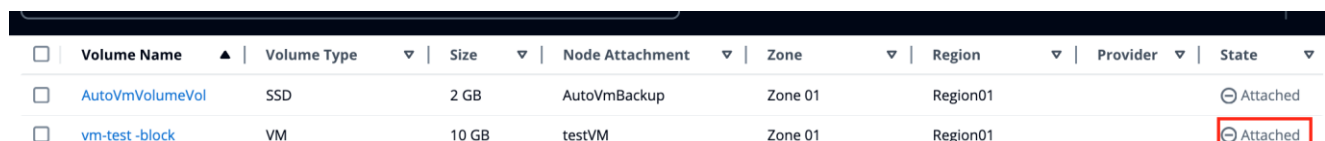


Figure: Block Volumes – Attached State

Step 3: Check the block volume on the dashboard to make sure it's successful. Instead of displaying as available, the state should now be attached.

14.0 Cloud Connect

Cloud Connect enables secure connectivity between MEC Manager zone subnets and public cloud subnets by establishing IPsec tunnels. This allows you to connect your MEC Manager subnets either to subnets created within your instance or to subnets in your own cloud accounts. The process leverages public cloud VPN endpoints to create secure connections between your distributed resources and public cloud regions.

Cloud Connect offers automation in the provisioning process, including options to reuse existing public cloud resources like VPN gateways and route tables, helping to streamline setup and minimize disruptions to your existing infrastructure. Depending on the type of subnet (managed or unmanaged), the platform handles the connection either automatically or through credentials that allow access to your public cloud resources for configuration.

14.0.1 Creating a Cloud Connect

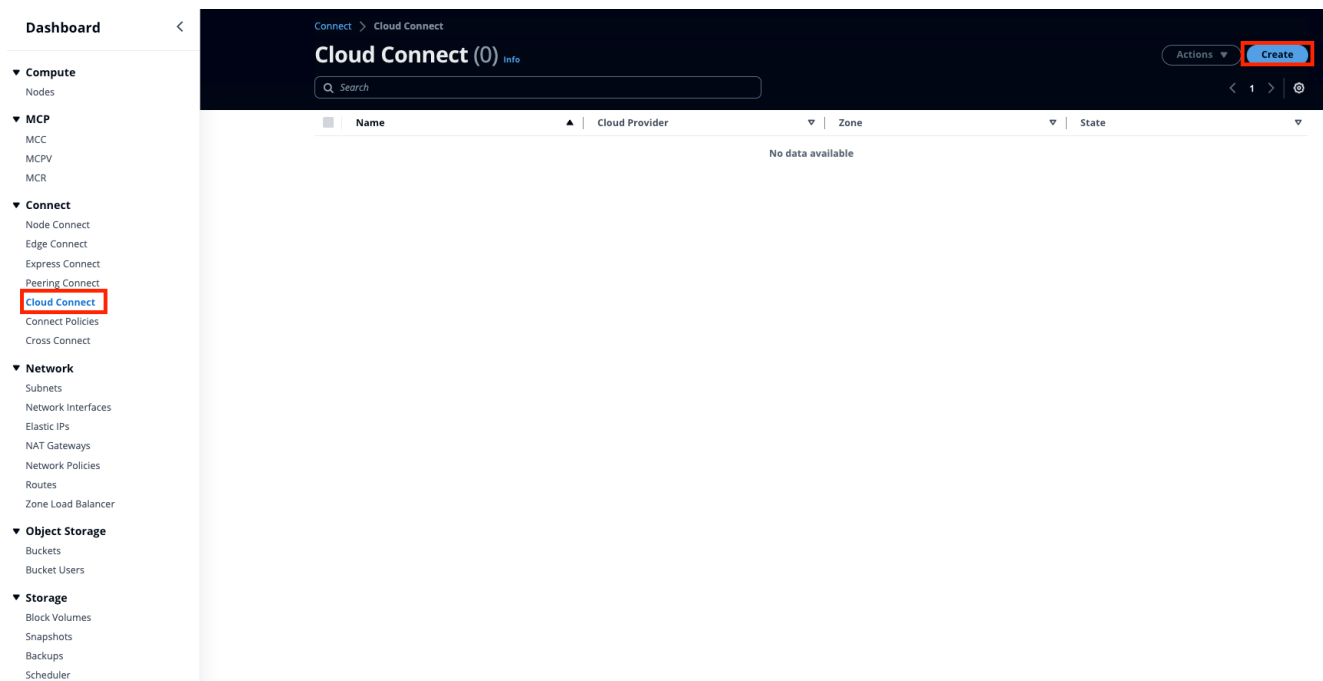


Figure: Cloud Connect – Creating a Cloud Connect

To begin, click onto the Cloud Connect label under the Connect section in the side menu, then click on "Create".

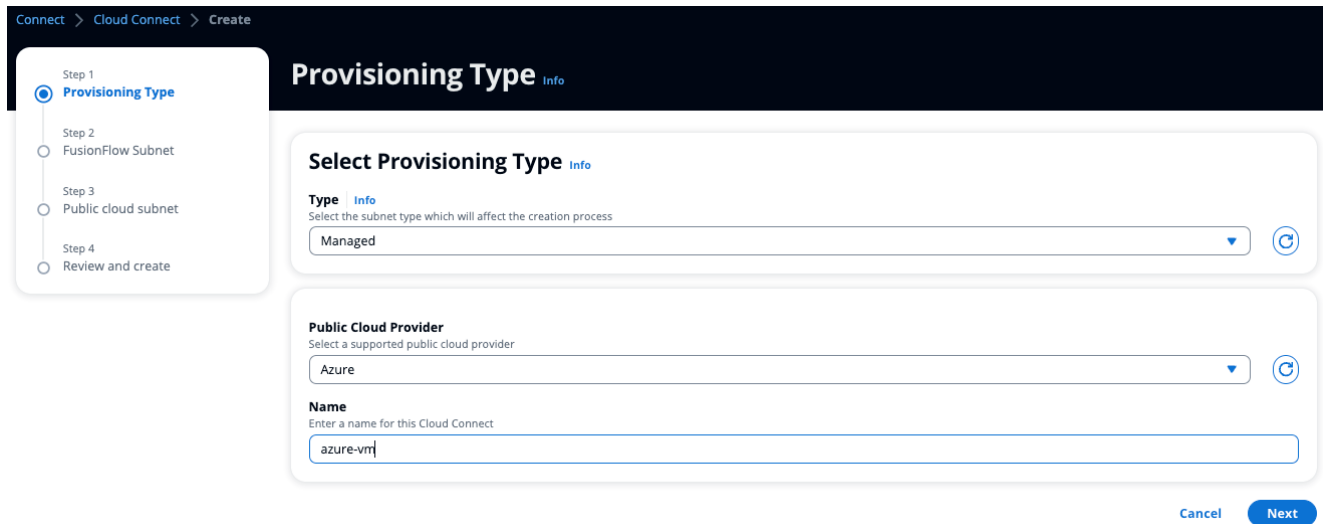


Figure: Creating a cloud connect – Select Provisioning type

Step 1: Select the type between Managed or Unmanaged.

Step 2: Select the supported cloud provider.

Step 3: Enter the preferred name for this cloud connect then click Next.

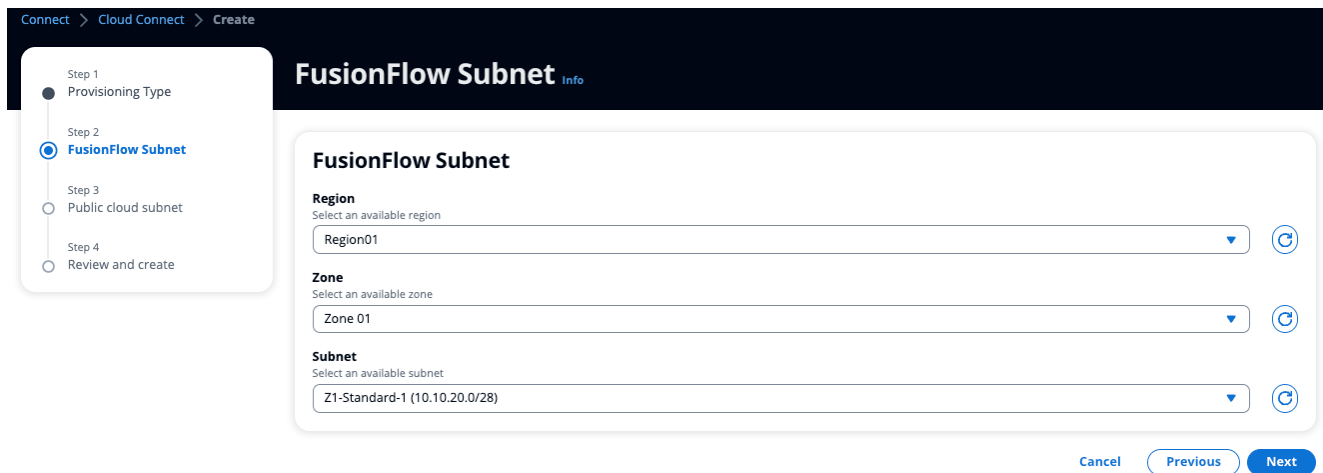


Figure: Creating a cloud connect – Subnet

Step 4: Select the region from the dropdown menu.

Step 5: Select the zone from the available list.

Step 6: Select the subnet then click Next.

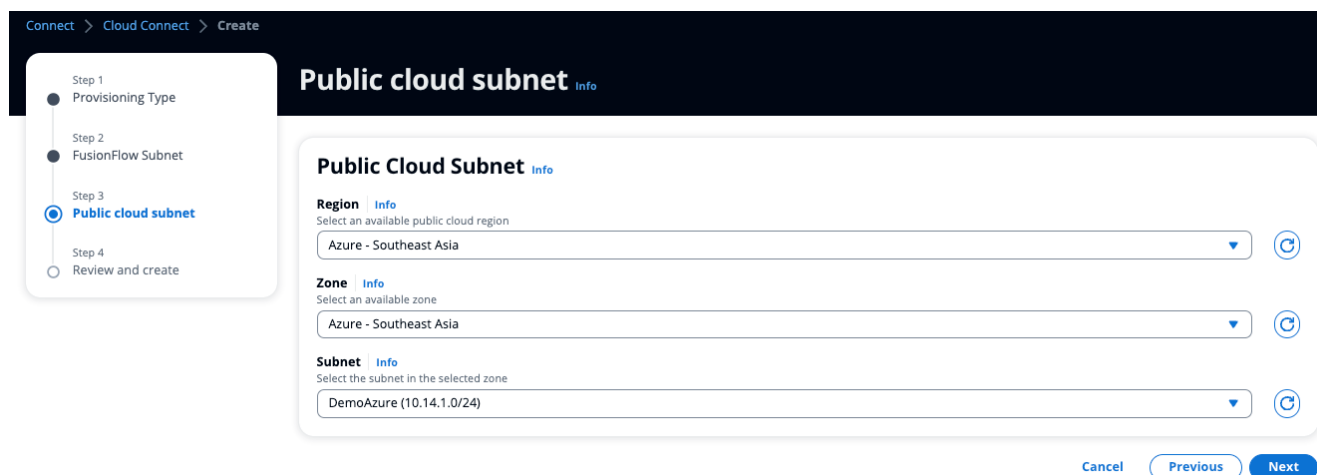


Figure: Creating a cloud connect – Public Cloud Subnet

Step 7: Select the region from the available list, this will populate the zone field as well.

Step 8: Select the subnet in the zone, then next.

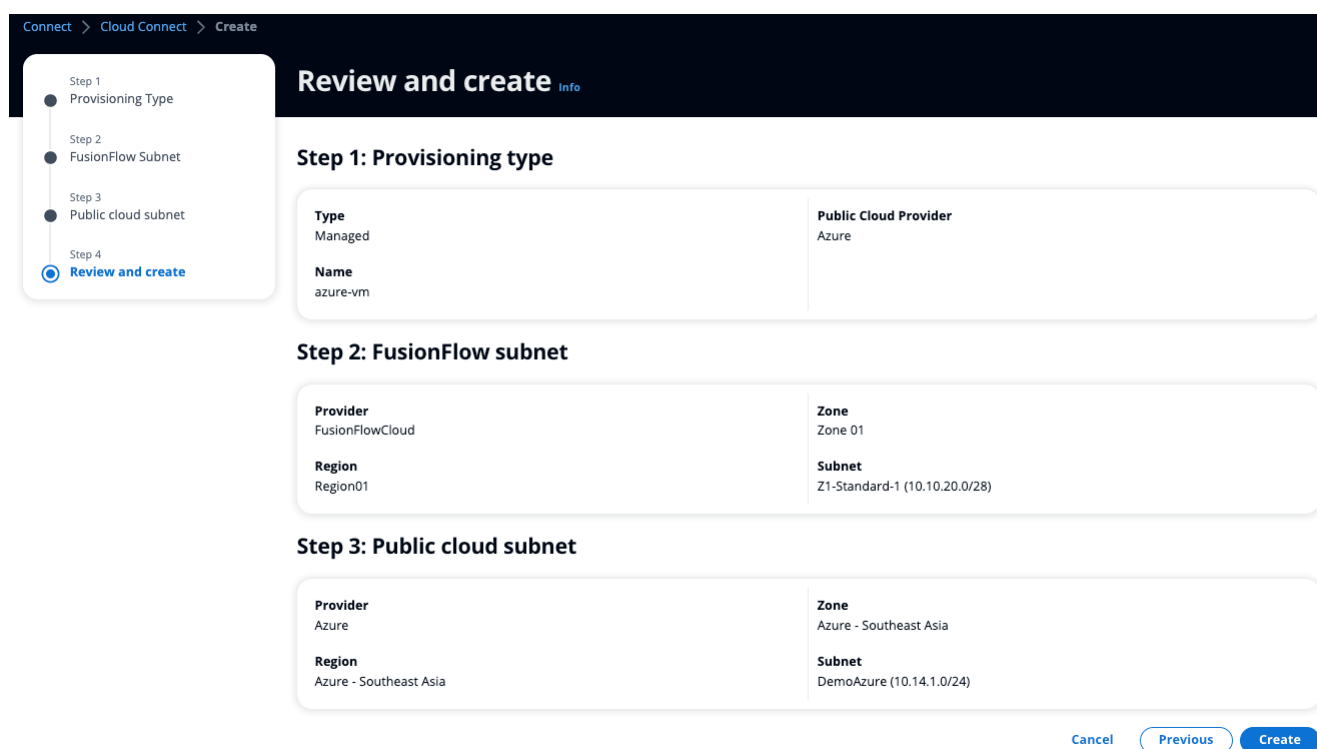


Figure: Creating a cloud connect – Review and create.

Finally, upon filling all the required fields for this creation, review the information for each phase and click on the “Create” button when all information provided is accurate as per your needs.

15.0 Cross Connect

Cross Connect allows users to link different cloud or express connections directly within a single region. Each connection creates one or more gateways (Cloud Gateways or Express Gateways), and

Cross Connect enables these gateways to communicate directly, allowing seamless connectivity between different cloud or express connections within the same zone.

If users have at least two zone-level connections (cloud or express), they can use Cross Connect to link them. This is done by adding the external networks associated with each connection into the Cross Connect, which then establishes direct communication between them.

For example, if users connect a customer site network and a public cloud network (like Azure), they enable these networks to communicate directly using private IP addresses. The existing infrastructure takes care of routing the traffic without impacting the subnets in the zone. This setup allows multiple connections to interact smoothly without disrupting any existing network paths.

By setting up Cross Connect, users make it easy for traffic to flow between networks, while still keeping the core zone setup unchanged.

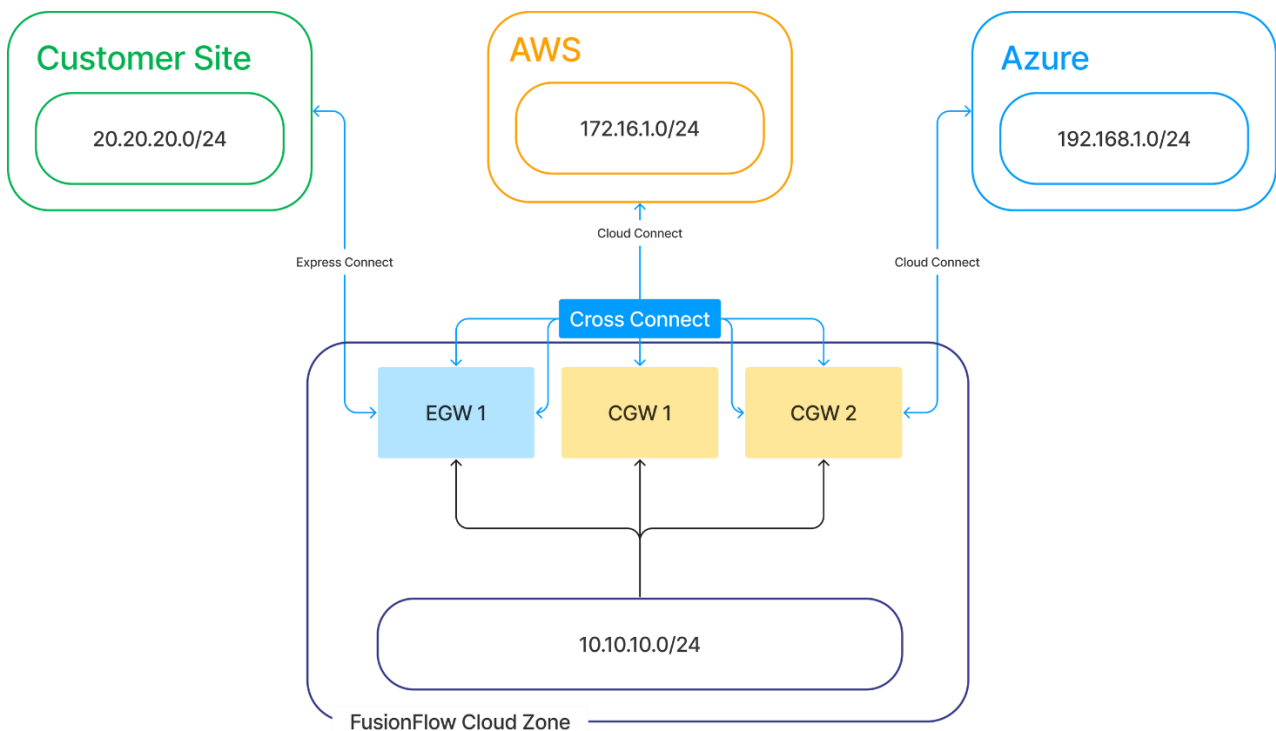


Figure: Cross Connect Flow Chart

15.0.1 Creating a Cross Connect

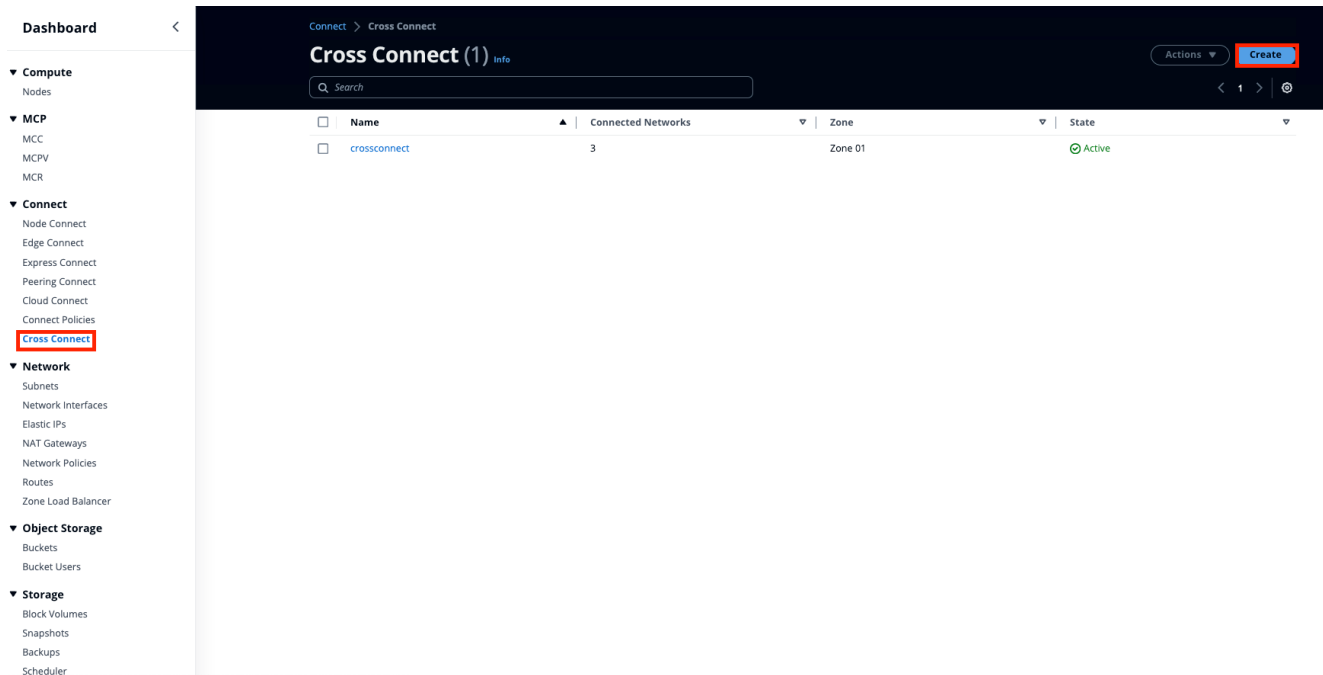


Figure: Cross Connect – Creating a Cross Connect

Step 1: Navigate to the side menu and expand the Connect menu, then click on Cross Connect.

Step 2: Click on the "Create" button.

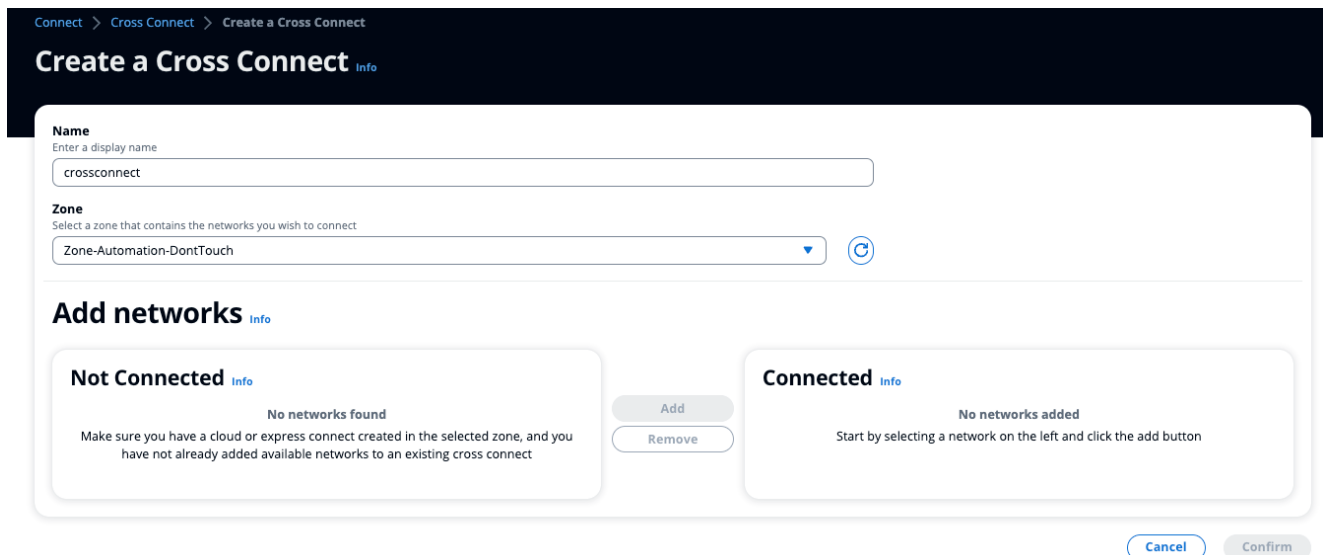


Figure: Creating a Cross Connect – Form input.

Step 3: Enter the display name for the cross connect.

Step 4: From the drop-down menu, choose a Zone that is open for use.

The list below ought to show you a list of networks that are "Not Connected."

If you are unable to see any networks, confirm that the zone has cloud or express connections accessible and that no other cross connects are using them.

Click the Add button in the center after selecting the pertinent networks from the "Not Connected" list. They ought to go to the list marked as "Connected."

Using the add and remove buttons, you can change the networks that are included in these lists.

To continue with the creation, at least two networks must be present in the Connected list.

Press the Confirm button. Once again, you should see a new entry in the list with the status of Creating when you are taken back to the list page. When it's finished, you'll get a notification and transaction. If the creation process was successful, the status ought to appear as Active.

16.0 MCP (Multi-Cloud Container Platform)

16.1.0 MCC (Multi-Cloud Container Cluster)

Managed Kubernetes clusters, or MCCs, are established using your instance. Although there are several standard features for automated cluster provisioning and management, these clusters' scalability sets them apart from the competition. It enables you to extend your cluster across multiple cloud providers and geographical locations, giving you the most flexibility in terms of where your services and applications can run within each cluster.

The volumes and load balancers are directly integrated into Multi-cloud container clusters, which are also compatible with regular Kubernetes management tools.

16.1.1 Creating a Multi-Cloud Container Cluster (MCC)

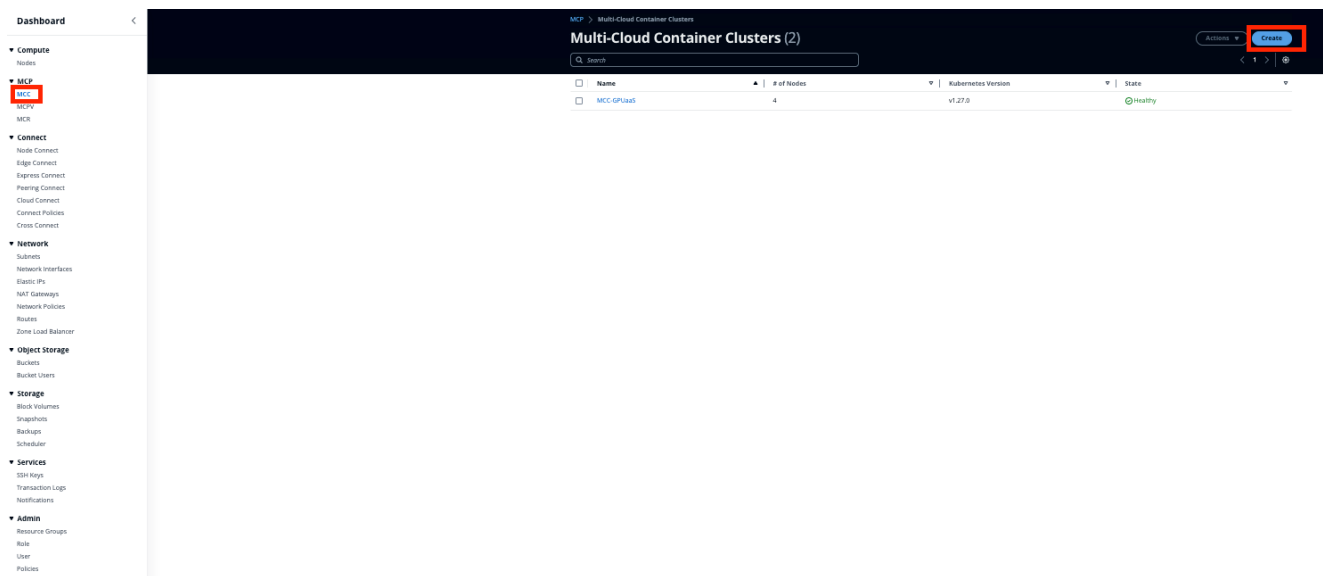


Figure: MCC – Creating a MCC

To begin, navigate to the MCP section on the side menu, and click on MCC.

Then, click on Create to proceed.

MCP > Multi-Cloud Container Clusters > Create MCC

Step 1

Enter details

Step 2

Review and create

Enter details

Deployment location
[Info](#)

The initial deployment of a Multi-Cloud Container Cluster must be within a FusionFlow Cloud Zone. You will be able to add workers to other zones after the initial creation

Provider
Select an available provider
Etisalat

Region
Select an available region
Region01

Core Zone
Select an available zone where the cluster will be provisioned
Zone 01

Cluster settings
[Info](#)

Name
Enter a name for this cluster. This will also be the name set in the cluster configuration
MCC-GPUaaS

External Domain
[Info](#)
Enter the external domain for cluster management. Default will use the cluster name provided
MCC-GPUaaS.tenant-bdcfce0e43ccb7e3.singtel5gplatform.com

Kubernetes Version
Select an available version
v1.27.0

Cluster High Availability
Enabling this toggle creates multiple master nodes which helps reduce downtime in this cluster
☒ Enable Cluster High Availability

Advanced
Customize the Kubernetes pod and service networks if you wish to use a different network than the provided defaults. The size of these networks will determine how many pods can be deployed

Internal Domain
[Info](#)
Enter a new internal domain for this cluster. Default will use the cluster name provided
MCC-GPUaaS.local

Kubernetes Pod Network CIDR
Define the Kubernetes pod network CIDR if you want to use something other than the default
10.231.0.0/23

Kubernetes Service Network CIDR
Define the Kubernetes service network CIDR if you want to use something other than the default
10.239.0.0/23

Figure: Creating a MCC – Form Inputs

Control plane settings Info

Set the control plane node sizing. The control plane and worker nodes need to be deployed in separate subnets within the selected zone. You can choose one of your existing subnets or a new one will be created for you.

Control Plane Node Size
Select a node size from the available list

Standard 4.8 (4 vCPU, 8 GB Memory) 

Network

The control plane nodes for this cluster must be in a dedicated subnet that is not shared with any workers nodes. Either a new subnet will be created automatically or you can select an existing one

Existing Subnet
Select an existing subnet in the deployment zone

Z1-ControlPlane-1 (10.10.20.16/28) 

Worker node settings Info

Set the worker node sizing and quantity. The worker nodes need to be deployed in a separate subnet from the control plane. You can choose one of your existing subnets or a new one will be created for you.

Worker Node Size
Select a node size from the available list

Standard 4.8 (4 vCPU, 8 GB Memory) 

Number of workers
Enter the number of worker nodes you want to provision in this cluster. Up to 10 nodes can be provisioned

3

Network

The worker nodes for this cluster must be in a separate subnet than the master nodes. Either a new subnet will be created automatically or you can select an existing one, provided it is different than the one above

Existing Subnet
Select an existing subnet in the deployment zone

Z1-Standard-1 (10.10.20.0/28) 

Additional options Info

Select any available options that will be applied during the cluster creation process

Crossplane Info
Automatically deploy Crossplane in this cluster by selecting this option

☒ Install Crossplane

Cancel
Next

Figure: Creating a MCC – Form Inputs Pt 2

Step 3: Select Provider, Region, and Core Zone:

Provider: Choose the cloud provider where you want to deploy your cluster (e.g., AWS, Azure, GCP).

Region: Select the geographical region for deployment.

Core Zone: This is where the control plane nodes will be initially deployed. It acts as the "core zone" for your cluster.

Step 4: Enter a Name for the Cluster:

Provide a unique name for your cluster.

Step 5: Enter the Desired External Domain:

This will be the domain used to access and manage your cluster. It will be appended as a sub-domain of your current instance domain.

Step 6: Select Kubernetes Version:

Choose the version of Kubernetes you want to use for your cluster.

Step 7: Enable Cluster High Availability (Optional):



Provision multiple control plane nodes (typically 3) for high availability. This creates a virtual IP for failover.

Step 8: Configure Advanced Options (Optional):

Customize internal Kubernetes domain, pod, or service networks if needed.

Step 9: Select Control Plane Node Size:

Choose the size/configuration of control plane nodes.

Step 10: Define Control Plane Subnet:

Create a new subnet or select an existing one for the control plane nodes. Ensure it's different from the worker node subnet but in the same zone.

Step 11: Select Worker Node Size:

Choose the size/configuration of worker nodes.

Step 12: Define Worker Node Subnet:

Create a new subnet or select an existing one for the worker nodes. Ensure it's different from the control plane subnet but in the same zone.

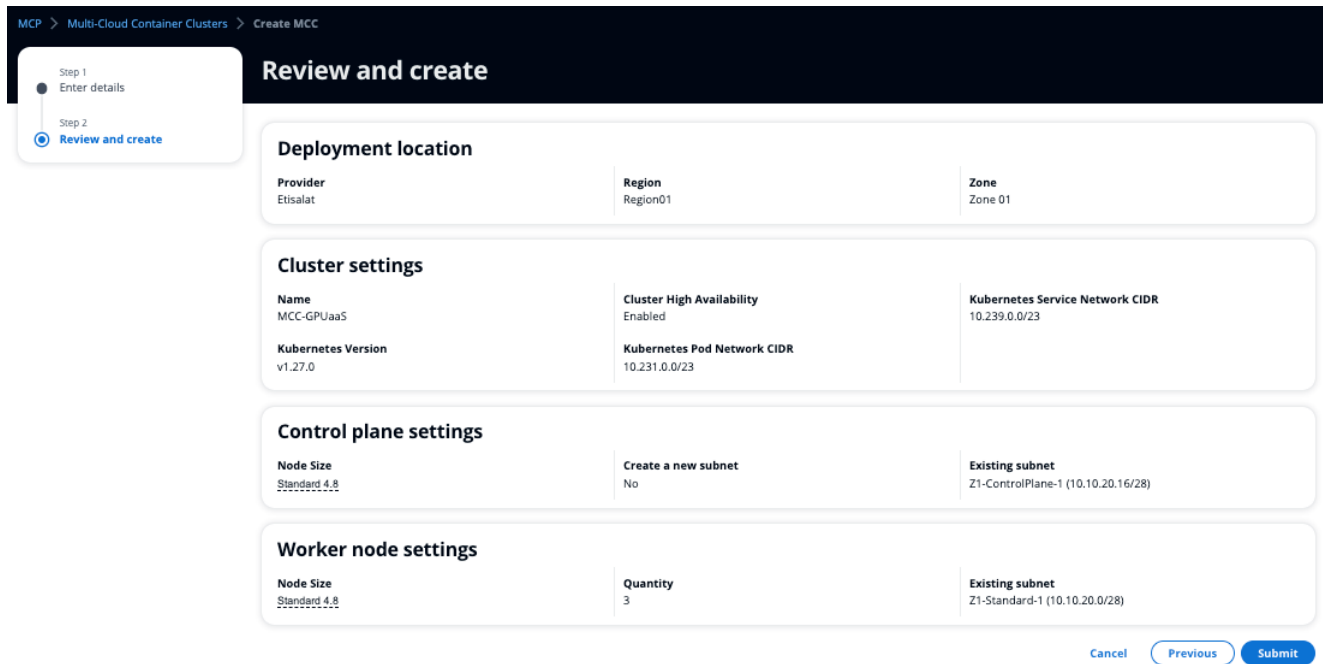
Step 13: Specify Number of Workers:

Choose the quantity of worker nodes you want to initially deploy (from 1 to 10).

Final Checks and Confirmation:

Review all selections and configurations before proceeding with the cluster creation.

These steps ensure that your MEC cluster is properly configured with the desired settings, network configurations, and high availability options.



Review and create

Deployment location

Provider Etisalat	Region Region01	Zone Zone 01
-----------------------------	---------------------------	------------------------

Cluster settings

Name MCC-GPUaaS	Cluster High Availability Enabled	Kubernetes Service Network CIDR 10.239.0.0/23
Kubernetes Version v1.27.0	Kubernetes Pod Network CIDR 10.231.0.0/23	

Control plane settings

Node Size Standard 4.8	Create a new subnet No	Existing subnet Z1-ControlPlane-1 (10.10.20.16/28)
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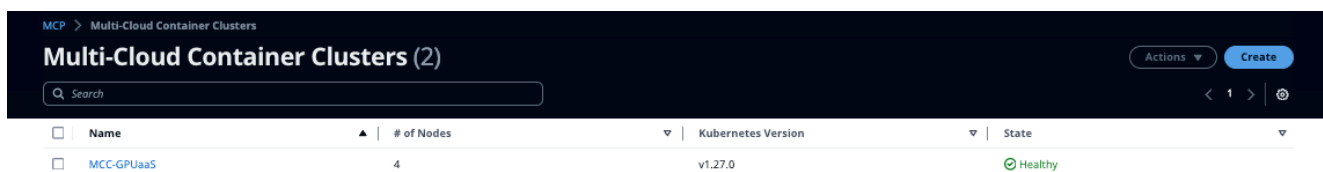
Worker node settings

Node Size Standard 4.8	Quantity 3	Existing subnet Z1-Standard-1 (10.10.20.0/28)
----------------------------------	----------------------	---

Cancel Previous Submit

Figure: Creating a MCC – Review and create

Step 14: Review all the information and then proceed to click on Submit.



Name	# of Nodes	Kubernetes Version	State
MCC-GPUaaS	4	v1.27.0	Healthy

Figure: MCC – Successful Creation

Step 15: The cluster would appear on the dashboard and the state will be reflected as healthy if the creation was successful.

16.2.0 MCR (Multi-Cloud Container Registry)

You may manage container images with the Multi-cloud Container Registry (MCR), a private container registry. In terms of image management, there are a few typical functions like pushing and pulling images, but the main unique features are as follows:

- The ability to easily set up a Registry Server.
- Native integration with container clusters for Multi-cloud computing (MCC). This makes it possible to use the images hosted within MCR to easily install applications on MCC.
- Secure and reliable picture management from several cloud providers.
- Reassurance that, by turning on the vulnerabilities scanning option, images are scanned and clear of vulnerabilities.
- Reduction of image size in the MCR to optimize the number of saved images.

16.2.1 Creating a Multi-Cloud Container Registry (MCR)

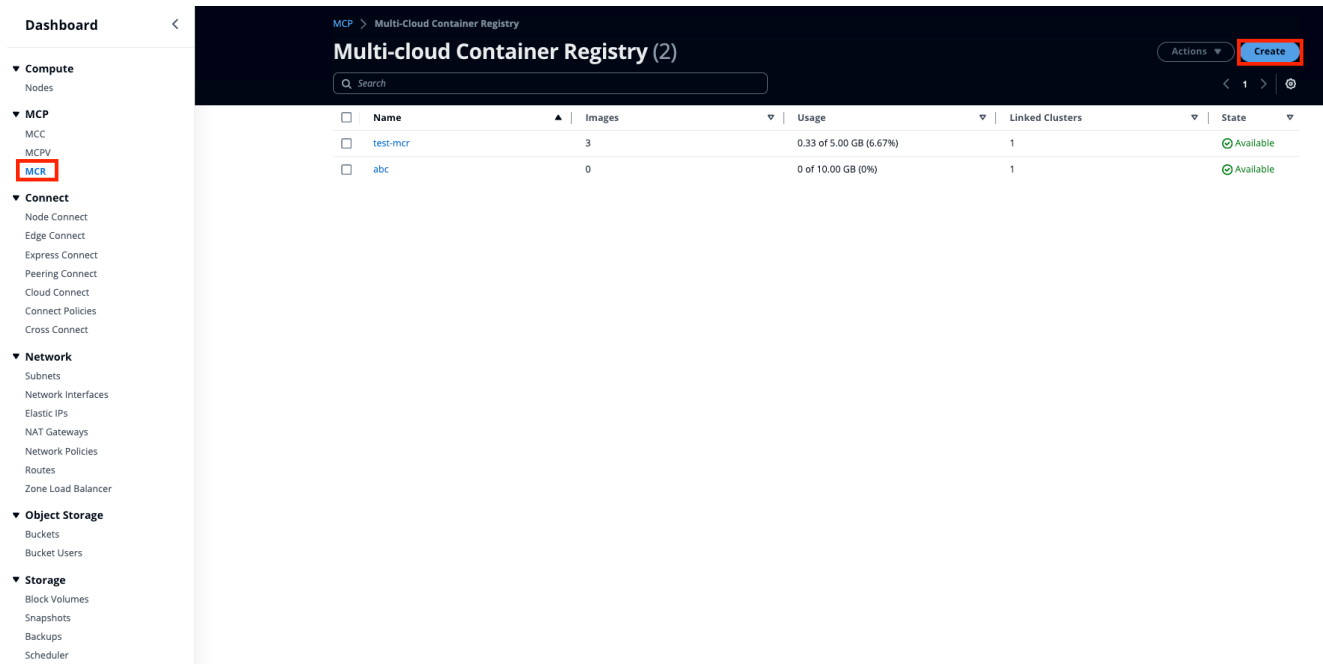
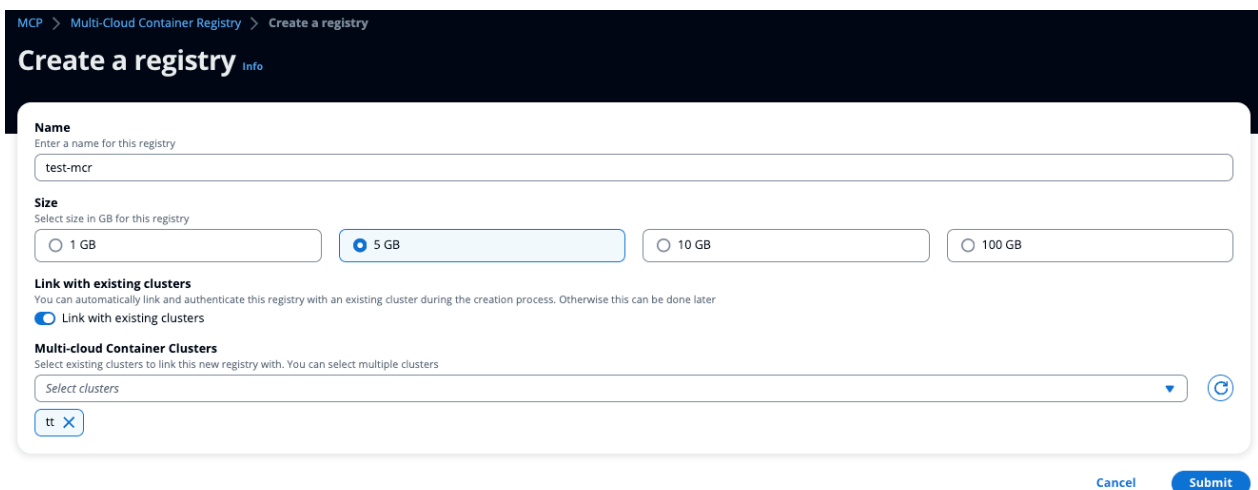


Figure: MCR – Creating a Multi-Cloud Container Registry

Step 1: To begin, navigate to the MCP section on the side menu, and click on MCR.

Then, click on Create to proceed.



Create a registry Info

Name
Enter a name for this registry
test-mcr

Size
Select size in GB for this registry
☐ 1 GB
 ☒ 5 GB
 ☐ 10 GB
 ☐ 100 GB

Link with existing clusters
You can automatically link and authenticate this registry with an existing cluster during the creation process. Otherwise this can be done later
☒ Link with existing clusters

Multi-cloud Container Clusters
Select existing clusters to link this new registry with. You can select multiple clusters
 Select clusters

Cancel Submit

Figure: MCR – Creating a Registry

Step 2: Give your MCR a display name.

Choose a Registry Size in Step Three.

Step 4: Please choose the MCC that you would like to link with this MCR if you already have one.

Step 5: Go over the information and press Submit to create.

Use the Registry URL to access your MCR after it has been created.

16.2.2 Provisioning Access for Multi-Cloud Container Registry

You must first create an Access in order to obtain the login credentials you'll need to connect using your CLI in order to access your MCR.

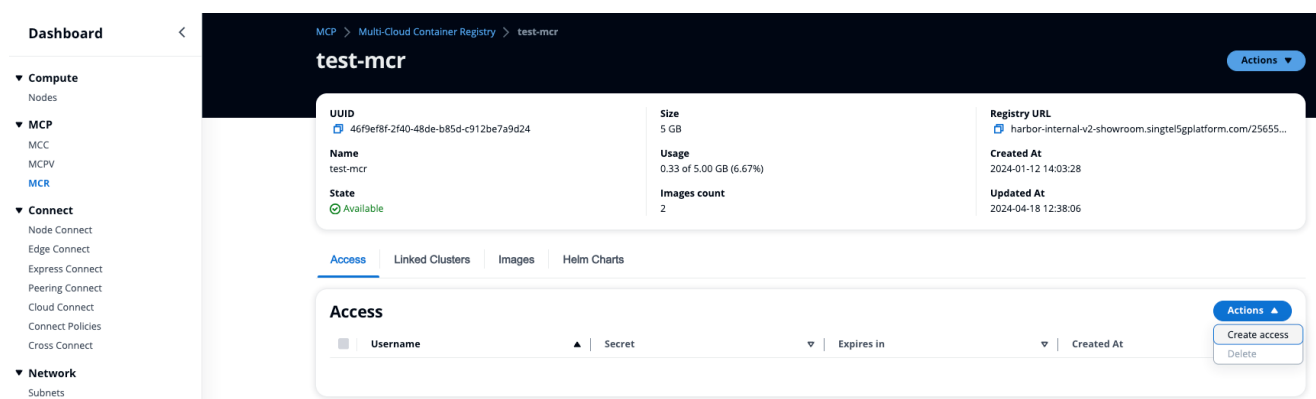


Figure: MCR – Creating Access for MCR

1. Select MCP then MCR.
2. Select the MCR for which you want to establish access.

Create Access

To push an image into this registry, you first need to create access to log in via the docker command

Name

Enter a user name for this access.

test-mcr

Expiration time

Enter the number of days this access will be expired. Enter -1 for unlimited.

-1

Cancel

Confirm

Figure: MCR- Providing Access Credentials

Click Actions then Create access after selecting the Access tab.

3. For this access, provide your username. Please be aware that the username you submit here may not exactly match the one used to log into your MCR.

4. Enter the duration in days for which this access will expire. Or, for unrestricted access, type "-1."
5. Press Confirm.

Access				Actions
☐	Username	☐ Secret	☐ Expires in	☐ Created At
☐	robot\$256559306_test-mcr+test-mcr	☐ k9u4lwc9gQCRquWV2jcuVeh9DrwMNdVC	Unlimited	2024-04-18 12:50:57

Figure: Access - Generated user credentials

The generated username and secret can be found on the Access tab after the access has been created.

Glossary of Terms, Abbreviations and Acronyms

Term	Definition
SSH	SSH (Secure Shell) is a network protocol that provides a secure way to access and manage remote devices or servers over a potentially unsecured network. It is commonly used for secure remote administration and file transfers.
Nodes	Node is a virtual machine that runs one or more service instances (instances of a specific version of service code) on a cluster within a service mesh.
VM	A virtual machine (VM) is a software emulation of a physical computer. It allows multiple operating systems to run on a single physical host computer, enabling efficient resource utilization and isolation.
Scheduler	A scheduler is a component responsible for managing and allocating resources, such as CPU time, memory, or network bandwidth, to different tasks or processes efficiently.
API	An API (Application Programming Interface) is a set of rules and protocols that allows different software applications to communicate and interact with each other. APIs are crucial for enabling the integration of diverse services and functionalities.
MEC	MEC (Multi-Access Edge Computing) refers to a network architecture that brings cloud computing capabilities closer to the edge of the network. It enables low-latency, high-performance computing for applications and services in proximity to end-users and devices.
Volume	A volume typically refers to a storage unit, which can be a logical partition, used for storing data.
Pods	In the context of containerization, a pod is the smallest deployable unit that can contain one or more containers. It is used in container orchestration systems like Kubernetes to manage and scale applications.
Snapshot	A snapshot is a point-in-time copy or image of a virtual machine or storage volume. Snapshots are often used for backup, disaster recovery, and system state preservation.
DNN	DNN typically refers to "Data Network Name." It's an identifier within mobile networks used to differentiate and route traffic for specific services or applications. DNNs are crucial in 5G networks, especially in the context of network slicing, where they help segregate and manage different services or applications efficiently by assigning each a unique DNN.
IMSI	IMSI stands for International Mobile Subscriber Identity. It's a unique identification number associated with a SIM card in mobile networks. The IMSI is used to identify



	individual subscribers and allows mobile networks to authenticate and authorize the connection of a device to their network.
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